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Explainable artificial intelligence for medical imaging systems using deep learning: a comprehensive review

Essam H. Houssein[1,2] · Amr M. Gamal[1] · Eman M. G. Younis[1] · Ebtsam Mohamed[1]

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Abstract

The world recently witnessed strong growth in artificial intelligence (AI) use across various sectors, driven by the digital revolution that began in 2016. Despite this progress, significant concerns persist regarding the black-box nature of AI. Intelligent systems provide decisions without explanations, which has raised pressing issues, particularly in critical domains such as medicine. In medicine, errors can lead to disastrous consequences, putting lives at risk. The "unexplainable" nature of AI is a heavily debated topic in biomedical informatics and computing. Many "black-box" algorithms and systems obscure the logic behind their decisions, leaving users and even developers in the dark about how results are derived. Researchers have developed the field of explainable artificial intelligence (XAI), which holds significant promise for fostering confidence and openness between AI systems and their users. Unlike traditional AI methods, such as deep learning (DL), XAI provides mechanisms for decision-making while offering explanations that are understandable to humans. Medical imaging (MI) plays a crucial role in diagnosing and monitoring a broad range of diseases, and advancements in computer vision, image processing, and the availability of medical image datasets have revolutionized automated MI analysis. However, trust in these systems remains a challenge. To gain the trust of clinicians, authorities, and patients, diagnostic methodologies must be transparent, interpretable, and explainable, clearly conveying the rationale behind specific decisions. This paper reviews the current landscape of XAI methods for medical imaging, including methodologies, techniques, and applications. It covers various types of explanations, such as visual explanations, textual justifications, and example-based reasoning, emphasizing their significance in medical imaging contexts. Furthermore, the paper presents a comparative analysis of XAI methods, evaluating their effectiveness, interpretability, and alignment with medical standards. By identifying research gaps and exploring potential advancements, this review aims to contribute to the development of robust, interpretable, and reliable XAI systems for critical applications like medical imaging, ensuring accountability and fostering trust in AI-powered healthcare systems.

Keywords Explainable artificial intelligence (XAI) · Artificial intelligence (AI) · Deep learning · XAI for medical imaging

1 Introduction

Artificial intelligence (AI) is crucial in image analysis, particularly in medical imaging systems. Diagnostic systems based on the analysis of medical images have been successfully implemented. However, the inexplicable nature of AI systems poses a barrier to trust between these systems and their users, as well as regulatory and legal frameworks. In the past, image analysis heavily relied on systems designed by human experts. However, developing deep learning methods has revolutionized the field by producing remarkable results [1]. AI uses technology and computers to replicate intelligent behavior and critical

✉ Essam H. Houssein
essam.halim@mu.edu.eg

Amr M. Gamal
amrmohamed435pg@fci.s-mu.edu.eg

Eman M. G. Younis
eman.younas@mu.edu.eg

Ebtsam Mohamed
ebtesam.mohamed@mu.edu.eg

1 Faculty of Computers and Information, Minia University, Minia, Egypt

2 Minia National University, Minia, Egypt

thinking like humans. The term "Artificial intelligence" was coined by McCarthy in 1956 to describe the engineering and science of creating intelligent systems [2]. Alan Turing's work on the Turing test, which aims to distinguish between humans and machines, highlighted the potential of machines to mimic human behavior and intelligence [3]. AI encompasses the process or science of developing machine intelligence [4]. While AI techniques such as deep learning (DL) have revolutionized medical imaging by automating tasks such as tumor detection and organ segmentation, their black-box nature raises significant concerns in high-stakes fields like healthcare, where transparency and trust are essential. This lack of explainability has hindered the adoption of AI in clinical settings, particularly when medical professionals cannot validate or understand the rationale behind AI-driven decisions [5, 6].

Machine learning (ML), a subset of AI, plays a vital role in evaluating radiological images, pathology slides, and electronic medical records (EMR) of patients, assisting in disease diagnosis and treatment, and enhancing clinicians' abilities [3]. ML,[4, 7], involves studying algorithms and mathematical models computer systems use to perform specific functions without explicit programming. These algorithms find applications in various tasks such as predictive analytics, image processing, data mining, and more [8]. ML algorithms learn by processing numerous samples for a specific activity, identifying statistical patterns, and eventually automating the process through established rules [9].Despite these advancements, traditional ML and DL techniques fail to provide interpretable outputs, which is a critical requirement in medicine. For instance, while a model may classify a lung X-ray as indicating pneumonia, it often does so without highlighting the specific regions or features that influenced its decision. This limitation has significant implications, particularly when clinicians must justify diagnoses to patients or regulators.

DL emerged as an extension of ML in 2006 and has revolutionized various domains of AI [10, 11]. DL utilizes artificial neural networks with multiple hidden layers. DL algorithms have achieved significant advancements by employing nonlinear transformations and high-level model abstractions on large datasets. In medical applications, both DL and conventional ML techniques have been extensively utilized [12–14]. However, their inherent complexity has earned them the label of "black-box" models, as even domain experts struggle to interpret the inner workings of these systems [1].This challenge is particularly critical in medicine, where explainability is not just a technical preference but a clinical necessity to foster trust, meet regulatory standards, and ensure ethical accountability.

Explainable Artificial Intelligence (XAI) has emerged as a solution to address these challenges. XAI bridges the gap between AI-driven insights and clinical decision-making by providing interpretable outputs, such as heatmaps that highlight key areas in medical images or textual justifications for predictions. For example, XAI techniques such as Grad-CAM enable models to visually explain their predictions by localized areas of interest, such as tumor regions in an MRI [15, 16]. Similarly, feature attribution methods like SHAP and LIME provide numerical insights into how specific features influence diagnostic outcomes. These tools enhance transparency, helping clinicians validate AI outputs and build confidence in their decisions.

The absence of explainability has led to real-world challenges, such as clinicians hesitating to adopt AI tools due to mistrust or regulatory bodies rejecting black-box models that fail to meet transparency requirements [5]. For instance, a case study reported in [6] highlights how a lack of transparency in a deep learning model for breast cancer detection led to its rejection by clinical trial reviewers. This underscores the critical need for XAI techniques to align AI decisions with clinical reasoning.

The field of XAI, or interpretable AI, has emerged to address these concerns. The "black box" nature of deep learning motivates the need for XAI to explain the decisions made by AI systems. This is particularly relevant in medicine, where decision-making carries substantial risks, and medical professionals are cautious about relying on the "black box" nature of deep learning models [17]. As an example of AI V.s XAI, the success of Reinforcement Learning (RL) in areas like AlphaGo, autonomous vehicles, and Atari video games led to its recognition as a top technology in 2017. However, RL algorithms often operate as black boxes, making it difficult for experts to understand their decision-making processes. For instance, AlphaGo's unexpected move during its game against Lee Sedol puzzled experts until they analyzed its calculations. To make RL systems accessible to non-experts, they must be able to explain their behavior [18, 19].

Key contributions of this review are as follows:

- This review paper provides a comprehensive review of XAI methods specifically focused on medical imaging systems using deep learning. It synthesizes and analyzes various research studies, highlighting the field's key concepts, frameworks, techniques, and evaluation methods.
- Three Perspectives of XAI: The study presents XAI from three perspectives. These perspectives include the frameworks for XAI (specific vs. agnostic, global vs. local, pre- vs. post-hoc model, Problem type and Explanation type), the techniques of XAI for medical imaging (visual, textual, and example-based explanation), and the evaluation methods for XAI.
- The review categorizes the publications based on the anatomical location of the body and medical imaging

modalities. It identifies the emphasis of most papers on specific areas such as the chest, brain, and MRI. By providing this classification, the paper offers insights into applying XAI techniques in different medical imaging contexts.
- The review highlights that most publications in the field of XAI for medical imaging focus on providing local explanations rather than global ones. It specifically points out the prevalence of saliency mapping approaches as suitable methods for Convolutional Neural Networks (CNNs).
- The review paper underscores the significance and urgency of research in XAI, particularly in medical imaging. It emphasizes that the effectiveness and acceptance of AI in the medical community depend on its ability to provide meaningful explanations for its outcomes.

By consolidating existing knowledge and identifying research trends and gaps, this paper contributes to understanding and advancing XAI in medical imaging systems using deep learning. It provides researchers, practitioners, and decision-makers with valuable insights to further explore and develop explainable AI models for improving trust, transparency, and effectiveness in medical imaging diagnostics.

1.1 Motivation

The motivation behind this review paper is to demystify the concept of XAI, which represents a modern scientific approach used in various domains, particularly in disease diagnosis through medical imaging. By reading this review paper, you will find answers to several important questions:

1. What is the relationship between ML and deep learning?
2. How do AI, ML, deep learning, and XAI intersect and relate to each other?
3. What is the significance of explainable artificial intelligence? How is it currently being applied?
4. What is the underlying structure of XAI, and how does it operate?
5. What XAI techniques are specifically employed in medical image diagnostics?
6. What evaluation methods are available for assessing the performance of XAI models?
7. What challenges exist in the implementation of XAI in the domain of medical imaging?

This review paper offers a distinct advantage by drawing upon a comprehensive collection of over 200 research studies and reviews focused on explainable artificial intelligence techniques. These sources have been meticulously categorized into tables based on the various artificial intelligence techniques utilized in different models. By engaging with this research, readers can initiate further exploration and development of various studies within the field of XAI.

1.2 Review methodology

A systematic and structured methodology was followed to comprehensively review XAI methods for medical imaging systems using deep learning. The following steps outline the review methodology employed in this study:

1.2.1 Identification of relevant literature

An extensive search of the existing literature was conducted to ensure comprehensive topic coverage. Multiple electronic databases, including PubMed, IEEE Xplore, Science Direct, and Google Scholar, were searched using appropriate keywords related to XAI, deep learning, medical imaging, and explainability. The search was focused on peer-reviewed journal articles, conference papers, and relevant review articles published within a specific time frame.

1.2.2 Inclusion and exclusion criteria

The identified articles were screened based on predefined inclusion and exclusion criteria. The inclusion criteria considered articles addressing XAI techniques applied to medical imaging systems using deep learning approaches. Studies that provided insights into deep learning models' explainability, interpretability, or transparency in medical image analysis were included. On the other hand, articles focusing solely on traditional machine learning techniques or not directly related to medical imaging were excluded.

1.2.3 Data extraction

Data extraction was performed on the selected articles to collect relevant information for the review. The extracted data included details about the authors, publication year, study objectives, deep learning models used, XAI techniques employed, medical imaging modalities, evaluation methods, and key findings related to explainability in medical imaging.

1.2.4 Data analysis and synthesis

The extracted data were analyzed and synthesized to identify common themes, trends, and patterns in the literature. The findings from the reviewed articles were categorized and organized based on the frameworks,

techniques, evaluation methods, and application areas of XAI in medical imaging. The analysis also identified research gaps, limitations, and challenges reported in the reviewed studies. Figure 1 shows the various study areas for XAI across all databases in this field and how diverse fields can be combined in biomedical research depending on data from the Web of Science databases. Additionally, based on data from Scopus, Fig. 2 displays several trends for all databases XAI in drug design from 2019 to 2024.

1.2.5 Quality assessment

High-quality studies published in reputable journals or conference proceedings were given priority. Peer-reviewed articles were considered to ensure the reliability and validity of the included studies.

1.2.6 Search strategy and study selection

The search strategy involves a sequence of steps that guide the search process, including the identification of digital libraries, formulation of search strings, execution of a pilot search, refinement of search strings, and retrieval of an initial list of relevant primary research articles [20]. Figure 3 depicts the entire process of searching and selecting studies. For this review, the Scopus and Web of Science databases were chosen as the platforms for conducting the literature search. Based on our search string, the initial search yielded 520 primary studies from Scopus and 350 from Web of Science (WOS). The literature search was restricted to studies published between 2019 and 2024 to ensure the inclusion of the most recent developments in XAI for medical imaging. This timeframe was chosen for several reasons; Since 2019, XAI techniques have gained substantial traction in medical imaging due to increasing concerns regarding AI transparency, regulatory requirements, and clinical deployment of deep learning models. Prior to this period, explainability research was in its early stages, with fewer applications in medical imaging as presented in Fig. 1. In addition, the latest deep learning architectures, such as Transformer-based models and generative explainability approaches, have been actively explored in medical imaging only in the last five years [21, 22]. Including older studies would not adequately reflect these advancements.

After applying a publication year filter (i.e., studies published between 2019 and 2024), the number of studies in Scopus was reduced to 500 (a 3.8% reduction), while the count of WOS studies was reduced to 340 (a 2.9% reduction). Next, we filtered by document type (i.e., article or conference paper), which reduced the Scopus studies to 400 (a 20% reduction) and the WOS studies to 280 (a

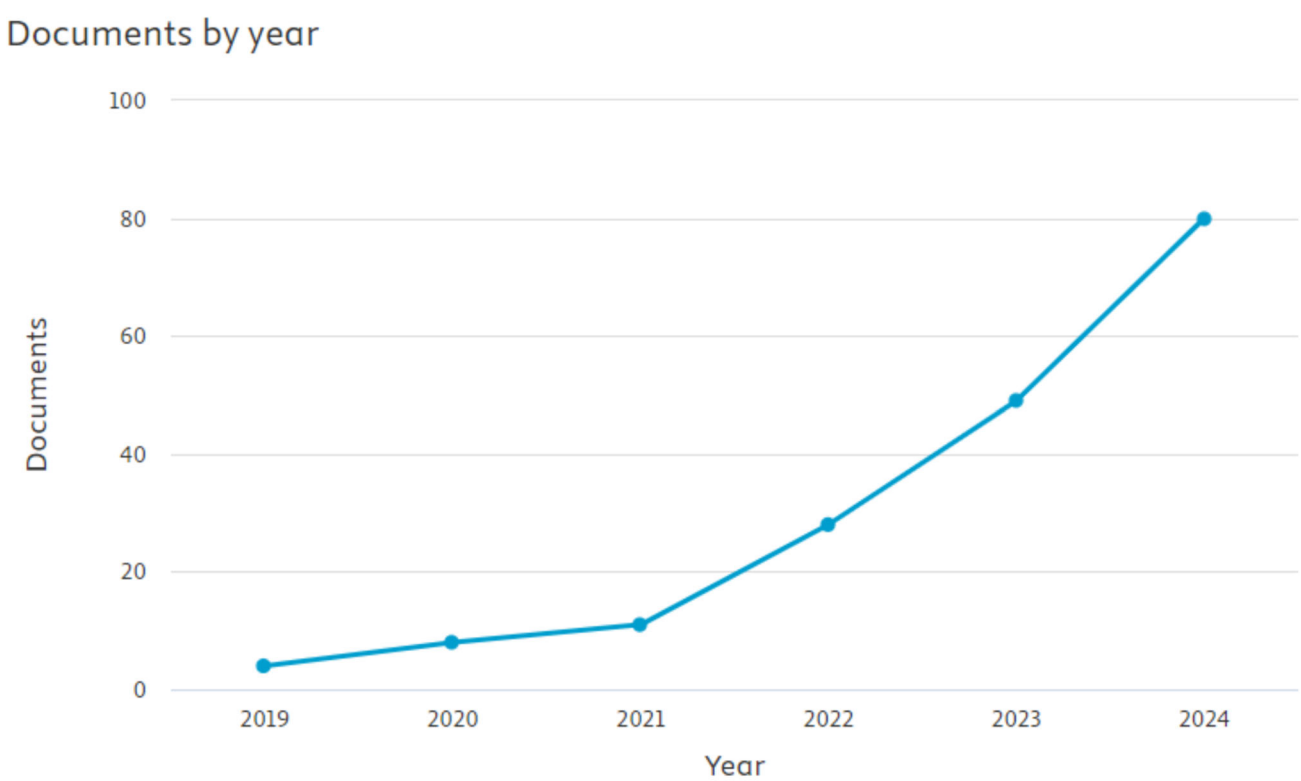

Fig. 1 Histogram of XAI for classification medical image system publications

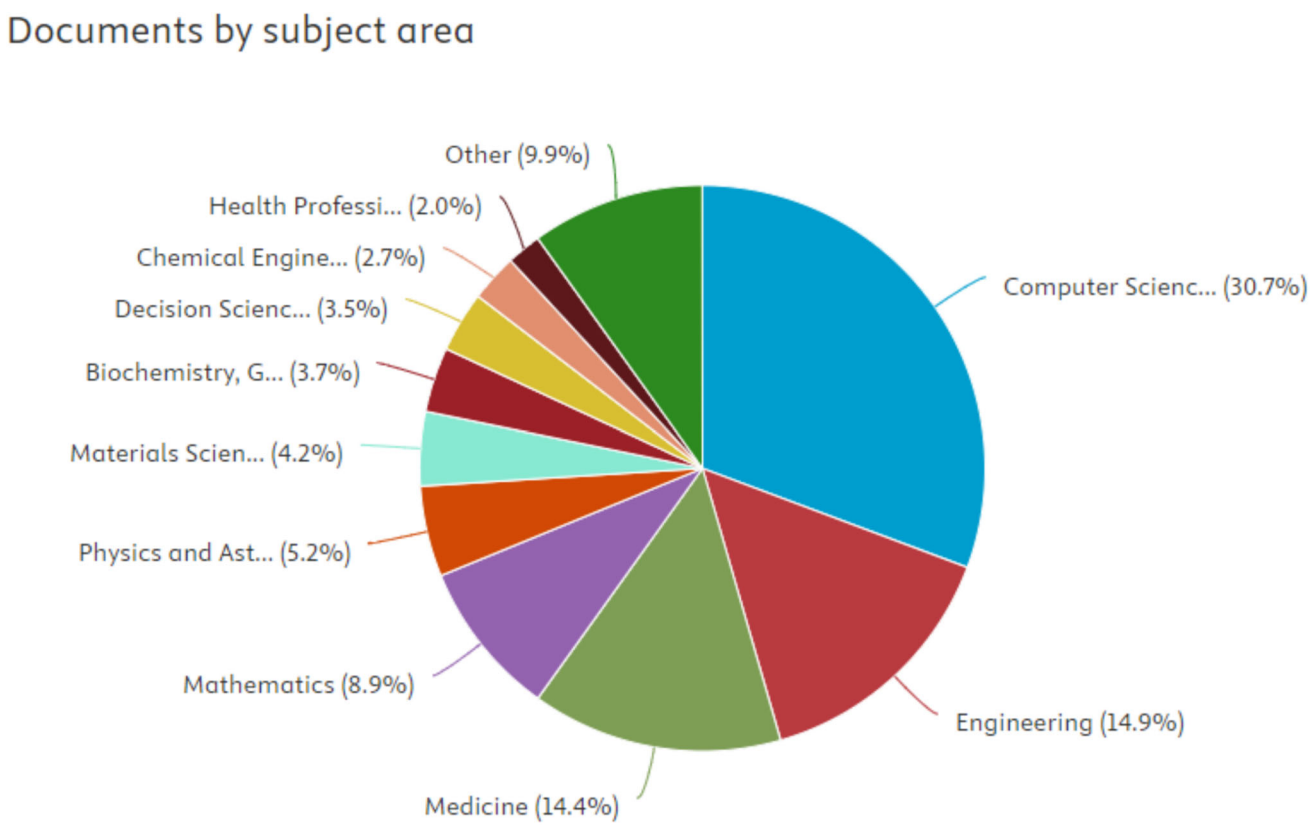

Fig. 2 Several graphical analyses for XAI research area

17.6% reduction). Further filtering by source type reduced the Scopus studies to 350 (an additional 12.5% reduction), while the WOS studies remained unchanged at 280. Finally, a language filter was applied, limiting the selection to studies in English. This reduced the Scopus studies to 240 (a 31.4% reduction from the previous step) and the WOS studies to 60 (a 78.6% reduction). After these steps, the final dataset comprised a total of 300 high-quality studies used in this review, as presented in Fig. 4.

1.3 Review limitations

It is essential to acknowledge the limitations of this review. The search strategy may have unintentionally excluded some relevant articles, although efforts were made to ensure a comprehensive search. The inclusion criteria may introduce bias by focusing only on specific aspects of XAI in medical imaging. Additionally, the interpretation of the reviewed articles is subjective to some extent, and the findings presented in this review may not fully represent the entire landscape of XAI in medical imaging.

Despite these limitations, this review provides a comprehensive overview of the current state of XAI methods for medical imaging systems using deep learning. It offers valuable insights into the frameworks, techniques, and evaluation methods employed in the field, highlighting the potential benefits and challenges of explainable AI in medical imaging. This review focuses on studies published between 2019 and 2024 to capture the most recent advancements in XAI techniques for medical imaging. While this selection ensures coverage of state-of-the-art methods, it may exclude earlier foundational work in explainability, which was more theoretical in nature.

1.4 Review structure

The remainder of this review is organized as follows. Section 2 will explain XAI, understand its main idea, and explain why we need XAI, followed by XAI's possible application and frameworks domains as presented in section 3. Section 4 discusses different techniques for analyzing the medical image for medical diagnosis. In addition, Sect. 5 presents some real-world applications for XAI in medical images. Furthermore, Sect. 6 presented Challenges in Medical Imaging With XAI. Section 7 illustrated the comparative analysis of XAI methods in medical imaging. Sect. 8 discuss the clinician interaction and usaboility of XAI in the daily practice. After that, Sect. 9 presents some feedback studies on XAI adoption. Moreover, Sect. 10 presented a comparison of XAI models

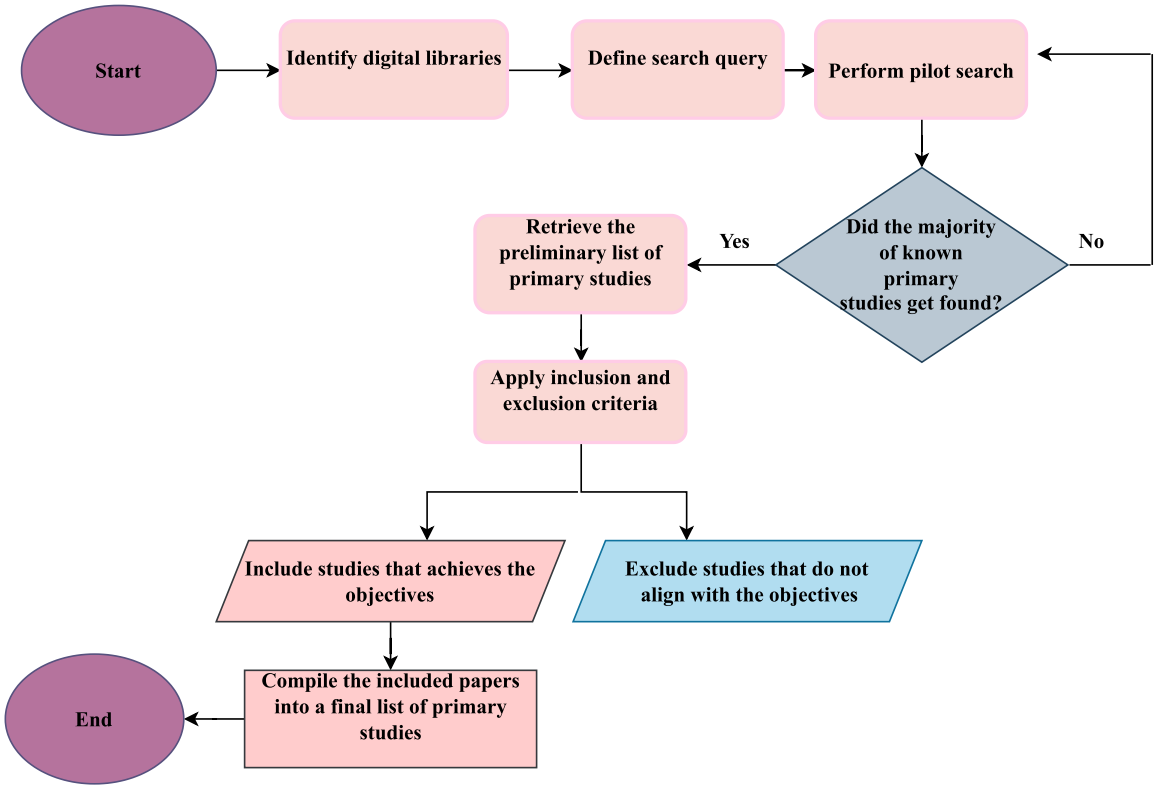

Fig. 3 Search and Study Selection Process (Publication year filtering ensures the inclusion of recent XAI advancements in medical imaging from 2019 to 2024)

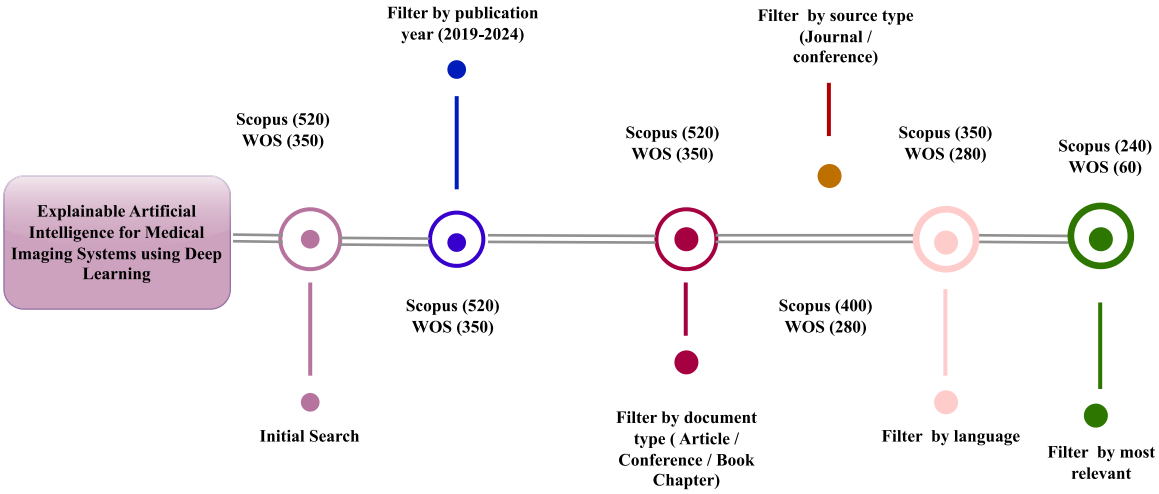

Fig. 4 Primary studies selection process according to inclusion and exclusion criteria, emphasizing studies published in the last five years to capture state-of-the-art methodologies

with traditional black-box models.Sect. 11 will provide a discussion of different Evaluation methods of XAI. The research gaps in XAI presented in Sect. 12. Section 13 provided various possible challenges and future perspectives. Finally, the research review will conclude under Sect. 14.

2 Explainable artificial intelligence (XAI)

XAI is a system that aims to provide insights into how an AI system makes decisions, makes predictions, and performs actions. It goes beyond the mere output of results and explains the decision-making process, highlighting the system's strengths, weaknesses, and future behavior [23].

The core focus of XAI is to enhance human understanding of the outcomes produced by AI systems [24]. The term "XAI" was initially used by Van Lent et al. in 2004 to describe the capability of a system in simulation gaming to illustrate the actions taken by artificially intelligent entities [25]. Although the expression "explainable artificial intelligence" has recently gained popularity, concerns regarding explainability have existed since the mid-1970s, when researchers first delved into the explanations provided by expert systems [26]. In practice, XAI does not possess a universally agreed-upon definition. However, all attempts to define XAI revolve around the common objective of alleviating fears and mistrust associated with AI systems by enabling a deeper understanding of their inner workings. The overarching goal is to remove barriers to comprehension surrounding AI systems.

According to DARPA, XAI aims to produce multiple explainable models while maintaining prediction accuracy, enabling humans to understand and manage the AI systems developed by successful partners in the field [27]. Similarly, the FAT-ML workshop perceives Explainable ML as the practice of ensuring that AI systems' decisions and the underlying data used for those decisions can be explained in non-technical terms, serving both end-users and decision-makers [28]. FICO, the organization responsible for organizing the Explainable Machine Learning Challenge, views XAI as a means to unveil the black box of machine learning, providing accurate models and techniques that can be comprehensively explained, thus enhancing customer confidence and meeting their needs. In [19], the author believes that XAI refers to the research area focused on creating explanations for AI systems that meet people's needs for understanding and accepting the decisions these systems make. Figure 5 illustrates an explainable AI system's key components and potential users. Explainable AI typically consists of two main parts: a machine learning model (often referred to as the black box model) and an XAI method. The machine learning model generates predictions from the training data, while the XAI method explains the model's internal processes and predictions. Therefore, an explainable AI system produces two outputs: predictions and explanations. These outputs are often displayed to the end user through a graphical user interface for better accessibility and understanding [29].

Based on what was mentioned above, AI is a broad field encompassing various subdomains, including ML and DL. ML enables systems to learn from data, while DL, a subset of ML, leverages neural networks for complex pattern recognition. However, traditional AI models often function as "black boxes," making their decision-making processes difficult to interpret. This has led to the rise of XAI, which aims to enhance transparency and trust in AI systems, particularly in critical domains such as healthcare. Figure 6 provides a visual representation of the hierarchical structure of AI and the role of XAI within this landscape.

2.1 Why we need XAI?

XAI enables consumers to comprehend, establish trust in, and effectively manage AI outcomes. Our exploration of the literature has revealed the fundamental importance of XAI in fulfilling the following key aspects: justification, control, improvement, and discovery [24].

2.1.1 Justification

When AI systems make decisions that require justification, particularly in critical areas like healthcare and disease

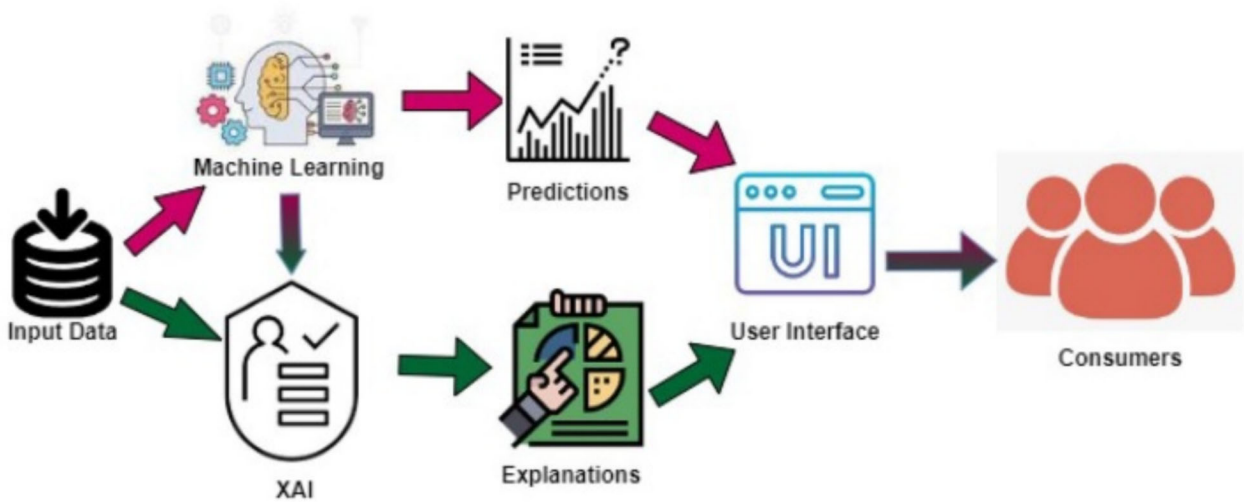

Fig. 5 The essential elements of an Explainable AI [29]. This diagram shows how Explainable AI (XAI) enhances machine learning by providing explanations for predictions, making results more transparent and understandable for users

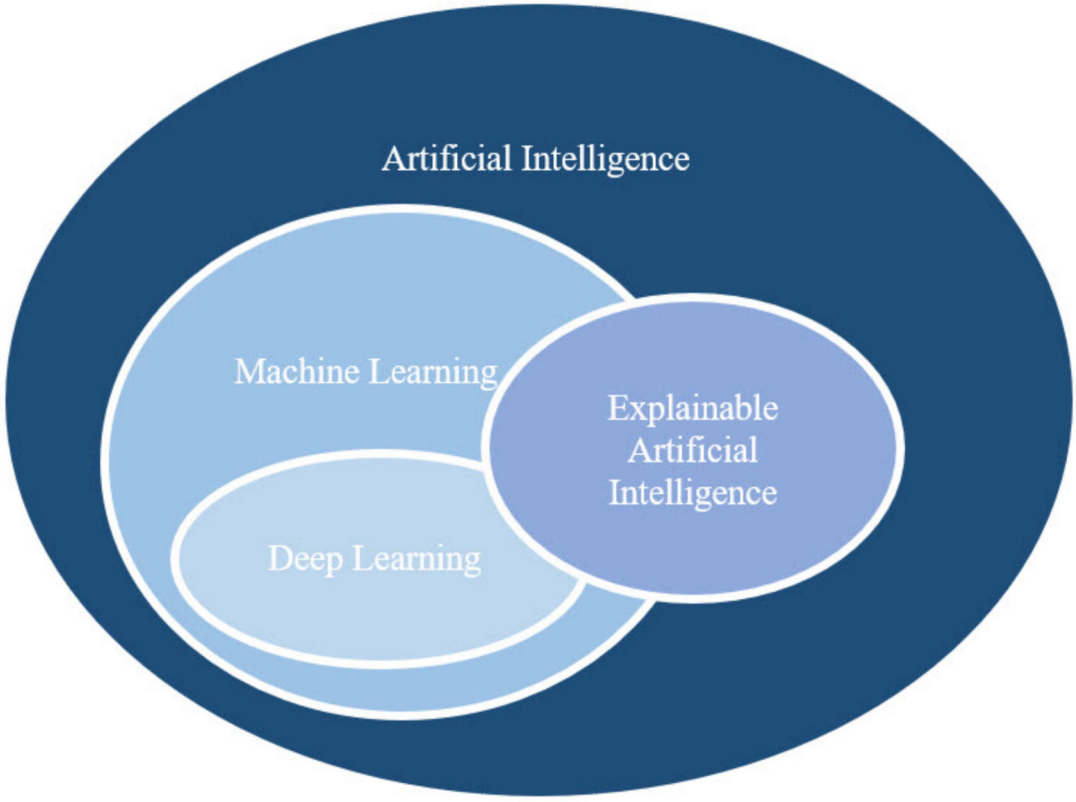

Fig. 6 The relationship between AI, ML, DL, and XAI. This Venn diagram shows how XAI overlaps with AI,ML, and DL

diagnosis, the deployment of XAI becomes imperative. XAI provides explanations for decisions, ensuring transparency and accountability. It becomes essential when unexpected decisions arise, enabling the defense of algorithmic conclusions in a verifiable and auditable manner, thereby fostering confidence. The "Right to explanation," a provision within the GDPR effective since 25 May 2018 across the European Union, further emphasizes the necessity of explanation.

2.1.2 Control

Explanations are not limited to decision-making alone but crucial in proactive control. By gaining a deeper understanding of system behavior, XAI facilitates the identification of potential vulnerabilities, flaws, and low-impact scenarios. This knowledge enables swift detection and resolution of issues, enhancing control over AI systems.

Given the tremendous and rapid development in the fields of artificial intelligence and the subsequent progress in the field of explainable artificial intelligence accompanied by the ability to control artificial intelligence systems, which is called controllable artificial intelligence, these systems designed with mechanisms that allow human users to influence, direct or restrict their behavior to ensure compliance with desired goals, ethical standards, and safety requirements. This model emphasizes human supervision, expected results, and the ability to modify or stop the system when necessary.

In essence, explainability provides the "why," and controllability provides the "how" for ensuring AI systems behave in ways that are transparent, predictable, and aligned with human values. Together, they build a framework for responsible AI deployment.[30, 31]

2.1.3 Improvement

Explanations are not limited to decision-making alone but crucial in proactive control. By gaining a deeper understanding of system behavior, XAI facilitates the identification of potential vulnerabilities, flaws, and low-impact scenarios. This knowledge enables swift detection and resolution of issues, enhancing control over AI systems.

2.1.4 Discovery

The pursuit of explanations holds the potential for discovery. By seeking clarifications and understanding, XAI facilitates accumulating knowledge, unearths new insights, and enriches expertise. Systems that can be explained

become valuable tools for knowledge expansion and exploration. As depicted in Fig. 7, it becomes evident that XAI is an all-encompassing tool that encompasses justifying AI decisions, controlling system behavior, improving system performance, and facilitating the discovery of new knowledge.

2.2 The necessity of XAI in healthcare

As AI tools become more prevalent in healthcare, the demand for XAI techniques grows to foster trust in these models for several reasons [29]:

1. Supporting Decision-Making: XAI is essential for justifying healthcare decisions. XAI enables stakeholders to comprehend the rationale behind decisions, thereby building trust and ultimately enhancing patient care and outcomes.
2. Ensuring Accuracy: In the medical field, precision is paramount, as incorrect decisions can have severe consequences. Accuracy is not optional but a necessity to safeguard human lives and well-being.
3. Continuous Improvement of AI Models: XAI is critical for advancing AI models by pinpointing their flaws. Recognizing these shortcomings facilitates ongoing improvements, making AI systems safer and more reliable, which is especially important in healthcare, where errors can lead to serious consequences.
4. Staying Updated with Current Knowledge: Access to the latest information and technologies is vital for effective disease diagnosis and treatment. Keeping abreast of current advancements allows healthcare professionals to utilize the newest tools and knowledge to provide the best possible care and outcomes for patients.

2.2.1 Necessity of XAI in medical imaging compared to other domains

Medical imaging is a high-stakes field where errors in diagnosis or treatment planning can lead to severe consequences, including misdiagnosis or delayed interventions, which can be life-threatening. This differentiates it from other domains such as finance, marketing, or autonomous systems, where errors, while impactful, typically have less immediate and irreversible consequences [16, 32].

Explainability is particularly critical in medical imaging for several reasons:

Trust and Adoption in Clinical Practice: Clinicians are reluctant to rely on AI models unless they can understand and validate the reasoning behind their predictions. For example, a model diagnosing lung cancer from a CT scan gains trust if it highlights specific tumor regions and explains why they indicate malignancy [15, 33].

Regulatory Compliance and Accountability: Medical AI systems are subject to stringent regulatory standards (e.g., FDA approval) that require transparency. XAI methods help bridge this gap by providing interpretable outputs, enabling models to meet the "right to explanation" requirement under frameworks like GDPR [34, 35].

Error Detection and Bias Mitigation: XAI allows clinicians to detect and correct errors or biases in AI predictions. For example, a black-box AI model might misdiagnose due to unbalanced training data, but an XAI model can highlight the features influencing the prediction, enabling verification and correction [16, 36].

Patient-Centric Communication: Unlike other domains, patient well-being is at the core of medical decisions. XAI enables healthcare providers to explain diagnoses and treatment plans to patients using interpretable visualizations, fostering trust and improving patient understanding [37, 38].

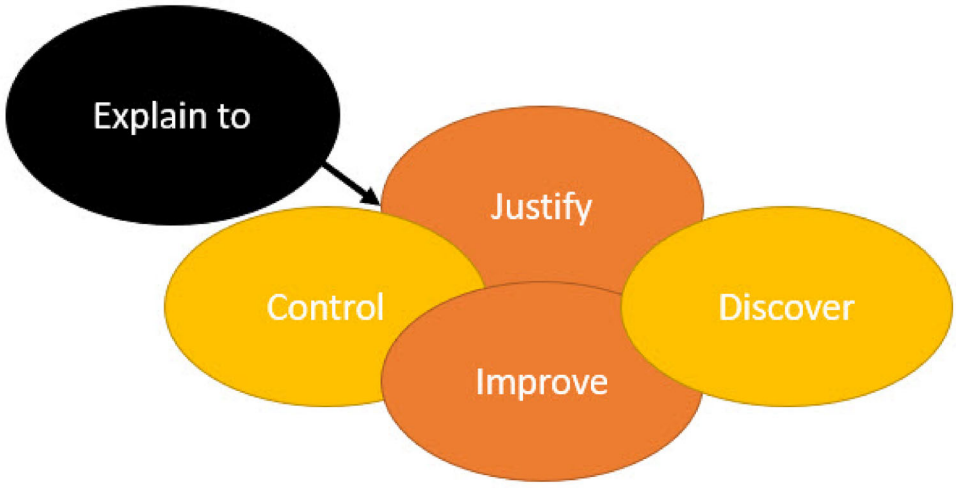

Fig. 7 Why is the need for Explainable artificial intelligence a necessity?

While XAI has applications in other fields, the direct impact of interpretability on patient safety and clinical workflows makes it indispensable in medical imaging. This is evidenced by the growing use of techniques like Grad-CAM, SHAP, and LIME in interpreting complex diagnoses such as identifying brain tumors, lung cancer, and retinal diseases [15, 36, 37].

2.3 Differentiating AI and XAI in medical imaging

AI and XAI represent fundamentally different paradigms in medical imaging analysis. Traditional AI methods emphasize predictive performance but operate as opaque systems, leaving clinicians unable to understand the rationale behind their predictions. Conversely, XAI offers a paradigm shift by emphasizing interpretability, making AI more suitable for high-stakes medical applications [32, 33].

Key distinctions between AI and XAI include:

Transparency: Traditional AI models, such as convolutional neural networks (CNNs), lack interpretability. XAI enhances transparency through techniques like Grad-CAM and SHAP, which provide visual and feature-based explanations for predictions [15, 37].

Trustworthiness: Clinicians are more likely to adopt XAI-based systems because they provide justifiable outputs. For example, a chest X-ray classified as positive for pneumonia by XAI will also include a heatmap indicating the affected lung region, enabling verification [16, 37].

Error Analysis: XAI techniques help identify biases or errors in model predictions by highlighting the influence of specific features, such as age or medical history, on diagnostic outcomes [32, 36].

3 XAI applications and frameworks

Through this section, we aim to illustrate the practical applications and foundational frameworks that drive the development and adoption of XAI. XAI has a broad and significant impact across various fields that rely heavily on AI systems. Understanding and implementing explainable models is crucial to enhance trust, transparency, and effectiveness in these applications. This section delves into the diverse application areas of XAI, highlighting its potential and importance in real-world scenarios. Furthermore, we provide an overview of the XAI methods used in deep learning, emphasizing the different taxonomies and classification approaches found in the literature.

3.1 XAI applications

The influence of Explainable AI (XAI) spans a variety of fields that depend significantly on AI systems. Figure 8 showcases the potential application areas where research on explainable models can be crucial. In the following subsections, we will explore some of these key domains:

– **Healthcare and Medical Diagnosis:** Explainable AI (XAI) has significantly enhanced medical diagnosis and decision-making by explaining AI-based predictions, allowing clinicians and healthcare professionals to gain insights into the reasoning behind specific diagnoses. This transparency fosters trust and enables better understanding, evaluation, and validation of AI-driven diagnostic systems. XAI can also aid in identifying potential biases, increasing interpretability, and ensuring ethical and accountable healthcare practices. Real-world applications of XAI in clinical workflows have demonstrated its potential to transform medical decision-making across various domains. For example, in cancer diagnostics, multimodal XAI systems integrate imaging data with genetic and clinical information to provide comprehensive explanations for tumor classification. Such systems not only improve diagnostic accuracy but also enhance clinician trust by contextualizing predictions [39]. Similarly, Grad-CAM has been successfully used in radiology for explaining chest X-ray classifications, such as detecting pneumonia or COVID-19, by generating heatmaps that highlight regions of interest. These visual explanations assist radiologists in validating AI outputs and making informed decisions [40]. In breast cancer diagnostics, XAI has further demonstrated its utility in enhancing treatment planning. Saliency mapping techniques such as Grad-CAM have been employed to highlight tumor regions in mammograms, assisting radiologists in identifying malignant tissues. Additionally, SHAP-based models have been applied to predict treatment outcomes by analyzing patient-specific features, such as tumor size and genetic markers, providing clinicians with actionable insights for personalized care [16]. XAI methods have also shown significant promise in pathology image analysis. Attention-based models have been used to analyze histopathological images for cancer diagnosis, highlighting regions with abnormal cell morphology. These methods not only improve diagnostic accuracy but also help pathologists validate model predictions by focusing on clinically relevant regions of interest. For example, recent studies have successfully integrated XAI into automated pathology systems, enabling pathologists to differentiate between benign and malignant samples with improved

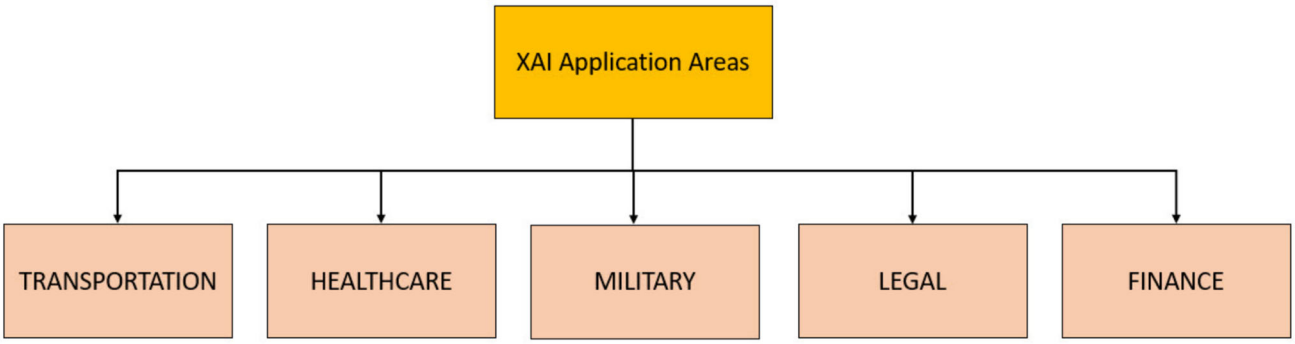

Fig. 8 XAI domains, this diagram highlights the main fields where explainable AI (XAI) is applied

confidence [41]. In diabetic retinopathy diagnosis, XAI techniques such as saliency maps have been applied to fundus images to highlight abnormal retinal regions. These visual explanations help ophthalmologists validate AI model predictions and prioritize cases for treatment. Real-world deployments of XAI-assisted tools have shown improved efficiency in screening programs, particularly in resource-limited settings, by enabling non-specialist clinicians to interpret AI outputs with greater confidence [42]. Another promising use case involves integrating XAI into surgical decision support systems. Attention mechanisms in AI systems have been applied to highlight critical anatomical regions during pre-operative planning, offering surgeons a clear understanding of potential risks. These examples demonstrate the practical utility of XAI in real-world medical workflows and emphasize the importance of developing domain-specific solutions tailored to clinician needs [43].

- Finance and Risk Assessment: The finance industry relies heavily on AI systems for risk assessment, fraud detection, and investment decisions. However, AI models' lack of transparency and interpretability can challenge regulatory compliance and understanding the factors driving financial predictions. XAI techniques can explain financial decision-making processes, assisting regulators, auditors, and stakeholders in comprehending the rationale behind AI-driven outcomes. Additionally, XAI can aid in identifying discriminatory or biased patterns and ensuring fairness and accountability in financial applications.
- Autonomous Vehicles and Robotics: The emergence of autonomous vehicles and robotics necessitates the need for XAI to ensure safe and reliable operation. XAI can explain the decision-making processes of autonomous systems, allowing humans to comprehend the reasoning behind their actions. This transparency is vital for building trust between humans and AI-powered machines, especially in critical scenarios such as self-driving cars and robotic-assisted surgeries. XAI can enable effective collaboration between humans and AI systems, enhancing safety and facilitating error detection and recovery.
- Legal and Regulatory Compliance: In legal and regulatory domains, AI systems are increasingly utilized for various tasks, including document analysis, contract review, and compliance monitoring. However, AI models' lack of interpretability and explainability poses challenges for legal professionals and regulatory bodies. XAI can aid in explaining AI-driven legal decisions, enabling lawyers, judges, and policymakers to understand the reasoning behind legal outcomes. This transparency is crucial for ensuring fairness, accountability, and legal and ethical standards compliance.

3.2 Lessons from other domains in adopting explainable AI

The adoption of Explainable AI (XAI) in various fields, such as finance, autonomous systems, legal technology, e-commerce, and education, offers valuable lessons for advancing its application in medical imaging. In finance, XAI is used to ensure transparency in tasks like fraud detection and credit scoring, where standardized evaluation frameworks, such as those supported by SHAP and LIME, enhance trust among stakeholders by providing clear and consistent interpretability metrics [16, 32]. Similarly, autonomous systems like self-driving cars rely on multimodal data and real-time explanations using techniques such as Grad-CAM and attention mechanisms, which balance operational complexity and interpretability, offering insights for integrating XAI into emergency medical imaging workflows [15, 44]. Legal technology emphasizes accountability by leveraging rule-based systems and counterfactual explanations to trace decisions back to their inputs, a practice that could improve the reliability and justification of XAI in clinical workflows [45, 46].

Furthermore, e-commerce platforms employ counterfactual explanations to personalize user recommendations, demonstrating the potential for tailoring XAI outputs to individual patients in medical imaging [47]. The education domain highlights the importance of accessibility, where simplified textual and visual explanations align AI outputs with users' cognitive workflows. These insights underscore the need for designing XAI systems in medical imaging that are intuitive and actionable for clinicians with varying technical expertise [48]. By adopting lessons such as standardized frameworks from finance, real-time explanations from autonomous systems, accountability mechanisms from legal tech, personalization techniques from e-commerce, and accessibility principles from education, XAI in medical imaging can overcome significant challenges in adoption, trust, and usability. These cross-disciplinary insights provide a roadmap for advancing XAI development and its seamless integration into clinical practice.

3.3 XAI frameworks

This subsection explores XAI methods in deep learning, which aim to make machine learning models more interpretable and trustworthy. The diversity of XAI techniques has resulted in various taxonomies for categorizing these methods. However, these classifications are not always consistent or absolute, as they often depend on the specific features of the methods. Consequently, taxonomies may group techniques differently, sometimes with overlapping categories. For an in-depth discussion on XAI taxonomies, readers can refer to [24, 49–52].

In this review, we examine and compare key XAI approaches, including Model-Specific, Model-Agnostic, Global Models, Local Models, Model-Based, and Post-Hoc Models, alongside other relevant techniques. Each approach is critically evaluated for its strengths, limitations, and ideal use cases. To provide a clear understanding, we include a summary table 1and two figures: one illustrating the comparison across key dimensions Fig. 9,-This figure is a bar chart that visualizes the comparison of different XAI approaches based on key metricsScalability,Precision,Applicability,Overall Strength,Flexibility,Fidelity,Interpretability,Use Case Coverage. Each XAI approach is scored across these metrics.The chart shows how each approach performs for each metric, with a score range from 1 to 5. Higher bars represent better performance.The chart helps to visually compare the strengths and weaknesses of these XAI methods across multiple evaluation criteria. the second figure Fig. 10 highlighting the distinctions between various XAI approaches,This figure helps to understand the different categories and types of XAI approaches based on their use cases, problem types, and the explanation they provide. These resources, together offer a comprehensive framework for analyzing XAI methods [1, 23, 29, 53].

3.3.1 Model-specific vs. model-agnostic methods

Model-specific: Model-Specific Approaches: These techniques are tailored to the unique variables of specific model architectures, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), or decision trees, and leverage their internal structure to provide detailed explanations. Examples include Grad-CAM and Integrated Gradients, which excel in tasks like image classification (e.g., tumor detection with CNNs), sequence analysis (e.g., sentiment analysis with RNNs), and decision-making processes (e.g., credit scoring with decision trees) [1, 53]. While these methods offer highly precise and architecture-aligned insights, their major limitation is their lack of generalizability, as they are constrained to compatible models. This specificity can also exclude other neural networks that might produce results better aligned with the input data. Despite this, they are invaluable in scenarios requiring deep model-specific understanding, such as medical imaging, natural language processing, and financial risk assessment.

Model-agnostic : These methods are architecture-independent, treating models as "black boxes" and relying solely on their inputs and outputs. By perturbing the input, they allow users to analyze changes in the model's output, helping identify which features or areas are most influential. Unlike model-specific methods, they do not access structural parameters or internal weights directly [1, 53]. Popular examples include SHAP and LIME, which are widely used to assess feature importance. These approaches are highly versatile and can be applied to any model, making them suitable for explaining ensemble models in tasks like medical imaging diagnostics, credit risk analysis, and natural language processing. However, they can be computationally expensive and are often less precise than model-specific methods.

3.3.2 Global models vs. local models

Global Models: These methods focus on providing a broad understanding of a model's overall behavior by leveraging insights from the model, its training, and related data. They offer dataset-level explanations, uncovering general relationships learned by the neural network. Techniques like Feature Importance Analysis identify which features contribute most significantly to the model's performance, while visualizations of learned filters reveal what features the model extracts and their relevance to the task at hand

Table 1 Detailed comparison of XAI approaches by key metrics

Aspect	Model-specific	Model-agnostic	Global models	Local models	Post-Hoc	Ante-Hoc
Scalability	Medium	High	Medium	Low	High	Low
Precision	High	Medium	Medium	High	Medium	High
Applicability	Limited to specific models	General	Dataset-wide	Instance-specific	General	Limited
Overall strength	High for specific models	High	Moderate	High for individual cases	High	Moderate
Flexibility	Low	High	Moderate	Moderate	High	Low
Fidelity	High (leverages model internals)	Moderate	Low	High	Moderate	High (interpretable design)
Interpretability	High	High	Moderate	High	High	Very High
Use case coverage	Moderate	High	High for dataset-level insights	High for critical cases	High	Low
Explanation type	Visual (e.g., heatmaps)	Visual, textual, tabular	Feature importance plots	Instance-specific visualizations	Variable	Rules or trees
Weakness	Limited generalizability	Computationally expensive	Lacks granularity	No dataset-wide insights	Post-training only	Limited scalability
Best use cases	Image tasks (e.g., Grad-CAM for MRI)	Explaining ensemble or black-box models	Feature importance analysis	Outlier detection, diagnostics	Retrospective validation	Simple, real-time decisions

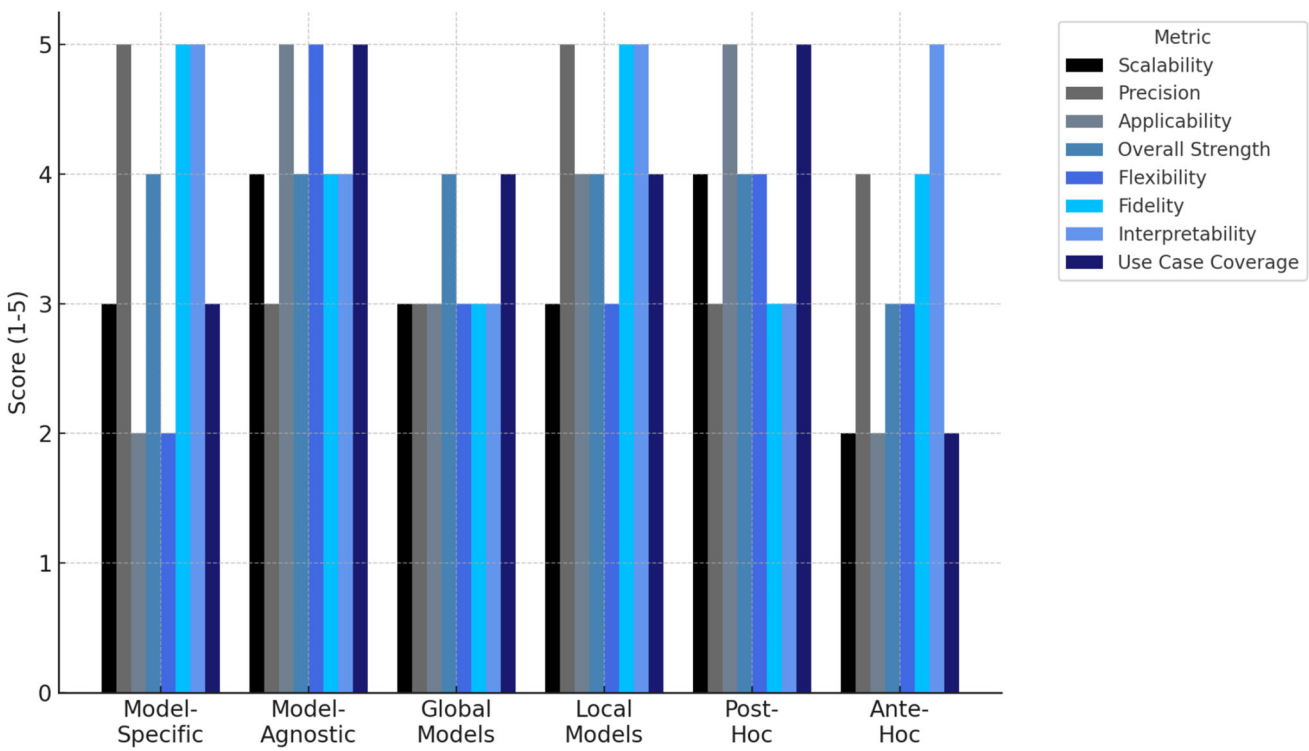

Fig. 9 Comparison of different XAI approaches across key metrics

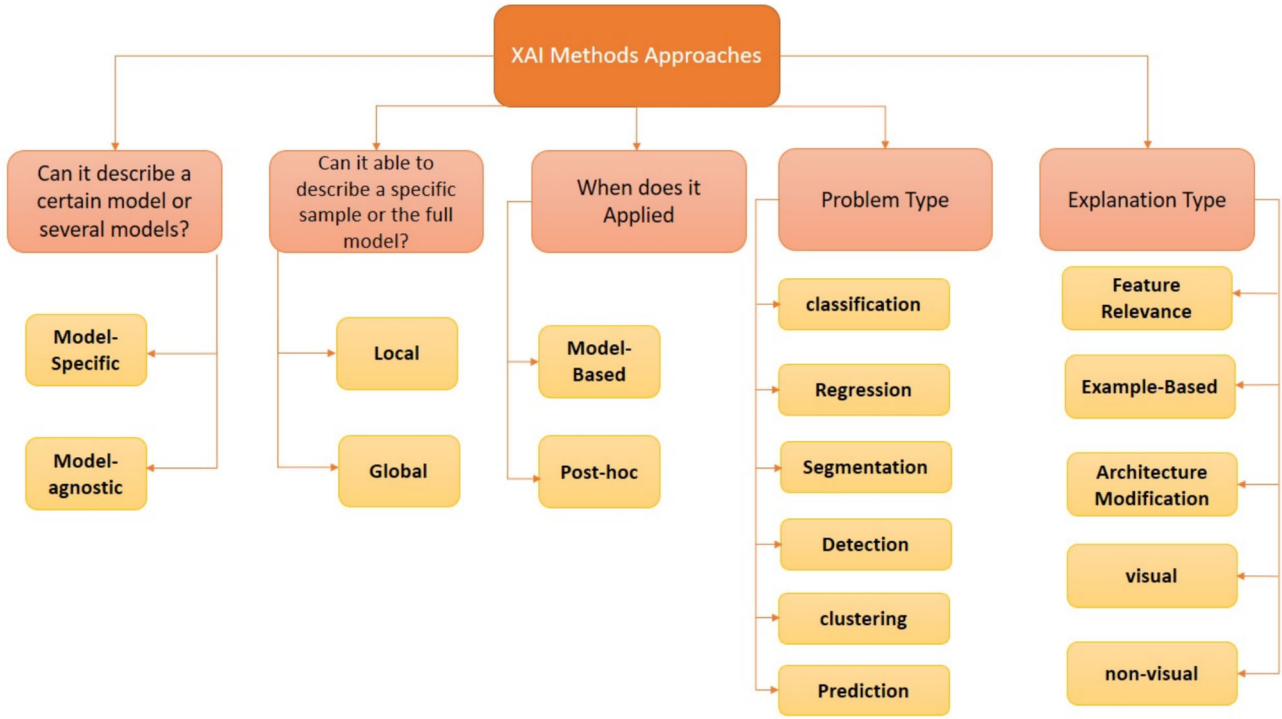

Fig. 10 Approaches of XAI

[37, 54, 55]. Other examples include Partial Dependence Plots (PDPs), which highlight the effects of specific features on predictions. Global approaches are valuable for identifying biases, understanding feature contributions in epidemiological studies, analyzing credit risk models, and optimizing resource allocation in logistics. While they provide a comprehensive view of model behavior, they lack the granularity required for explaining individual predictions.

Local Models: These methods are designed to explain individual predictions, focusing on specific outcomes of the model rather than its overall behavior. For example, in a cardiac risk assessment, local methods can analyze how blood pressure contributes to predicting the likelihood of a cardiac event for a particular individual, while global methods provide a broader understanding of the relationship between blood pressure and cardiac risks across the dataset [37, 54, 55]. Typical techniques include Counterfactual Explanations, which explore how slight changes in input could alter the prediction, and Saliency Maps, which highlight influential features in an instance. Local approaches excel in providing detailed, instance-specific insights, making them highly valuable for high-stakes applications like patient diagnoses in radiology, loan approval decisions in banking, and fraud detection in financial systems. However, their limited generalizability across datasets remains a key drawback.

3.3.3 Model-based vs. post-hoc model

Model-based explanation: These are the models themselves, like linear regression models or support vector machines. These simple and straightforward models are smart enough to match an input–output connection successfully [49]. These are frequently known models of machine learning.Because of focusing on deep learning methods for XAI, the model-based explanation is completely unfeasible here. Usually, thousands to millions of weights are used in deep neural networks, so humans can't know precisely how it works or how it takes its decision.

Post-hoc explanation: Post-Hoc vs. Ante-Hoc Models: Post-hoc explanations are generated after training a model to understand the learned relationships, particularly in complex systems like deep neural networks. These methods, such as Grad-CAM and SHAP, enable interpretability for black-box models by providing insights into their decision-making processes. They are particularly valuable for retrospective model validation, such as analyzing predictions in clinical diagnostics. However, post-hoc explanations may not fully capture the true internal logic of the model, potentially leading to oversimplifications.

In contrast, ante-hoc models, like decision trees or rule-based systems, are inherently interpretable by design. Their transparent structure makes them easy to understand and ideal for scenarios requiring real-time decision-making, such as healthcare systems or fraud detection. While ante-

hoc models excel in interpretability, they often struggle with scalability and predictive performance in complex tasks. Both approaches play complementary roles, balancing the need for interpretability with the complexity of modern machine-learning applications.

A key contrast between these two models is that the first one causes the neural network to be explainable, whereas the second trains an NN and then seeks to explain the ambiguity in the model result. Many methods provide a post-hoc explanation, e.g., (interaction, inspection, and importance) of learned features [54, 56, 57], and those include a visual explanation by saliency maps [15, 33, 37, 58, 59].

3.3.4 Problem type

XAI methods can be classified based on the type of problem they address, such as classification, regression, segmentation, detection, clustering, and prediction [18]. Classification assigns data points to predefined categories based on their attributes, with techniques like SHAP being particularly effective in this domain [60]. Regression predicts continuous numerical values using input data, commonly applied in forecasting tasks [61]. Segmentation divides datasets or images into meaningful regions, often used in image processing and medical imaging [62, 63]. Detection focuses on identifying specific objects or events within data, such as recognizing tumors in medical scans or detecting anomalies in system logs. Clustering, an unsupervised learning approach, groups similar data points based on inherent patterns, aiding in tasks like customer segmentation. Prediction estimates future values or outcomes based on historical data, useful in applications like demand forecasting or risk assessment. These tailored XAI techniques enhance interpretability and effectiveness across diverse problem domains, enabling insights into both supervised and unsupervised learning tasks.

3.3.5 Explanation type

Explanations in XAI can be classified based on their delivery method into categories such as feature relevance, example-based, architecture modification, visual, or non-visual explanations [18]. The delivery method influences the user's ability to understand, interpret, and act on the information provided by the AI system[64].Feature-Based Explanations: This approach identifies the most significant input features that influenced a specific prediction or decision by assigning a score to each attribute, indicating its importance and contribution to the outcome. Examples include feature importance, which highlights key variables in models like decision trees or regression; saliency maps, which visualize critical pixels affecting decisions in image classification; and relevance analysis, which helps understand the factors driving AI decisions in fields such as medical imaging or credit scoring [64, 65]. Example-Based Explanations: This approach uses specific examples from the training data to clarify the outcomes of machine learning models. By presenting instances similar to the input query, it provides context for the prediction and highlights patterns within the underlying data distribution. Examples include case-based reasoning, which uses nearest neighbors to explain decisions, and prototypical examples, which emphasize typical cases representing a class. This method is particularly effective in tasks like fraud detection, where specific instances help validate the system's decision [66, 67]. Architecture Modification in XAI refers to designing AI models with built-in interpretability, allowing them to provide explanations alongside predictions. This can involve self-explaining models that generate their own explanations, attention mechanisms that highlight important input features, and hybrid models that combine interpretable structures (e.g., decision trees) with more complex models like neural networks. Additionally, modular designs break the model into smaller, interpretable components. The main benefits of this approach include enhanced transparency, increased trust, easier debugging, and better compliance with regulations. However, there are challenges, such as the potential reduction in model performance and the increased complexity of balancing interpretability with accuracy. Ultimately, architecture modification aims to integrate explainability directly into the model rather than relying on external interpretation tools[68]. Visual Explanations in XAI use graphical representations to communicate the rationale behind a model's decisions. These explanations help make complex AI behaviors more interpretable by highlighting relevant input features or areas that influenced the decision. Examples include saliency maps, which visualize the importance of specific input features, such as pixels in an image; heatmaps, which show which regions of an image were most significant for classification; and activation maps, which indicate the regions of input data that activate specific neurons within a network. These visual methods allow users to intuitively grasp how the model operates, making them especially useful in domains like computer vision, where interpreting visual data is essential [37, 64]. Non-visual explanations can be categorized into auxiliary, case-based, and textual explanations, depending on the expected outcome for the user. Textual explanations provide written descriptions ranging from basic features to detailed medical research and can include methods like concept attribution testing, image captioning, and pictorial explanations. Auxiliary explanations offer additional information, such as statistical indicators, and can be presented in tabular or graphical forms[69].

3.4 Unique contributions of XAI

XAI significantly contributes to improving trust, transparency, and accountability in medical imaging diagnostics. Unlike traditional AI, XAI techniques enable clinicians to validate predictions by visualizing key regions or features influencing the model's decision. Saliency Maps: Techniques like Grad-CAM allow radiologists to focus on areas of medical images most relevant to a diagnosis, such as tumor regions in MRIs or lung abnormalities in CT scans [15, 36]. Feature Attribution Methods: SHAP and LIME provide numerical and visual explanations, showing how individual features (e.g., lesion size or shape) contribute to model outputs, ensuring alignment with clinical expertise [16, 37]. Practical Case Studies: XAI applications, such as classifying skin cancer using Deep SHAP, demonstrate a 97% accuracy rate while providing interpretable heatmaps for clinical validation [9, 36]. To further emphasize the advantages of XAI over traditional AI in medical imaging, Table 2 provides a detailed comparison. It highlights key aspects such as transparency, trust, accountability, error detection, adaptability, and clinical adoption, supported by examples that illustrate XAI's practical impact.

These frameworks comprehensively understand how XAI operates in medical imaging systems. As discussed in the following sections, they form the foundation for exploring specific techniques that implement these frameworks.

4 XAI techniques for medical image diagnosis

Explainable Artificial Intelligence (XAI) has become a rapidly growing field with a diverse range of techniques emerging over recent years. This rapid evolution has introduced innovative approaches that enhance the interpretability of AI systems, particularly in the domain of medical image diagnosis. For example, multimodal XAI techniques, which integrate visual, textual, and tabular data, have shown promise in improving diagnostic accuracy for complex cases like cancer detection [70]. Additionally, new evaluation metrics focus on quantifying the usability and trustworthiness of explanations, ensuring that XAI tools align with clinical needs [71]. Despite these advancements, a significant challenge persists for practitioners: bridging the widening gap between existing XAI approaches and their practical applications. The field's rapid growth has made choosing the "most optimal" method for a specific use case increasingly difficult.

To address this challenge, Retzlaff et al. [72] introduced a decision framework that guides practitioners in selecting appropriate XAI approaches based on their requirements

Table 2 Comparative analysis of traditional AI and XAI in medical imaging

Aspect	Traditional AI	Explainable AI (XAI)
Transparency	Operates as a "black-box" system; outputs are predictions without explanations	Provides interpretable outputs, such as visual explanations (e.g., Grad-CAM highlights regions in images) and feature-based insights (e.g., SHAP explains how lesion size impacts the result)
Trust	Limited clinician trust due to lack of insight into how decisions are made, especially in critical cases	Builds trust by allowing clinicians to verify model outputs through visualizations (e.g., lung regions in pneumonia diagnosis) and textual justifications (e.g., reports summarizing model reasoning)
Accountability	Hard to justify decisions, especially in legal or ethical scenarios, as no rationale for outputs is provided	Offers auditable and justifiable outputs; for example, XAI tools can show which features led to cancer detection, providing documentation for regulatory compliance
Error Detection	Biases or errors often go unnoticed, leading to potential risks in sensitive applications like healthcare	Identifies biases or errors by explaining feature contributions (e.g., SHAP or LIME revealing biases towards certain demographics) Example: Detecting bias in predicting stroke risk based on age
Adaptability	Models are difficult to adapt as understanding their internal workings is limited	Facilitates refinement and debugging; for example, XAI techniques can show where models fail in segmenting tumor regions, guiding dataset improvements
Clinical Adoption	Often resisted by clinicians due to lack of interpretability and inability to validate predictions	Encourages clinical adoption by providing interpretable explanations, such as Grad-CAM heatmaps or textual insights, enhancing collaboration between clinicians and AI tools
Examples	A deep learning model classifies an MRI scan as "positive for a tumor" without indicating the region or rationale	Grad-CAM highlights the specific region of the tumor in the MRI, and SHAP explains that tumor size and density were significant factors in the diagnosis

and constraints. This framework distinguishes between post-hoc explanations, applied after model training, and ante-hoc explanations, which are inherently interpretable by design. These insights are particularly valuable in medical imaging, where the choice of the XAI method can significantly impact model interpretability, usability, and trustworthiness. Different XAI techniques, categorized into visual, textual, and example-based approaches, will be discussed in this section. These methods demonstrate the diversity of XAI solutions available for medical imaging and highlight their ongoing evolution to address field-specific challenges. However, their effectiveness and reliability depend on robust evaluation metrics, which will be examined in the subsequent section.

To provide a clearer understanding of the capabilities and limitations of various XAI methods, as you can see in figure 11, this section includes a detailed comparison of their performance across different application areas in medical imaging. Techniques such as Grad-CAM, SHAP, Integrated Gradients, and others are evaluated based on their effectiveness in analyzing anatomical regions (e.g., brain, cardiovascular, skin) and imaging modalities (e.g., MRI, CT, photography). This comparison highlights the strengths of each technique in improving diagnostic accuracy, interpretability, and clinical trust, offering a practical guide for researchers and practitioners to select the most suitable XAI method for their specific use case.

4.1 Visual explanation

The analysis of medical images often relies on a widely used XAI technique known as Class Activation Mapping (CAM) or saliency map. CAM highlights the crucial points in the image that are significant in making decisions. Backpropagation methods are commonly employed in CAM, although some variations incorporate perturbation-based or multiple-instance learning-based approaches.

4.1.1 Backpropagation-based approaches

Backpropagation-based approaches have been commonly used in XAI to generate saliency maps, highlighting the pixels that influence the model's decision the most. Two early methods, namely deconvolution and guided backpropagation, have been widely employed for this purpose. Table 3 presented research methods based on Backpropagation and deconvolution techniques for different diseases. Deconvolution and guided backpropagation techniques have been utilized to produce saliency maps by visualizing the partial derivatives of the output at the pixel level [33]. In this study, the author presented two visualization

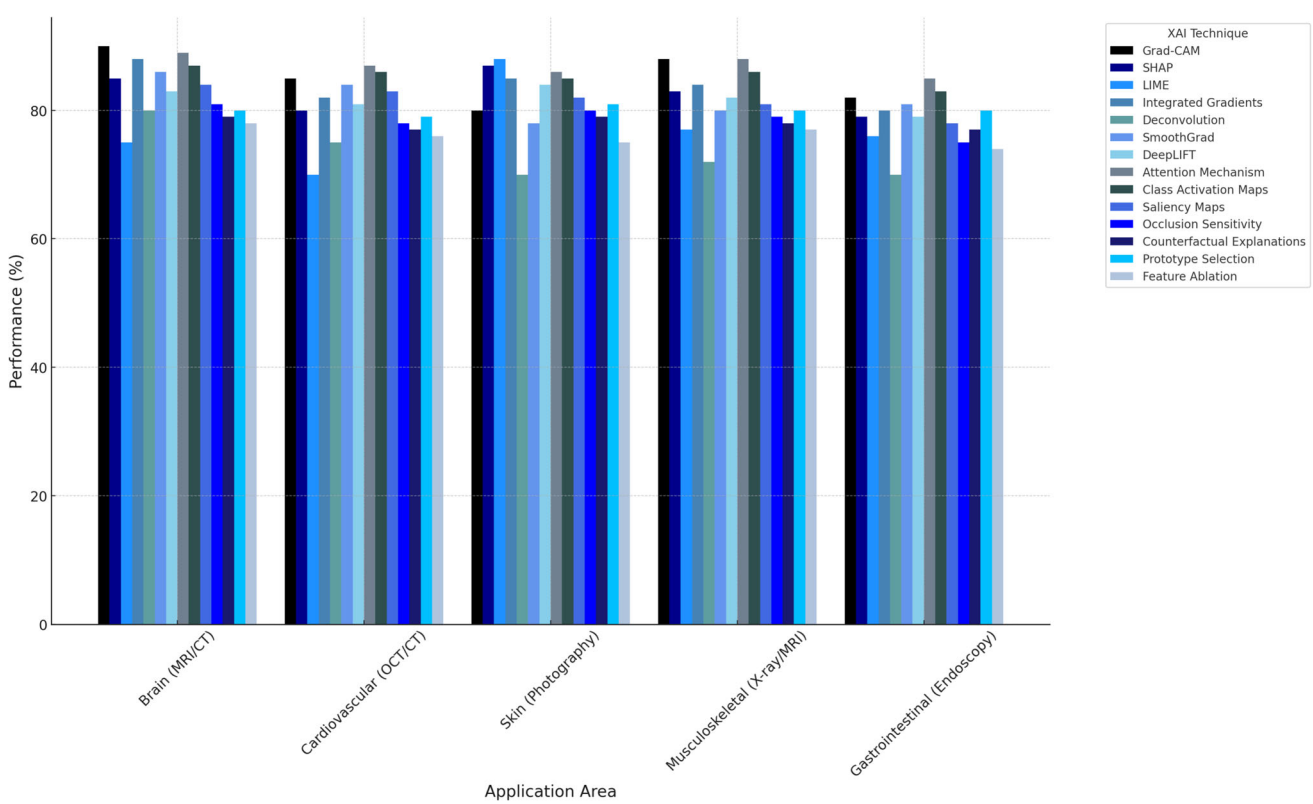

Fig. 11 Comparison of XAI techniques by application area

Table 3 Gradient-based methods for different diseases

Class of explainability	Main XAI techniques	Anatomical location	Form of data	References
Gradient-based methods	Deconvolution	Brain	MRI	Gao et al. [75]
		Cardiovascular	OCT	de Vos et al. [76]
	Backpropagation /guided backpropagation			Gessert et al. [77]
		Brain	MRI	Dubost et al. [78]
				Rezaei et al. [79]
				Wang et al. [80]
				Wei et al. [81]
			CT	Fuchigami et al. [82]
		Eye	Fundus photography	Quellec et al. [83]
		Gastrointestinal	Ultrasound	Lee et al. [84]
			Endoscopy	Wickstrøm et al. [85]
		Musculoskeletal	MRI	Jamaludin et al. [86]
			X-ray	Kim et al. [87]
		Skin	Photography	Bian et al. [88]

techniques. The first technique involves creating an image that optimizes the score, while the second technique identifies discriminative regions in a given image for a specific class by computing an image-specific class saliency map. Several authors have proposed another unique visualization method called deconvolution [37] as shown in Table 3. This method focuses on analyzing the performance of the classifier and intermediate feature layers in CNNs trained for image categorization.

Guided backpropagation has also been extensively explored for object recognition in CNNs [58]. The authors introduced a novel variation of the deconvolution technique for displaying features learned by CNNs. This technique enables the study of the network and can be extended to various network architectures. These techniques, along with other explanation methods discussed in Sect. 3.3 [1], offer diverse avenues for interpreting model decisions. These techniques have found applications in medical image diagnosis as well. For instance, in a study focused on hyperacute stroke lesion segmentation using CT scans [73], a unique 3D U-Net architecture incorporating prior knowledge was proposed. The authors applied a gradient-based method to visualize the contribution of each voxel to the segmentation outcomes, leading to improved segmentation accuracy. The Dice similarity coefficient increased from 0.44 to 0.54, sensitivity improved from 0.80 to 1.00, and specificity improved from 0.90 to 0.98. In another study [21], a neural decoder utilizing deep transfer learning (DTL) was constructed for behavior task classification. The model integrated components from state-of-the-art networks pre-trained on the ImageNet dataset and connected them with adaptive layers. Furthermore, Deconvolution was employed in a study [74] to identify the specific areas within a CT image slice that influenced the decision-making process when estimating the quantity of coronary artery calcium per heart or chest CT image slice.

4.1.2 Class activation mapping (CAM)

An innovative approach known as Class Activation Mapping (CAM) was introduced in [59]. CAM replaces the fully connected layers at the end of CNNs by employing global average pooling on the last convolutional feature maps. This allows classification-trained CNNs to learn object localization without relying on bounding box annotations. CAM provides various explanation methods, including local, model-specific, and post hoc explanations. CAMs have also been employed in the interpretation of medical images. Table 4 and Table 5 provide different CAM research. For instance, in a study focused on detecting Parkinson's Disease using 3D MRI analysis [89], a 3D convolutional neural network was utilized. The network analyzed full brain 3D MRI scans to comprehend complex patterns in subcortical areas related to Parkinson's Disease. 3D class activation maps (CAMs) were used to explain the model's decision-making process, revealing that the model focuses on the substantia nigra region (SNc) to detect Parkinson's Disease from a given MRI scan. Another study [90] compared different state-of-the-art architectures trained on a private dataset and evaluated on the publicly available Messidor-2 dataset. The paper also examined localization scores of various designs using class activation maps (CAMs). The results indicated a trade-off between performance in classification and localization.

Table 4 Class activation mapping (CAM) based techniques for brain, breast, and chest diseases

Main XAI techniques	Anatomical location	Form of data	References
CAM	Brain	MRI	Ahmad et al. [92]
			Baumgartner et al. [93]
			Ceschin et al. [94]
			Chakraborty et al. [95]
			Gao et al. [96]
			Li et al. [97]
			Ng et al. [98]
			Shinde et al. [99]
			Shinde et al. [100]
		PET/CT	Choi et al. [101]
		CT	Lee et al. [102]
		Ultrasound	Lin et al. [103]
	Breast	X-ray	Huang et al. [104]
			Lee and Nishikawa et al. [105]
			Xi et al. [106]
			Yi et al. [107]
		Ultrasound	Kim et al. [108]
			Qi et al. [109]
			Zhou et al. [110]
		MRI	Luo et al. [111]
	Chest	X-ray	Ausawalaithong et al. [112]
			Dunnmon et al. [113]
			Huang and Fu et al. [114]
			Khakzar et al. [115]
			Liu et al. [116]
			Rajpurkar et al. [117]
			Sedai et al. [118]
			Tang et al. [119]
			Wang et al. [120]
			Paul et al. [121]
		CT	Guo et al. [122]
			Kumar et al. [123]
			Lei et al. [124]
			Tang et al. [125]

Models with more parameters performed better in classification, with NASNet achieving the highest accuracy and typical precision, recall, and F1-scores of approximately 95%. VGG19 outperformed other models in localization, with a mean Intersection over Minimum (IoM) of 0.45.

4.1.3 Multi-scale CAMs

Multi-scale CAMs have been proposed to address the presence of information at different scales in medical images. In a study by [91], CAMs were extracted at various resolutions to accurately highlight interpapillary capillary loop patterns in endoscopic images, which are relatively smaller than the overall image. Another research work [22] introduced a clinically interpretable Convolutional Neural Networks (ConNets) architecture called EAMNet, aiming for enhanced interpretability by emphasizing different regions identified by the network. The model focused on providing an accurate glaucoma diagnosis. EAMNet resolved the interpretability issue associated with CNN-based glaucoma diagnosis CAD systems. The model utilized M-LAP and ResNet, incorporating three global average pooling (GAP) layers corresponding to different resolutions. This significantly improved the resolution of EAMNet while preserving classification performance.

Table 5 Class activation mapping (CAM) based techniques for different diseases

Main XAI techniques	Anatomical location	Form of data	References
CAM	Eye	Fundus photography	Ahmad et al. [126]
			Jiang et al. [127]
			Kumar et al. [128]
			Liao et al. [129]
			Liu et al. [130]
			Narayanan et al. [131]
			Shen et al. [132]
			Tu et al. [133]
		OCT	Wang et al. [134]
			Perdomo et al. [135]
			Zhou et al. [136]
		CT	Wang et al. [137]
	Female reproductive system	Histology	Sun et al. [138]
	Gastrointestinal	Endoscopy	Everson et al. [139]
			García-Peraza
			Herrera et al. [140]
			Wang et al. [141]
		Histology	Heinemann et al. [142]
			Kiani et al. [143]
			Yan et al. [144]
	Musculoskeletal	MRI	Bien et al. [145]
			Chang et al. [146]
		X-ray	Paul et al. [147]
			Zhao et al. [148]
	Prostate	Histology	Silva-Rodríguez et al. [149]
		MRI	Yang et al. [150]
	Skin	Dermatoscopy	Li et al. [151]
		Photography	Xie et al. [152]
	Skull	X-ray	Kim et al. [153]
	Thyroid	Ultrasound	Wang et al. [154]
	Bladder	Histology	Woerl et al. [155]

4.1.4 Gradient-weighted CAM (Grad-CAM)

Selvaraju introduced gradient-weighted Class Activation Mapping (Grad-CAM) and guided Grad-CAM [15] as extensions of CAM. Unlike CAM, grad-CAM does not rely on global average pooling and can provide post hoc local explanations using any CNN. Guided Grad-CAM combines Gradient-CAM with guided backpropagation through element-wise multiplication. These techniques have been widely applied in the analysis of medical images. For instance, a study by Hilbert et al. [156] proposed a Deep Learning technique for predicting reperfusion results and functional outcomes in patients with acute ischemic stroke using CT angiography data. The researcher found that pre-training with auto-encoders significantly improved the prediction without needing expert annotation of the images. Grad-CAM was employed to visualize the imaging data and aid in model interpretation. In another example, Kubach et al. [157] trained a CNN to classify dysplasia type IIb and cortical tuber of tuberous sclerosis complex and visualize the results. The researcher also proposed a new web-based deep learning application to confirm pathology interpretations. The CNN utilized morphological features to distinguish between the two entities, and guided Grad-CAMs were generated to visualize these features.

In [158], the authors propose an enhanced computer-aided diagnosis (CAD) system for brain tumor classification, leveraging XAI techniques to improve accuracy and interpretability. Traditional machine learning models often lack transparency, posing challenges for clinicians. This

study employs the Grad-Cam algorithm to create an interpretable CAD system that achieves high classification accuracy and provides physicians with valuable insights into the decision-making process, thereby boosting their confidence in the system's recommendations. The model is evaluated using a large, diverse dataset and compared with modern DL and traditional CAD systems. Results indicate that the proposed CAD system with XAI extensions offers superior accuracy and valuable explanatory insights, potentially enabling doctors to make more accurate diagnoses.

Moreover, Lamba et al. [159] introduce a novel approach to brain-computer interaction (BCI) that enhances transparency and builds confidence by applying XAI techniques, specifically using Grad-CAM. The study employs the Xception architecture, pre-trained on the ImageNet database via transfer learning, to extract significant features from magnetic resonance imaging (MRI) datasets. A linear support vector machine is then used to classify the distinct features. Grad-CAM is utilized to create informative heatmaps highlighting the spatial localization of features crucial for the model's predictions. The proposed model delivers accurate diagnostic outcomes for identifying abnormal brain tissues and ensures transparency in its predictions, mitigating overfitting issues. The model's effectiveness is validated through hyperparameter tuning and performance metrics on unseen brain MRI scans. In [160], the authors propose an advanced CAD tool for classifying primary brain tumors using MRI, addressing the limitations of traditional deep learning models by incorporating Vision Transformer (ViT) networks. These networks improve feature extraction through self-attention, capturing long-range image dependencies. The study employs explainable AI techniques like Grad-CAM, LIME, and SHAP to enhance model transparency and build clinician confidence. Using T1-weighted contrast-enhanced MRI sequences, the ViT model is trained and evaluated on a benchmark dataset, achieving a test accuracy of 91.61%, outperforming traditional CNN models with an accuracy of 83.37%. The results suggest the ViT-based CAD tool's potential for effective and interpretable use in clinical diagnosis routines.

Furthermore, in a study by [161], Grad-CAM was employed to visualize histological lymphadenopathy sections and determine the presence of distant metastasis tissue based on a classifier's decision. Grad-CAM provided insights into the regions of interest for accurate classification. Table 6 and Table 7 summarize different research based on the Grad-CAM technique.

While saliency mapping techniques, such as Grad-CAM, are widely adopted for their intuitive visual explanations, they are not without limitations. Saliency maps are often sensitive to noise and input perturbations, leading to inconsistent outputs that can affect clinical trust. Moreover, they primarily offer localized insights without addressing global patterns or relationships in the data. Emerging methods such as SHAP and attention mechanisms address these challenges by providing feature-level attributions and enhanced robustness. SHAP quantifies the importance of input features globally, allowing clinicians to understand broader trends in model decision-making. Similarly, attention mechanisms identify critical regions while being less affected by noise, offering consistent and interpretable results. These methods demonstrate the potential to complement or improve upon saliency mapping for more robust and clinically meaningful explanations [162, 163].

4.1.5 Layer-wise relevance propagation (LRP)

Layer-wise relevance propagation (LRP), introduced by Bach et al. [217], is a technique that iteratively back-propagates the neural network's output, typically a classification score, throughout the network. In each iteration, LRP assigns relevance scores to the input neurons of the previous layers. The conservation law ensures that the sum of these propagated relevance values equals the overall score of the neuron that generated them. For example, Hägele et al. [218] demonstrated two key advantages of using LRP in DL-based methods. Firstly, LRP enables qualitative and quantitative comparisons between end-to-end learned features and labels provided by domain experts. This helps in validating the interpretability of the model. Secondly, LRP facilitates the identification of latent biases present in the dataset. In their study, pixel-wise heatmaps computed using LRP provided appropriate visualizations of the model's decision-making process for three tumor entities: skin melanoma (TCGA-Skin Cutaneous Melanoma), breast carcinoma (TCGA-Breast Invasive Carcinoma), and lung carcinoma (TCGA-Lung Adenocarcinoma). These visualizations showed significant overlap with expert descriptions. Another application of LRP was demonstrated by Böhle et al. [219] in the context of Alzheimer's disease detection from brain MR scans. By comparing the saliency maps generated by guided back-propagation with LRP, the researchers found that LRP was more accurate in pinpointing the areas associated with Alzheimer's disease.

LRP offers an interpretable framework for understanding the contribution of different regions in the input to the network's output. Its ability to propagate relevance scores throughout the network aids in visualizing the decision-making process and identifying relevant features or areas of interest. Other techniques are presented in Table 8

Table 6 Grad class activation mapping (Grad-CAM) based techniques for brain, breast, cardiovascular, and chest diseases

Main XAI techniques	Anatomical location	Form of data	References
Grad-CAM	Brain	MRI	Hilbert [164]
			Kim [165]
			Liao [166]
			Natekar [167]
			Pereira [168]
			Pominova [169]
			Windisch [170]
		Histology	Tang [171]
			Kubach [172]
			Guided Grad-CAM
		Ultrasound	Xie [173]
		CT	Ye [174]
	Breast	MRI	El Adoui [175]
		Histology	Obikane [176]
	Cardiovascular	CT	Candemir [177]
			Huo [178]
		X-ray	Cong [179]
		Ultrasound	Patra [180]
	Chest	X-ray	Brunese [181]
			Chen [182]
			Mahmud [183]
		Histology	He [184]
		CT	Hosny [185]
			Humphries [186]
			Ko [187]
			Paul [188]
			Philbrick [189]
			Xu [190]
		PET/CT	Qin [191]
		Histology	Teramoto [192]
		Ultrasound	van Sloun and Demi[193]

4.1.6 Deep shapley additive explanations (Deep SHAP)

Deep Shapley Additive explanations (Deep SHAP), introduced by Lee and Lundberg [16], offer a unified approach to explain predictions using Shapley values. Shapley values, derived from game theory, quantify the marginal contribution of each feature to the model's output. However, computing Shapley values can be computationally expensive due to evaluating numerous feature combinations. To address this limitation, Lundberg and Lee [16] proposed Deep SHAP, which combines DeepLIFT (Deep Learning Important Features) with Shapley values to approximate their computations efficiently for CNNs. Deep SHAP has been applied in the analysis of medical images.

Moreover, in [225], the authors propose a hybrid model, "DeepXplainer," for lung cancer detection using clinical data. This model combines a convolutional neural network with XGBoost, allowing for accurate predictions and explanations of these predictions. By integrating the SHAP explainable AI technique, the model provides both local and global levels of interpretability, aiding healthcare professionals in making more informed decisions. Various visualizations are used to illustrate the reasoning behind the predictions. The model's performance is compared to state-of-the-art techniques, demonstrating superior accuracy, sensitivity, and F1-score results. The "Survey Lung Cancer" dataset is used for evaluation, where the proposed method achieves an accuracy of 97.43%, sensitivity of 98.71%, and F1-score of 98.08. In the same context, the authors in [226] introduce an automatic tuberculosis (TB) detection system utilizing advanced deep learning models for the analysis and classification of chest X-rays (CXRs)

Table 7 Grad class activation mapping (Grad-CAM) based techniques for different diseases

Main XAI techniques	Anatomical location	Image type	References
Grad-CAM	Dental	X-ray	Vila-Blanco [194]
	Eye	Fundus photography	Kim [195]
			Martins [196]
			Meng [197]
			Zhang et al. [198]
			Jang [199]/ Guided Grad-CAM
		OCT	Thakoor et al. [200]
			Wang et al. [201]
	Female reproductive system	Histology	Gupta et al. [202]
			GV and Reddy [203]
	Gastrointestinal	CT	Chen et al. [204]
			Rajpurkar et al. [205]
		Endoscopy	Itoh [206]
		Histology	Korbar [207]
			Kowsari [208]
		MRI	Wang [209]
	Lymph nodes	Histology	Ji [210]
	Musculoskeletal	X-ray	Cheng [211]
			Gupta [212]
			Zhang [213]
			von Schacky [214]
	Skin	Photography	Zunair and Hamza [215]
	Thyroid	CT	Lee [216]

Table 8 LRP-based techniques for brain, breast, cardiovascular, and chest diseases

Main XAI techniques	Anatomical location	Form of data	References
LRP	Brain	MRI	Böhle et al. [220]
			Dang and Chaudhury[221]
			Eitel et al. [222]
			Grigorescu [223]
			Xu [224]

into TB and non-TB categories. Recognizing that a large portion of a CXR image is dark and uninformative, they employ sophisticated segmentation networks to isolate the regions of interest. These segmented images are then processed by CNN models. The system uses explainable artificial intelligence to highlight TB-infected lung areas to aid subjective assessment. In tests with three publicly available CXR datasets, the EfficientNetB3 model achieves the highest accuracy of 99.1%, a receiver operating characteristic of 99.9%, and an average accuracy of 98.7%. The findings demonstrate that segmented lung CXR images yield better performance than raw lung CXR images.

Furthermore, the authors in [227] propose an advanced XAI-based CAD system named "Skin-CAD" for classifying dermoscopic photographs of skin cancer (SC). Skin-CAD accurately categorizes the images as benign or malignant and classifies them into seven SC subclasses. The system employs four CNNs with different topologies, extracting features from each CNN's final pooling and fully connected layers. To reduce dimensionality and complexity, Skin-CAD combines principal component analysis (PCA) on the pooling layer features with fully connected features. It integrates these dual-layer features from all four CNNs, and then a feature selection step identifies the most important attributes, streamlining the classification process. For interpretability, the local interpretable model-agnostic explanations (LIME) approach is used to provide visual interpretations of the predictions. The system is validated using two benchmark datasets, "Skin Cancer: Malignant vs. Benign" and "HAM10000,"

achieving maximum accuracies of 97.2% and 96.5%, respectively.

In the context of skin cancer classification, Young et al. [228] explored the reliability of GradCAM and Kernel SHAP by building a collection of 30 models trained on the publicly accessible HAM10000 dataset. They investigated the domain space and utilized GradCAM and Kernel SHAP to explore a random sample of models from their suite. This study represents initial progress in balancing the trade-off between model accuracy and explainability in the classification of skin cancers. Zhu et al. [229] proposed an Additive explanation framework based on guideline-based approaches to understand Machine Learning-based CAD systems. This framework aligns with medical guidelines to standardize diagnostic choices. The Guideline-based Additive Explanation (GAX) framework consists of rule-based segmentation and anatomical regularities to generate anatomical features based on guidelines, followed by an analysis based on perturbations to determine the values of each feature. GAX provides comprehensive explanations by examining complete datasets and producing a chart outlining key metrics. The GAX framework was tested on a dataset of lung CT images, demonstrating its potential to offer interpretable explanations and highlight data biases to improve models in clinical practice. Van et al. [230] derived volumetric breast density from breast MRI using a regression CNN. This technique allowed the identification of image regions that positively or negatively contributed to the density estimation.

Deep SHAP provides a robust framework for explaining model predictions by combining Shapley values and deep learning techniques. Its applications in the analysis of medical images demonstrate its potential to enhance interpretability and provide insights into feature importance and model behavior. Moreover, other approaches are presented in Table 9

4.1.7 Perturbation-based approaches

Perturbation-based approaches involve modifying the input image to assess the importance of specific regions for the task. Several studies have employed these techniques, as summarized in Table 10. Table 10 presents a compilation of papers that have utilized perturbation-based explanations, sorted based on the techniques employed in each study. These techniques aim to provide insights into different image regions' contribution to the model's decision-making process.

1. **Occlusion sensitivity:** An attempt to determine the extent to which the coverage of certain areas in the image is related to the correct image classification. Studies have proven that unmistakably, the model localizes the items within the picture because of the likelihood that the model correctly identified the class when an object was obscured sharply. A significant decrease in activity in the feature map was observed when the occlude covers the picture region displayed in the visualization [37]. Shahamat and Abadeh [242] showed that A 3D-CNN and a genetic algorithm-based brain masking (GABM) method as a visualization technique in the form of Occlusion of some parts that give fresh perspectives on how the 3D-CNN works were proposed to classify brain MRI scans into two predefined groups. Brain MRI datasets from ADNI (which included 140 patients for the classification of Alzheimer's disease) and ABIDE (which included 1000 subjects for Autism) were used to assess the proposed framework. In the ADNI dataset, the suggested GABM approach has extracted 6 to 65 knowledge-related brain areas and 15 to 75 knowledgeable brain regions in the ABIDE dataset. These areas are interpreted as the parts of the brain from which the 3D-CNN primarily extracts features for classifying brain diseases. According to experimental findings, the suggested GABM technique has, in certain situations, improved the classification model's ultimate performance in terms of model parameters in addition to improving model interpretability. In [243], a deep-learning-based framework was used to establish a diagnostic tool by employing a transfer learning algorithm. They implement a generic AI platform for diagnosing and referring diabetic macular edema and choroidal neovascularization found in neovascular age-related macular degeneration (AMD), two principal reasons for significant loss of sight.AN Occlusion testing was utilized to determine the critical areas the model considered when diagnosing. The main advantage of an occlusion test is that it provides insights into neural network decisions, which are notoriously known as "black boxes" because they lack transparency. This test de-mystified the algorithm without changing outcomes because it was run after training. The occlusion

Table 9 SHAP-based techniques for brain, breast, cardiovascular, and chest diseases

Main XAI techniques	Anatomical location	Form of data	References
SHAP	Breast	MRI	van der Velden[231]
	Chest	CT	Zhu and Ogino [232]
	Skin	Dermatoscopy	Young [233]

Table 10 Perturbation-based techniques

Main XAI techniques	Anatomical location	Form of data	References
Occlusion sensitivity	Brain	MRI	Dubost [234]
			Shahamat and Saniee Abadeh [235]
	Breast	Histology	Gecer [236]
Prediction difference analysis	Brain	MRI	Seo [237]
			Zintgraf [238]
	Skin	Photography	Li [239]
LIME	Chest	X-ray	Rajaraman [240]
	Gastrointestinal	Endoscopy	Malhi [241]

test verified that the network's judgments were based on specific distinguishing factors that could be communicated to a medical expert.

2. **Local interpretable model-agnostic explanations (LIME):** Ribeiro introduced this explanation model [32]. LIME provides a local explanation by substituting simpler models for more complicated ones where necessary, for instance, adjusting a Linear model to A CNN. Complex model outputs are altered as the input data are changed. LIME learns the relationship between output changes and input data when input data is perturbed using a simpler model. To ensure that explanations provided by fundamental models with disturbed inputs have less influence on the concluding explanation, a weight based on how similar the perturbed input is to the original input is used. In medical imaging, Rajaraman et al. [244] presented an AI system based on deep learning to identify pneumonia in pediatric chest X-rays (CXRs) to speed up diagnosis, enable early treatment, and enhance clinical decision-making. The reviewer employed LIME to explain the model predictions by region of interest (ROI) localization in chest x-rays for pneumonia diagnosis, which is more pertinent to making decisions and essential for successful interaction between humans and computers. Malhi [245] to explain where bloody regions were found in gastral endoscopy images, LIME was used by Malhi.
3. **Meaningful perturbation:** To discover changes in the predictions of a trained neural network, Vedaldi, and Fong [246] altered the input image to initiate that kind of perturbation. Instead of employing occlusion sensitivity or other perturbations that obscure certain portions of the image, they proposed replicating realistic or believable effects, resulting in more significant perturbations and, subsequently, more meaningful explanations. They chose three types of local perturbations-a constant value, noise, or blurring-. Akselrod-Ballin [247] uses A linked set of digital mammography images and electronic health data to introduce an algorithm that combines machine and deep learning strategies for early breast cancer diagnosis. Because The ML-DL model does not yet offer localization of the finding, only a global probability for the entire breast, they follow Fong and Vedaldi [246] approach by inferring the areas that most contributed to the global prediction. In contrast to Fong and Vedaldi, Uzunova [248] indicated that the suggested perturbations weren't appropriate for medical imaging. Uzunova [248] sees that replacing portions of a medical image with a constant value is improbable because medical images typically have noise and blur. They suggested utilizing a variational autoencoder (VAE) to swap out diseased regions for their healthy tissue equivalents. They demonstrated how the VAE perturbations might be used to identify diseased regions in various imaging tests, including brain MRI and optical coherence tomography pictures of the eye. Additionally, they demonstrated that utilizing a VAE produced better pathology localization than using straightforward blurring or constant-value perturbations. Following Uzunova's [248] approach, Lenis [249] replaced diseased regions with healthy tissue equivalents using inpainting. Using the Hausdorff distance between the pixel-level ground truth labels and thresholded heatmaps obtained from the saliency maps, they showed that inpainting's perturbations outperformed backpropagation and Grad-CAM in identifying masses on chest X-rays caused by TB and breast mammography.
4. **Prediction difference analysis:** Robnik-Šikonja [250] described a method for explaining predictions that produce explanations for each specific event. The approach is based on the breakdown of model predictions. These decompositions are utilized to determine the specific contribution of each characteristic and can be thought of as local gradients. Prediction difference analysis measures how predictions change when pixels are considered unknown. Each pixel is assigned a

significant value if the pixel is supposed to be a feature in an image. Zintgraf [251] for generating saliency maps Zingraf adapted the prediction difference analysis by using multivariable analysis, implying they examined the connected pixels in the same patches rather than individual pixels, and conditional sampling, which limited their research to pixels that are difficult to forecast by just examining neighbors. They provided a comparison of brain MRI results between HIV patients and healthy controls to explain the classifier's choice. In [252] Proposed a model to classify skin lesion dermoscopic images using the saliency features (biomarkers) found by the DL model. They conducted a prediction difference analysis to interpret the features of the deep learning classifier used to categorize skin lesions. Seo [253] employed superpixels (or supervoxels for 3D) and prediction difference analysis on several scales. Since they follow image edges, these multiscale supervoxel-based saliency maps offered a visually appealing explanation as described by the authors. The saliency maps described which areas provided helpful information to the classifier to discriminate healthy controls from patients with Alzheimer's.

4.1.8 Multiple-instance learning-based approaches

Multiple-instance learning-based approaches have been employed for visualization explanations. In multiple-instance learning, training sets are composed of bags of instances, as described by Dietterich [254]. While the individual instances within the bags may not be labeled, the bags themselves are labeled. Cheplygina [255] reviewed a patch-based approach using multiple-instance learning in the context of medical image analysis. In this approach, the bag can be represented as an image, and the instances are represented as patches extracted from that image. Several researchers have utilized this strategy to identify the instances of the bag that contribute to the classification decision.

Saab [256] utilized large datasets labeled at either the full-scan level or the individual slice level to develop a model for identifying intracranial hemorrhage (ICH) in skull computed tomography (CT) scans. Their attention-based multiple-instance learning method enabled the generation of relevant signals at different levels of granularity, even with scan-level supervision. By leveraging this doubly weak supervision strategy, models trained with less than 10 h of clinician labeling time achieved an average Area Under the Receiver Operating Characteristic Curve (ROC-AUC) score of 0.91, comparable to models trained on larger, manually labeled datasets. Schwab et al. [257] applied a patch-based approach to localize important findings in chest X-rays. Each image patch was assigned a prediction, which was then superimposed on the original image to visually indicate the regions where the classifier made judgments.

Furthermore, Araújo [258] employed multiple-instance learning to explain the regions of significance for diabetic retinopathy in fundus images. Using a patch-based methodology, visual explanation maps were generated for each grade of diabetic retinopathy. These techniques are summarized in Table 11, which provides an overview of the papers that have utilized multiple-instance learning-based approaches, sorted according to the techniques employed in each study. Papers utilizing multiple-instance learning-based approaches were sorted by the techniques employed.

4.2 Attention mechanisms explanations

Attention mechanisms represent a global approach to XAI that assigns importance weights to different input features, enabling the model to explain its focus during the decision-making process. In medical imaging, attention-based methods have been used in tumor detection and segmentation tasks, where they highlight image regions that contributed most to the model's predictions. For instance, attention mechanisms can identify specific areas of a CT scan associated with lung abnormalities or emphasize regions in an MRI scan indicative of a brain tumor. These techniques complement local explanation methods, such as saliency mapping, by providing a broader perspective on model behavior [44, 266].

However, attention mechanisms may occasionally assign importance to regions that are not clinically relevant, requiring careful validation by clinicians.

4.3 Global explanation models

Global explanation models, such as attention mechanisms and interpretable decision trees, provide insights into how AI systems make predictions across datasets, offering a broader understanding of model behavior. These models are particularly useful for identifying trends and patterns in medical imaging, such as correlations between anatomical regions and disease outcomes, by enabling interpretable visualizations of global patterns. For example, attention-based global explanations can highlight relationships between specific anatomical areas and disease severity. While global models enhance overall interpretability, they are often less effective at explaining individual predictions, making their integration with local methods, such as saliency maps, essential for a comprehensive interpretability framework. By combining both

Table 11 Multiple instances learning-based techniques

Main XAI techniques	Anatomical location	Form of data	References
Multiple instance learning	Brain	CT	Saab [259]
	Breast	MRI	Maicas [260]
	Chest	X-ray	Li [261]
			Schwab[262]
	Eye	Fundus photography	Araújo [263]
			Costa [264]
	Gastrointestinal	CT	Shapira [265]

global and local approaches, clinicians gain a more complete understanding of AI system behavior, with global models offering a high-level overview and local methods providing detailed case-specific explanations[44, 50].

4.4 Counterfactual explanations

Another promising XAI technique is counterfactual explanations, which provide actionable and interpretable insights by illustrating how slight modifications to input features could alter a model's output. In medical imaging, counterfactual methods are particularly beneficial in exploring diagnostic "what-if" scenarios, enabling clinicians to understand the causal relationships between imaging features and predictions. For instance, a counterfactual model may demonstrate that reducing the size of a detected mass would change a diagnosis from "positive" to "negative" for malignancy, aligning with clinical decision-making processes. Despite their promise, counterfactual methods face challenges in generating realistic and clinically meaningful counterfactuals, particularly in medical applications where the input features represent complex physiological patterns [46, 267].

4.5 Hybrid techniques explanation

Hybrid XAI Approaches combine multiple explanation methods to provide richer interpretability and address the limitations of individual techniques. For instance, integrating saliency mapping (e.g., Grad-CAM) with textual justifications enables clinicians to visualize key regions of an image while simultaneously receiving a contextual explanation of the decision. These hybrid methods are particularly valuable in scenarios where both local and global interpretability are required, such as explaining rare or complex medical conditions. Studies have shown that hybrid approaches improve clinical trust and usability by catering to diverse information needs of healthcare professionals [32, 268].

However, combining methods can introduce additional computational complexity, which needs to be addressed for real-time applications.

4.6 Textual explanation

This is an XAI form that gives textual descriptions. These descriptions could be simple feature reports up to specialist reports. Three textual forms will be discussed in this section below.

4.6.1 Image captioning

Using a CNN with a recurrent neural network (RNN)-specifically a long-short term memory net (LSTM) for encoding of the image and textual encoding, respectively, Vinyals [269] presented using a framework based on image captioning to give textual explaining for images. To describe training, they used sentences generated by humans as a fact and the bilingual evaluation understudy (BLEU) metric to evaluate this. For example, Gasimova [270] used previously obtained radiological reports from the hospital archiving system to propose a model to provide automatically organized reports from the image depending on image captioning for a chest x-ray, using this in the clinical decision-making system. Combining a new technique to show visual output for the network's most nearby alternative diagnosis with a textual explanation. Spinks [271] Propose a new technique to explain a neural network's diagnosis of medical images. In terms of explanation, The image cation technique used in this method runs admirably on personal examination and outperforms traditional saliency maps methods. These techniques were tested in a human professional opinion study and on recent single-task baseline results.

4.6.2 combining image captioning-visual explanation

Zhang [272] proposed a system that employed dual attention for text and images. They took a similar method to image captioning, employing an encoder for the image and an LSTM for the text, but they added dual attention. This enabled high-level interactions between picture and text predictions, resulting in visual attention maps in Histology images corresponding to textual explanations. Using a similar technique, Jing [273] Introduced a multitask

learning framework that can predict tags and generate descriptions simultaneously. Additionally, they presented a co-attention technique that can investigate visual and semantic information at the same time to pinpoint and describe aberrant locations accurately. They created a hierarchical LSTM network to handle long-range semantics more efficiently and produce slightly elevated long texts. According to current radiology practices, it consists of a visual examination followed by a textual report.

Kashyap [274] introduced a new approach using deep learning to determine the places for diagnosis in radiology images. This method provides a way to improve this theory by producing a mark on the image related to an explanation in the text. This is accomplished by automatically extracting location-specific labels from textual reports and learning the correlation of expected locations to labels via a hybrid combination of Bi-Directional Long Short-Term Memory RNNs (Bi-LSTM) and DenseNet-121. Lee [275] used visual explanations with image captions to show breast mammograms. They confirmed that the provided explanations correspond to the proper language of breast mammography reports by adding a visual word constraint loss to the text-generating LSTM.

4.6.3 Testing with concept activation vectors (TCAV)

Kim [276] proposed the Concept of Activation Vectors (CAVs), which provide a human-friendly interpretation of a neural network's internal state. The Main notion is to see a neural network's high-dimensional internal state as an assistant, not an impediment. The author uses CAVs as part of a technique, testing with concept activation vectors (TCAV) that employ directional derivatives to assess the importance of a user-defined notion in a classification result. For example, how sensitive is a zebra's prediction to the presence of stripes? Or ' spiculated mass' to determine if an image had malignancy. These notions can be presented as a post-hoc analysis after training the neural network. The authors demonstrated the viability of TCAV on medical image processing for diagnosis. Cine-MRI was able to detect heart illness by Clough [277] categorizing a VAE's latent space. The clinically known indicators associated with heart illness were demonstrated using TCAV. They also added the CAV to the latent space to rebuild pictures with low peak ejection rates, which may be a sign of heart illness. Graziani [278] introduced regression concept vectors, which enhanced TCAV. The primary change was that regression concept vectors were continuous-valued measures of a concept, whereas TCAV represented the presence or absence of binary ideas. This can be helpful when examining a continuous concept, such as tumor size. They might be able to clarify, for instance, why the network identified one region of a breast histopathology scan as cancerous and another as healthy: The idea of "contrast," which refers to a clinically used system for judging cell size, was highly valued in both parts of the image, while the concept of "nuclei area," which distinguishes between healthy and malignant regions, was not.

4.7 Example-based explanation

This is an XAI technique called an example-based explanation that offers examples relevant to the data point examined at the time. This is relevant to human reasoning and can be helpful when attempting to explain how a neural network concluded. For instance, knowing the outcome of an earlier biopsy may help a pathologist make a better clinical choice when examining a biopsy from a patient who exhibits similarities to another patient they previously evaluated. The latent space, one of the hidden layers deep in the neural network, is frequently optimized by an example-based explanation such that similar points are close to one another, and dissimilar ones are farther apart.

4.7.1 Triplet network

Three identical networks with shared parameters make up a triplet network. These networks receive three input samples and output two values representing the L2 distances between the input samples' embedded representations (i.e., representations in the latent space). This enables the unsupervised comparison of data to learn useful representations. When studying a data point, this embedded representation's neighbors will show instances of data points compared to the one being studied, which can explain how the network arrived at its conclusion Hoffer and Ailon [279]. Yan [280] extracted 32,000 clinically significant lesions from the complete body using a radiological image archiving and communication system (PACS). To learn appropriate lesion embeddings, they trained a triplet network containing three supervision cues-lesion size, lesion anatomical location (such as the lung, liver, or kidney), and relative coordinate of the lesion in the body. These embeddings distinguished liver lesions from lung lesions based on anatomical location and successfully retrieved an example-based explanation from a test set. Codella [281] utilized a triplet loss as well, but they merged it with CAM's global average pooling method. As a result, they could extract example-based explanations, search result activation maps, and query activation maps. Or, to put it another way, a visual explanation demonstrated where the network used the area of the input image to produce the example-based explanation. They used dermatological photos of melanoma to show this technique.

4.7.2 Influence functions

In their proposal, Wei Koh and Liang [282] suggested using influence functions to describe which inputs from a training set a decision relied on. They did this by examining what would occur if one input from the training set was unavailable or altered. Due to the high cost, they offered an effective approximation using influence functions. This impact function implementation and SHAP are similar in enabling accurate computation of feature importance. Wang [283] utilized influence functions to clarify which radiological parameters were associated with which categories of liver lesions on multiphase MRI. This comprehensive description sheds light on the functioning of the neural network. For instance, the radiological finding "thin-walled mass" was most frequently related to the category "benign cyst." This justification could increase user confidence in the network's output because the network produced the class label and the related radiological properties.

4.7.3 Prototypes

According to Chen et al. [284] proposal, which they referred to as "this-looks-like-that" prototypes, typical examples should be used to explain concepts. The approach replicated human case-based reasoning. For instance, when someone explains why a photo has a car in it, they may internally conclude it is a car since it resembles a car they have previously seen. The neural network was given a prototype layer that categorized training inputs into classes in the latent space. For each class, a prototype that included a representative member was chosen. The method used portions of the test image that resembled these trained prototypes throughout testing. The result was a weighted combination of these prototypes' commonalities. As a result, the explanation was based on an accurate neural network calculation rather than a post hoc guess. To explain why a neural network identified certain patches of histology images as cancer or as non-cancer, Uehara et al. [285] employed prototypes. The network could recognize the areas of the image on which it based its judgment and the degree to which those areas of the image resembled prototypical examples discovered from the training set.

4.7.4 Examples from the latent space

Sarhan et al. [286] suggested employing a residual adversarial VAE with a total correlation constraint to learn disentangled representations of the latent space. This adversarial VAE improved the reconstruction's fidelity and offered more thorough explanations of the data's underlying generative properties. They demonstrated that, compared to a conventional VAE, their approach produced reconstructions more accurate to human-interpretable variables, including lesion size, lesion eccentricity, and skin color. Sabour et al. [287] demonstrated that they could encode high-level aspects of images by substituting vectorized representations (i.e., capsules) for the scalar feature maps from convolution neural networks. In a layer, capsules were essentially smaller groups of neurons. In the following layers, these were connected to smaller groups of neurons, forming a capsule network. Using dynamic routing, this capsule network was made more efficient. In other words, higher-level capsules are activated if their equivalent lower-level capsules are active. Routing coefficients, which added up to one for each capsule, were used to explain this correspondence. In response to new input data being fed into the capsule network, the coefficients were iteratively (i.e., dynamically) modified. They discovered that these capsules learn human-interpretable properties like scale, thickness, and skew for the MNIST digits dataset. LaLonde [288] employed capsules to diagnose lung cancer and predict visual characteristics such as sphericity, lobulation, and texture. They modified the dynamic routing technique to consider that these visual characteristics were not mutually exclusive, as they were in MNIST (a digit cannot be both a two and a nine simultaneously). Notably, the routing coefficients were not required to add up to one in their implementation. According to the previous, lung nodule malignancy might be predicted as well as these visual characteristics.

4.8 Multimodal XAI

Recent advancements in XAI have introduced multimodal techniques that integrate data from diverse sources, such as medical images, clinical notes, genetic markers, and patient histories. By combining multiple modalities, these methods provide comprehensive and interpretable explanations that align closely with the decision-making processes of healthcare professionals. For example, in cancer diagnostics, multimodal XAI systems can analyze imaging data alongside genetic markers and patient histories, offering holistic explanations that enhance diagnostic accuracy while ensuring interpretability across various data types within the broader clinical context.[39]. Despite their potential, multimodal XAI techniques require significant computational resources and robust integration frameworks, which are ongoing areas of research.

4.9 Emerging technologies and their impact on XAI in medical imaging"

The field of XAI in medical imaging is evolving rapidly, driven by the integration of emerging technologies that

address critical challenges such as data scarcity, interpretability, scalability, and adoption in clinical workflows. Innovations such as generative AI, multimodal models, and advanced computational frameworks are transforming the way XAI systems are developed, deployed, and utilized in healthcare. These technologies not only enhance the robustness and generalizability of XAI models but also create opportunities to align AI systems more closely with the complex demands of clinical decision-making and regulatory compliance [289–291].

By leveraging generative AI, synthetic medical imaging datasets can be created to overcome the limitations of small and imbalanced datasets, while maintaining patient privacy [289]. Multimodal approaches enable the integration of diverse data sources, such as imaging, textual reports, and genetic profiles, to provide layered and clinically relevant explanations [291]. Meanwhile, advanced frameworks like federated learning, edge computing, and blockchain technology are addressing scalability and data security concerns, paving the way for widespread adoption of XAI in high-throughput medical imaging environments [290, 292].

This section explores how these emerging technologies are reshaping XAI, focusing on their contributions to overcoming existing limitations and driving its adoption in real-world clinical practice. By highlighting the synergies between XAI and these transformative technologies, this discussion provides insights into the future directions of explainable AI in medical imaging. Later in the section, we will discuss several technologies that can emerge with artificial intelligence.

- **Generative AI:**Generative AI models, such as Generative Adversarial Networks (GANs) and diffusion models, play a pivotal role in addressing data scarcity and improving model robustness. By creating high-quality synthetic datasets, these techniques enable training on diverse and representative data, including rare or edge-case scenarios. This not only enhances the generalization capabilities of XAI models but also supports privacy-preserving collaborations by eliminating the need to share sensitive patient data. Additionally, counterfactual visualizations generated by generative models provide actionable and interpretable insights, fostering greater trust among clinicians [289].
- **Multimodal Models:** Multimodal XAI techniques integrate various data sources, such as imaging, clinical notes, and genetic information, to deliver comprehensive and context-aware explanations. These models are particularly valuable in medical imaging, where combining data modalities can enhance diagnostic accuracy and precision. For instance, integrating radiological imaging with genomic profiles has shown promise in cancer diagnostics, offering clinicians layered insights that align with multidisciplinary decision-making processes [291]. The ability to provide enriched, multimodal explanations makes these models highly adaptable to complex healthcare environments.
- **Foundation Models:** Foundation models, such as transformers and large pre-trained architectures, introduce scalability and generalizability to XAI in medical imaging. These models leverage pretraining on diverse datasets, reducing the dependency on task-specific labeled data. They enable fine-tuning for specific applications, ensuring robust performance across various imaging modalities. Moreover, the scalability of foundation models accelerates the adoption of XAI by providing consistent and standardized solutions for diverse clinical workflows [293].
- **Advanced Computational Frameworks:** Distributed computing and federated learning frameworks are critical enablers of scalable and privacy-preserving XAI systems. Distributed systems facilitate the real-time processing of large-scale imaging datasets, enabling the deployment of XAI methods in high-throughput environments. Federated learning supports collaborative model development across institutions while ensuring compliance with privacy regulations like GDPR and HIPAA. These frameworks address the computational challenges of XAI, making it more feasible for clinical adoption [290].
- **Federated Learning:** Federated learning enables collaborative training of XAI models across multiple institutions without sharing sensitive patient data. This privacy-preserving approach ensures compliance with regulations like GDPR and HIPAA while allowing the development of robust XAI systems trained on diverse datasets. By enabling decentralized learning, federated frameworks enhance the generalizability and adoption of XAI in medical imaging [290].
- **Edge Computing:** Edge computing supports the deployment of XAI models on portable and real-time systems by performing inference and explanation generation at the point of care. This reduces latency and enhances scalability, making XAI outputs accessible for bedside diagnostics and emergency use cases. For example, edge devices integrated with XAI models can provide immediate, interpretable results from handheld ultrasound or wearable imaging devices [294].
- **Blockchain Technology:** Blockchain enhances XAI by ensuring the integrity and traceability of medical imaging data used for training and validation. Immutable records maintained through blockchain can validate the provenance of XAI-generated explanations, fostering greater trust in AI systems. Additionally,

blockchain enables secure and transparent data sharing for collaborative research settings [292].

- **Internet of Medical Things (IoMT):** IoMT integrates real-time data streams from medical devices, such as imaging systems and patient monitors, with XAI models. This combination allows continuous monitoring of patient conditions with explainable outputs. IoMT-enabled XAI solutions improve early diagnostics, such as detecting cardiac events or abnormalities in imaging data, and align well with real-time clinical workflows [295].
- **Reinforcement Learning (RL):** RL enhances XAI by dynamically adapting explanations to user feedback. RL-based XAI models learn optimal policies for generating tailored explanations that improve clinician usability and trust. In medical imaging, RL combined with XAI can support real-time decision-making in dynamic environments like intensive care units (ICUs) [296].
- **Neuro-Symbolic AI:** Neuro-symbolic AI combines the statistical learning of neural networks with the logical reasoning capabilities of symbolic systems. This hybrid approach produces inherently interpretable models that provide logical explanations alongside data-driven predictions. In medical imaging, neuro-symbolic AI can integrate imaging data with clinical guidelines to offer rule-based, transparent insights [45].
- **Natural Language Processing (NLP):** NLP enhances XAI by generating textual explanations of model outputs, making explanations more accessible to non-expert users. For instance, XAI models integrated with NLP can convert saliency maps or feature importance scores into natural language summaries, improving clinician understanding. NLP also supports interactive querying of XAI outputs, enabling seamless integration with radiology reports or electronic health records [44].

4.10 Challenges and conflicting perspectives in XAI

While XAI techniques have significantly advanced the interpretability of AI models in medical imaging, several challenges and conflicting findings remain. Saliency maps, one of the most popular methods for local explanations, have been criticized for their potential to highlight spurious correlations instead of clinically relevant features. For example, recent studies have shown that saliency maps may emphasize irrelevant areas in an image, leading to misinterpretations in diagnostic tasks [15].

In contrast, example-based explanations, such as case-based reasoning, offer an alternative by identifying similar cases in the training dataset to justify predictions. However, these methods are computationally intensive and may struggle with scalability when applied to large datasets [70]. Similarly, counterfactual explanations, while providing actionable insights by illustrating how input changes affect predictions, often face challenges in generating realistic counterfactuals in the medical domain [71].

These conflicting findings underscore the need for comparative studies that evaluate the suitability of different XAI methods for specific medical imaging tasks. Balancing interpretability, reliability, and computational efficiency remains a critical area for future research.

4.11 Ethical considerations in XAI for medical imaging

XAI holds the promise of transforming medical imaging by enhancing transparency, trust, and interpretability. However, to ensure its reliable adoption in clinical practice, XAI must adhere to critical ethical principles: fairness, accountability, and compliance with data privacy regulations [297].

4.11.1 Fairness

Ensuring fairness is vital for XAI systems to operate equitably across diverse patient populations. This involves addressing biases in datasets and algorithms that can lead to disparate outcomes, particularly for underrepresented demographics. Imbalanced datasets, for example, may result in unequal treatment of patients based on age, gender, or ethnicity. To mitigate these issues, techniques such as curating diverse datasets and employing fairness-aware algorithms are essential [43, 298, 299]. Additionally, rigorous testing of XAI tools is necessary to verify consistent performance across varied groups, ensuring that all patients benefit equally from these systems and promoting equitable healthcare outcomes.

4.11.2 Accountability

Accountability ensures that decisions made by XAI systems are auditable and explainable. Tools like SHAP and Grad-CAM provide clinicians with clear visualizations or numerical justifications for predictions, fostering trust and enabling validation. Additionally, accountability mechanisms such as decision logs and model versioning allow for thorough audits of AI outputs, ensuring adherence to medical and regulatory standards [300].

4.11.3 Compliance with data privacy regulations

In medical imaging, Safeguarding sensitive patient data is critical for XAI systems, particularly in fields like medical

imaging, where privacy concerns are paramount. Adherence to data privacy regulations such as GDPR and HIPAA is essential to ensure the protection of personal health information during data collection, processing, and sharing. Privacy-preserving techniques, such as federated learning and differential privacy, offer innovative solutions by enabling collaborative model training without sharing raw data [290, 300]. These approaches not only uphold patient confidentiality but also ensure ethical responsibility and legal compliance, paving the way for the secure and trustworthy deployment of XAI tools.[298, 299]

5 Real-world applications of explainable AI (XAI) in medical imaging

Explainable AI (XAI) is transforming medical imaging by increasing the transparency, interpretability, and trustworthiness of AI-driven diagnostic systems. The integration of XAI in clinical workflows allows healthcare professionals to validate AI decisions, identify potential biases, and ensure that model predictions align with human expertise. While black-box AI models have demonstrated high accuracy, their lack of interpretability has been a significant barrier to widespread clinical adoption. XAI mitigates this challenge by providing human-understandable explanations that enhance clinician confidence and facilitate regulatory compliance. This section explores real-world applications of XAI across radiology, pathology, dermatology, cardiology, and emergency medicine, demonstrating its impact on diagnostic accuracy, patient safety, and medical decision-making.

1. Radiology: Enhancing Trust in AI-Based Image Interpretation Radiology has been one of the earliest adopters of AI-driven diagnostic tools, given the large volume of medical images analyzed daily. AI models are increasingly used for disease detection, segmentation, and risk assessment in imaging modalities such as X-rays, CT scans, and MRIs. However, one of the primary challenges has been ensuring that these models make decisions based on clinically relevant features rather than confounding artifacts. A key application of XAI in radiology is pneumonia detection from chest X-rays, where Grad-CAM (Gradient-weighted Class Activation Mapping) is used to generate visual heatmaps highlighting regions of interest in the lungs [34]. These heatmaps help radiologists verify that the AI model correctly identifies inflamed lung regions, ensuring that the decision aligns with medical knowledge. Similarly, in brain tumor segmentation from MRI scans, XAI methods such as SHAP (Shapley Additive Explanations) and Deep Taylor Decomposition have been used to identify specific tumor regions, increasing the reliability of AI-driven diagnoses [301]. Another crucial advancement in radiology is fracture detection from X-ray imaging. AI-based fracture detection systems use LIME (Local Interpretable Model-agnostic Explanations) to highlight skeletal abnormalities that influence the model's classification. A study comparing LIME, SHAP, and Grad-CAM for bone fracture detection in musculoskeletal X-rays found that models trained with XAI-generated feedback performed better in real-world settings, leading to increased clinical adoption [38]. Impact on Clinical Workflows: Studies indicate that when AI-generated explanations align with radiologists' expectations, clinicians report a higher level of trust and acceptance of AI-assisted diagnostics, leading to improved diagnostic accuracy and reduced second-opinion requests [302].
2. Pathology: Improving Diagnostic Accuracy in Tissue Analysis Pathology involves the microscopic examination of tissue samples to detect diseases such as cancer, infections, and inflammatory conditions. AI-powered pathology tools enhance efficiency and accuracy, but their use in clinical settings requires high interpretability to ensure that the AI's decision-making process aligns with pathologists' expertise. A major application of XAI in pathology is breast cancer detection from biopsy slides, where AI models using Grad-CAM and Class Activation Mapping (CAMs) generate heatmaps that highlight malignant cell clusters [38]. These explanations help pathologists correlate AI predictions with cellular morphology, ensuring that the model focuses on relevant histopathological structures. Additionally, counterfactual explanations are being used to improve cancer grading. In automated prostate cancer diagnosis, XAI techniques allow pathologists to understand why a biopsy sample was classified as aggressive or benign. By modifying key features and observing changes in AI predictions, clinicians can validate the robustness and reliability of AI models [303]. Impact on Clinical Workflows: AI-powered pathology models with XAI explanations have been shown to reduce inter-observer variability among pathologists, leading to more consistent and reproducible cancer diagnoses [304].
3. Dermatology: Explainability in Skin Lesion Classification In dermatology, AI-driven models are used to classify skin lesions, moles, and melanoma risks based on images captured using dermoscopy and mobile imaging technologies. The lack of explainability in early AI models led to skepticism among dermatologists, as there was no way to verify which features influenced the model's decisions. To address this, SHAP-based explainability models have been

integrated into skin lesion classification models trained on the HAM10000 dataset [304]. SHAP explanations provide pixel-wise feature importance, allowing dermatologists to visualize which parts of the lesion contributed most to the AI's classification. In comparative studies, SHAP-based models outperformed Grad-CAM in dermatology applications, as dermatologists preferred more granular attributions rather than broad heatmaps. Similarly, LIME-based explanations have been utilized in automated melanoma detection systems to assess whether the AI model focuses on relevant lesion characteristics. A study on AI-assisted dermatology showed that dermatologists were 30% more likely to trust AI decisions when LIME-based feature attributions were included [303]. Impact on Clinical Workflows: AI-based dermatology models enhanced with XAI explanations have demonstrated higher diagnostic accuracy, reducing the rate of unnecessary biopsies while improving early melanoma detection rates [302].

4. Cardiology: AI-Assisted ECG and Echocardiography Interpretation In cardiology, AI-assisted diagnostic tools analyze electrocardiograms (ECGs) and echocardiograms to detect arrhythmias, heart failure, and structural abnormalities. Due to the high-stakes nature of cardiac diagnoses, explainability is crucial for ensuring that AI predictions align with electrophysiologists' interpretations. A significant application of XAI in cardiology is atrial fibrillation detection from ECG signals. AI models using Deep SHAP interpretations generate feature importance plots, allowing cardiologists to identify which waveform segments contributed to the AI's classification [34]. Studies have found that clinicians prefer SHAP-based explanations over traditional black-box AI outputs, as they allow for manual verification of irregular heart rhythms. Similarly, explainable recurrent neural networks (RNNs) are used in heart failure risk stratification to analyze long-term changes in cardiac function. XAI techniques such as temporal attention mechanisms help highlight key diagnostic features across multiple time steps, improving clinicians' ability to track disease progression [301]. Impact on Clinical Workflows: AI-based cardiac models integrated with XAI explanations have led to higher clinician confidence in automated ECG analysis, resulting in increased adoption of AI-based decision support tools in cardiology [38].
5. Emergency Medicine: Real-Time XAI in Critical Decision-Making In emergency medicine, AI-powered diagnostic tools are increasingly used to support real-time decision-making where speed and interpretability are critical. One application is stroke detection from CT scans, where AI models using Grad-CAM overlays assist radiologists in quickly identifying ischemic and hemorrhagic stroke patterns [305]. These models reduce diagnosis time in emergency settings, improving patient outcomes. Another area is sepsis prediction in ICU patients, where AI models use SHAP-based interpretability to provide real-time risk scores for sepsis development. Studies show that hospitals adopting AI-based sepsis prediction systems have achieved earlier intervention and reduced mortality rates [306]. Impact on Clinical Workflows: Real-time XAI-enhanced diagnostic systems have improved triage accuracy and early intervention rates, reducing misdiagnosis and improving patient care in emergency departments [307].

6 Unique challenges in medical imaging with XAI

Medical imaging data presents several unique challenges, including noise, high dimensionality, variability in imaging protocols, data privacy concerns, and the need for clinical usability. These challenges must be addressed to ensure the reliability and effectiveness of XAI systems in real-world applications.

- **Noise in Medical Imaging Data:**Medical images often suffer from noise and artifacts, which can arise due to limitations in imaging hardware, patient movement, or environmental factors during acquisition. For example, low-resolution X-rays or motion artifacts in MRI scans can obscure critical anatomical details, reducing diagnostic accuracy. XAI techniques, such as saliency maps and Grad-CAM, help mitigate this challenge by highlighting the most influential regions in an image, enabling clinicians to distinguish diagnostically relevant patterns from noise [15].
- **High Dimensionality:**Medical imaging data, especially from 3D CT or MRI scans, is inherently high-dimensional, making interpretation and computational efficiency significant challenges. XAI methods like SHAP and LIME address this issue by focusing on the most relevant slices or regions within the data, reducing the dimensionality of interpretability without compromising diagnostic accuracy [16, 32].
- **Variability in Imaging Protocols:**Imaging protocols vary across institutions, scanners, and practitioners, leading to inconsistencies in image quality and format. For instance, different hospitals may use varying resolutions or contrast settings in MRIs, complicating the development of generalizable AI models. XAI methods can identify dataset-specific biases, aiding

researchers in harmonizing features across datasets and improving cross-institutional model performance [290].
- **Data Privacy and Sharing Restrictions:**Stringent privacy regulations, such as HIPAA and GDPR, limit access to diverse datasets essential for training robust AI models. Federated learning, combined with XAI techniques, offers a solution by enabling collaborative model training without sharing sensitive data. XAI methods further enhance this by revealing and mitigating biases introduced during localized training [290].
- **Clinical Usability and Trust:**Clinicians require AI systems to provide clear and actionable insights that align with their decision-making processes. Black-box models often fail to meet this requirement, leading to skepticism and low adoption rates. XAI models address this by offering interpretable outputs, such as heatmaps and feature attribution scores, which enhance usability and foster trust in AI-assisted decision-making [38].

7 Comparative analysis of XAI methods in medical imaging

In this section, we provide a comparative analysis of the various XAI methods discussed in the preceding sections. By highlighting the strengths, weaknesses, and suitability of each method, this analysis aims to guide researchers and practitioners in selecting the most appropriate approach for their specific medical imaging applications.

XAI methods differ significantly in mechanisms, computational complexity, interpretability, and applicability across different medical imaging modalities. Some methods, such as Backpropagation-based approaches and Class Activation Mapping (CAM), excel in generating visual heatmaps for identifying critical regions in medical images. These techniques are particularly useful in radiology applications (e.g., X-ray, CT, and MRI analysis), as they align well with clinicians' expectations for visual explanations. However, these methods may struggle with noise and may not offer sufficient granularity for complex tasks such as tumor segmentation or multi-organ analysis [15, 308].

Techniques like LIME and Deep SHAP offer model-agnostic explanations, making them versatile across different architectures and applicable to a wide range of medical imaging tasks, including histopathology, dermatology, and ophthalmology [65, 302]. Despite their flexibility, these techniques have higher computational costs, which can hinder real-time decision-making in settings such as emergency medicine or intensive care units (ICUs).

Perturbation-based approaches, such as Occlusion Sensitivity and Feature Attribution methods, provide robust insights into feature importance by analyzing how changes in input data affect model predictions. These methods are particularly useful in high-stakes diagnostic applications (e.g., stroke detection in MRI scans) but suffer from high resource requirements and interpretability issues when applied to high-dimensional data such as whole-slide pathology images [309].

Advanced approaches, including Testing with Concept Activation Vectors (TCAV) and Influence Functions, bridge technical explanations with domain-specific concepts, enabling deeper insights into decision-making processes [276, 282]. These techniques are particularly beneficial for explaining AI-driven risk prediction models in cardiology and oncology, as they allow for semantic-level explanations. However, they require extensive preprocessing and domain expertise, limiting their scalability in routine clinical practice.

While all XAI techniques aim to enhance interpretability, their effectiveness varies depending on the type of medical imaging modality and clinical use case. In radiology, which includes CT, MRI, and X-ray imaging, techniques such as Grad-CAM, Layer-wise Relevance Propagation (LRP), and Integrated Gradients are particularly effective. These methods provide pixel-wise feature attribution, making them well-suited for visual explanations in tumor detection, pneumonia classification, and stroke identification [310]. Their ability to highlight relevant regions within medical scans enhances clinician trust and facilitates decision-making in radiological assessments.

For histopathology, which deals with whole-slide images and biopsy samples, techniques like SHAP, LIME, and Counterfactual Explanations offer a more granular level of interpretability. These methods allow for precise feature attribution at the cellular level, making them highly valuable for identifying abnormal cell structures and diagnosing cancerous tissues. However, the application of these techniques requires substantial computational power, given the high resolution and complexity of histopathological data [34].

In the field of cardiology, where AI models are frequently applied to ECG and echocardiogram interpretation, SHAP and Attention Mechanisms have emerged as leading XAI methods. These techniques allow for detailed explanations of waveform-based AI predictions, helping cardiologists identify abnormalities such as arrhythmias, heart failure risk, and myocardial infarction indicators. By emphasizing which segments of the ECG contribute most to a given diagnosis, these XAI techniques improve clinician confidence in AI-assisted cardiac diagnostics [303].

Similarly, in ophthalmology, particularly in retinal imaging and optical coherence tomography (OCT), Grad-CAM and Attention Mechanisms have proven to be

effective. These techniques assist in diabetic retinopathy and glaucoma detection by highlighting affected retinal areas, allowing ophthalmologists to make more informed and interpretable assessments [305]. Their ability to provide spatially relevant feature explanations aligns well with the visual analysis requirements of eye disease diagnostics.

For ultrasound imaging, which includes point-of-care imaging and obstetric ultrasound, techniques such as LIME and Example-Based Explanations are particularly useful. Unlike structured imaging modalities like CT and MRI, ultrasound data is often highly variable due to probe positioning, patient movement, and real-time acquisition artifacts. XAI methods that allow for flexible, case-specific explanations are crucial in this setting, helping clinicians improve trust in AI-assisted fetal health assessment and organ imaging [311]. These methods provide context-aware insights, which are essential for interpreting AI predictions in dynamic and patient-specific ultrasound scans.

The choice of XAI technique should be guided by the specific medical imaging modality, interpretability requirements, and computational constraints of the application. While visual feature attribution methods such as Grad-CAM and LRP are widely used in radiology and ophthalmology, concept-based and perturbation-based explanations like SHAP and TCAV are more effective in histopathology and cardiology, where fine-grained feature importance is crucial for diagnostic accuracy. This comparative analysis highlights the trade-offs between interpretability, computational efficiency, and domain-specific applicability across XAI methods. Selecting the appropriate technique depends on factors such as model type, desired level of interpretability, and available computational resources, ensuring optimal alignment with clinical workflows and decision-making processes [312–316].

The following table 12will provide a more detailed comparison of these methods based on the discussed criteria, offering a clear overview of their strengths, limitations, and suitability for different applications.

8 Clinician interaction and usability of XAI in daily practice

Explainable AI (XAI) has the potential to enhance clinical decision-making by providing interpretable insights into model predictions. However, clinician interaction and usability remain critical challenges in the successful adoption of XAI systems in healthcare. Despite advancements in explainability techniques such as SHAP, LIME, Grad-CAM, and counterfactual explanations, studies indicate that clinicians face significant barriers in integrating XAI into their daily practice. This section explores clinician trust, usability evaluations, and key adoption barriers, offering insights into how XAI can be optimized for real-world healthcare settings.

1. Clinician Trust and Interpretability Challenges A major factor influencing the adoption of XAI in clinical practice is trust. Clinicians often struggle to trust AI-generated explanations, particularly when they do not align with their medical intuition or established diagnostic workflows. Studies have shown that explanations generated by XAI models must be interpretable, reliable, and contextually meaningful to be accepted by healthcare professionals [317]. For example, in radiology, Grad-CAM heatmaps for pneumonia detection in chest X-rays have been found to improve interpretability. However, when these heatmaps highlight irrelevant regions or fail to align with expected anatomical markers, clinicians may reject AI recommendations despite high accuracy [302]. Moreover, research has shown that even when AI-generated explanations align with clinical reasoning, clinicians may still override AI decisions due to concerns about liability and accountability. A study analyzing AI-assisted diagnosis in ICUs and emergency medicine found that clinicians preferred making independent final decisions even when AI predictions were highly accurate [306]. This underscores the need for explainability techniques that not only provide clarity but also build long-term clinician confidence in AI recommendations. furthermore, the concept of explainability varies across medical specialties. While radiologists prefer visual explanations (e.g., heatmaps in Grad-CAM), cardiologists dealing with ECG signals may require temporal feature importance scores (e.g., SHAP interpretations highlighting waveform segments) [318]. This discrepancy highlights the need for specialty-specific XAI solutions that cater to the unique interpretation needs of different medical domains. In addition to trust issues, clinician training on AI and explainability remains a significant gap. Many medical professionals receive little to no formal education on how to interpret AI-driven insights, leading to inconsistencies in XAI adoption across hospitals and specialties. Several AI-focused continuing medical education (CME) programs have emerged to address this, offering training on how to incorporate XAI explanations into clinical workflows [303].
2. Usability Studies: How Clinicians Use XAI in Diagnostics Empirical studies have investigated how clinicians interact with different XAI techniques in real-world medical settings. In radiology, Grad-CAM explanations have been evaluated for detecting pneumonia in chest X-rays and brain tumors in MRI scans.

Table 12 Comprehensive comparative analysis of XAI techniques

Technique	type	Model compatibility	Interpretability	Computational Cost/ scalability	Strengths	Limitations	Use cases
SHAP	Post-hoc Model-agnostic	Works with any model	High (local explanations)	High/ Medium	Model-agnostic, clear feature attribution	Computationally expensive	Tabular, structured datasets
LIME	Post-hoc Model-agnostic	Works with any model	High (local explanations)	High/ Medium	Intuitive for individual predictions	Instability with different data samples	Tabular, text, and image data
Grad-CAM	Post-hoc Model-specific	CNN	Medium (visual heatmaps)	Medium/ Low	Visual explanations, useful for CNNs	Limited to CNN architectures	Image classification, segmentation
Integrated Gradients	Post-hoc Model-specific	Differentiable models	Medium (attribution-based)	Medium	Provides pixel-level attributions	Requires baseline for comparisons	Image and text models
DeepLIFT	Post-hoc Model-specific	Differentiable models	Medium (attribution-based)	Medium	Considers neuron activation differences	Limited to neural networks	Medical imaging, neural networks
Saliency Maps	Post-hoc Model-specific	Differentiable models	Medium (visual overlays)	Low	Simple and efficient	Prone to noise and low detail	Image classification
LRP	Post-hoc Model-specific	Differentiable models	Medium (visual and numerical)	Medium/ Low	Provides pixel-level relevance for decisions	Requires well-optimized neural networks	Image analysis, deep learning diagnostics
Textual	Intrinsic/ Post-hoc	NLP models, multimodal	High (natural language)	Medium	Human-readable outputs	Limited to text-based domains	Clinical NLP, radiology reports
Example-Based	Post-hoc Model-agnostic	Works with any model	High (contextual examples)	Medium/ High	Provides interpretable examples for insights	May struggle with edge cases	Medical diagnostics, personalization
Multimodal XAI	Intrinsic/ Post-hoc	Multimodal	High (integrated outputs)	High/ Medium	Combines multiple data types (text, image)	Computationally intensive	Cancer diagnostics, precision medicine
Hybrid Techniques	Intrinsic, Hybrid	Symbolic-neural hybrid models	High (logical reasoning + interpretability)	Medium	Combines logical reasoning and learning	Requires specialized architectures	Knowledge-based diagnostics, rule-based AI
Counterfactual	Post-hoc, Model-agnostic	Works with any model	High (intuitive, actionable)	High/ Medium	Actionable, interpretable insights	Computationally expensive	Personalization, decision support
Attention Mechanisms	Intrinsic, Model-specific	Models with attention layers	High (contextual explanations)	Medium/ High	Inherent interpretability	Limited to models with attention layers	NLP, multimodal AI
Prototype Selection	Post-hoc, Model-agnostic	Works with any model	High (example-based)	Medium/ High	Provides examples for interpretability	May miss outliers	Medical diagnostics, case-based reasoning

Table 12 (continued)

Technique	type	Model compatibility	Interpretability	Computational Cost/ scalability	Strengths	Limitations	Use cases
Feature Ablation	Post-hoc Model-agnostic	Works with any model	High (quantitative feature impact)	Medium	Quantifies feature importance accurately	Requires multiple model runs	Tabular data, clinical decision support
SmoothGrad	Post-hoc, Model-specific	Differentiable models	Medium (noise reduction)	Medium	Reduces noise in visual explanations	Sensitive to parameter settings	Image models
(CAM)	Intrinsic Model-specific	CNN	Medium (visual heatmaps)	Medium	Class-specific visual explanations	Limited to CNNs with specific architectures	Image models
Occlusion Sensitivity	Post-hoc Model-agnostic	Works with any model	Medium (feature-specific)	High/ low	Identifies important regions in input	Computationally expensive	Image and structured data
Rule-Based Models	Intrinsic, Model-specific	Decision trees, rule lists	High (explicit rules)	Low/ High	Inherently interpretable	Limited to simple tasks	Tabular data, risk assessment
Neuro-Symbolic AI	Intrinsic, Hybrid	Works with symbolic-neural models	High (logical reasoning)	Medium	Combines logical reasoning and neural learning	Requires specialized architectures	Knowledge-based systems
Attention Rollouts	Post-hoc, Model-specific	Transformers, attention models	High (contextual explanations)	Medium	Explains decisions in attention layers	Limited to transformer-based models	NLP, multimodal systems

Research shows that radiologists are more likely to trust AI decisions when visual explanations highlight disease-affected areas that align with human annotations [304]. In pathology, SHAP-based attributions have been found to improve diagnostic confidence in tumor grading and histopathological analysis. A study comparing SHAP vs. Grad-CAM for breast cancer detection found that SHAP's pixel-wise feature attribution was more effective for pathologists, as it provided fine-grained explanations rather than broad heatmaps [38]. However, usability studies also revealed that some XAI explanations are too complex or computationally expensive, limiting their real-world applicability in high-volume diagnostic settings. Interestingly, longitudinal studies tracking clinician adoption of XAI over time suggest that trust and usability improve with continued exposure to explainability tools. A multi-hospital study found that after six months of AI-assisted diagnosis, clinicians were significantly more likely to accept AI recommendations compared to initial trials [319]. This indicates that familiarity and iterative feedback loops can enhance clinician trust in AI In emergency medicine, real-time decision-making requires XAI models that are both fast and interpretable. Studies on AI-assisted stroke detection from CT scans found that LIME-based perturbation models were too inconsistent, leading to poor clinician adoption [305]. Instead, hybrid XAI models combining Grad-CAM heatmaps with structured textual explanations improved usability and adoption rates in emergency departments.

3. Barriers to XAI Adoption in Clinical Workflows Despite its advantages, the integration of XAI into clinical practice faces multiple barriers that hinder widespread adoption. These challenges can be broadly categorized into cognitive load, time constraints, and regulatory concerns.

- Cognitive Load and Information Overload Clinicians are already burdened with interpreting large volumes of patient data. The introduction of complex AI-generated explanations can increase cognitive load rather than reduce it [303]. A usability study on AI-driven sepsis prediction found that clinicians often ignored AI-generated feature importance scores because they required additional mental effort to interpret [306]. To improve adoption, XAI must be designed for simplicity,

presenting only the most relevant insights rather than overwhelming clinicians with unnecessary details.

– Time Constraints in High-Stakes Environments In critical care and emergency medicine, clinicians lack the time to analyze detailed AI explanations. A study on AI-assisted ECG analysis found that SHAP-based feature attributions took significantly longer to process than black-box predictions, making them impractical for real-time triage settings [318]. To address this issue, researchers have proposed confidence-aware XAI systems, where AI models provide high-certainty predictions autonomously while deferring uncertain cases to human experts, optimizing time efficiency without sacrificing interpretability.
– Legal and Regulatory Concerns Clinicians remain legally responsible for patient outcomes, creating hesitation in fully relying on AI-driven diagnoses. Regulatory bodies such as the FDA and EU AI Act emphasize the need for explainability and accountability in AI systems [307]. However, without standardized guidelines for evaluating the reliability of AI-generated explanations, many clinicians prefer to rely on traditional diagnostic methods [319]. The development of clinician-in-the-loop AI frameworks, where medical professionals participate in AI training and validation, has been proposed as a solution to increase trust and accountability in medical AI applications.

9 User acceptance studies and clinician feedback on XAI adoption

Despite the increasing integration of explainable AI (XAI) into healthcare, clinician acceptance remains a critical factor influencing adoption and real-world impact. Several studies have explored how medical professionals interact with XAI models, assessing their trust, usability, and effectiveness in clinical workflows.

1. Clinician Perception and Trust in XAI Systems A study by Tonekaboni et al.[317] surveyed 200 physicians across multiple specialties to evaluate their trust in AI-generated explanations. The results indicated that clinicians were more likely to trust AI decisions when the explanation was aligned with their diagnostic reasoning. However, excessively complex or ambiguous AI explanations reduced trust, especially when the outputs conflicted with human intuition. Similarly, a usability study in radiology found that Grad-CAM heatmaps increased clinician confidence in AI-assisted pneumonia detection, provided that the highlighted areas aligned with expected anatomical markers [302]. However, in cases where the heatmap focused on non-relevant regions (e.g., image artifacts), radiologists ignored the AI's prediction, reinforcing the need for task-specific XAI validation. In addition to interpretability, clinician accountability concerns influence trust in AI models. Studies have shown that even when XAI-generated explanations align with clinical reasoning, many clinicians prefer to override AI-based recommendations to retain full legal and ethical responsibility for patient outcomes [306]. This underscores the need for XAI systems that not only provide explanations but also establish trust through consistent, validated, and regulatory-compliant decision support.
2. Real-World User Acceptance Studies Across Specialties Multiple studies have evaluated XAI user acceptance in radiology, pathology, dermatology, and cardiology: Radiology: A study on AI-assisted stroke detection from CT scans found that LIME-based explanations were often dismissed by emergency physicians due to their inconsistent feature attributions [305]. Instead, structured textual explanations combined with Grad-CAM heatmaps improved adoption rates. Pathology: In histopathological AI models, SHAP-based attributions were preferred over Grad-CAM, as they provided pixel-level insights into cancerous regions, leading to higher diagnostic confidence among pathologists [38]. Cardiology: A study on AI-assisted atrial fibrillation detection from ECG signals found that SHAP-based waveform explanations were more interpretable than black-box predictions, leading to faster clinician decision-making [318]. Longitudinal studies on XAI adoption suggest that clinician trust improves with continuous exposure and refinement of AI models. A multi-hospital study tracking clinician adoption of XAI over 12 months found that initial skepticism toward AI explanations decreased by 45% as clinicians became more accustomed to interacting with XAI interfaces [319]. This highlights the importance of iterative AI training programs and real-world feedback loops to enhance trust and usability over time.
3. Factors Influencing Clinician Adoption of XAI Clinician feedback has identified several key factors that influence user acceptance and real-world implementation of XAI models: Interpretability and Simplicity: Explanations must be concise, clear, and directly useful for decision-making. Alignment with Clinical Workflow: AI models that require additional cognitive effort or disrupt existing workflows are less likely to be adopted. Time Efficiency: In emergency settings, slow or complex XAI explanations hinder adoption-faster, confidence-aware models are preferred [306]. Regulatory and Legal Considerations: AI models must meet FDA and GDPR explainability requirements before

clinicians feel comfortable relying on them [307]. Another critical factor affecting clinician adoption of XAI is the lack of standardized training on how to interpret AI explanations. Many medical professionals lack formal AI education, making it difficult for them to critically assess XAI-generated insights. Recent efforts in medical AI education, such as AI-focused continuing medical education (CME) programs, have attempted to bridge this gap by introducing clinicians to explainability methods and AI-assisted decision-making [303].

Impact on XAI Model Development: Clinician feedback has been used to refine XAI models, leading to the development of hybrid AI systems that combine multiple explanation techniques (e.g., Grad-CAM + SHAP) to improve usability [304]. Additionally, collaborative AI training frameworks, where clinicians actively participate in labeling and refining AI outputs, have shown promise in improving both model accuracy and clinician trust. AI-assisted diagnostic platforms that allow real-time clinician feedback on AI recommendations are emerging as a practical approach to aligning AI systems with real-world medical workflows [43].

10 Comparison of XAI models with traditional Black-Box models

While traditional "black-box" AI models, such as deep neural networks, have demonstrated exceptional accuracy in medical imaging tasks, their lack of interpretability has been a significant barrier to adoption in clinical workflows. In contrast, XAI models aim to balance accuracy with transparency, enabling clinicians to understand and validate AI-generated predictions.

Comparative studies reveal the trade-offs between these approaches. For instance, Lundberg and Lee (2017) introduced SHAP, a method that retains high accuracy while providing interpretable feature importance scores. When applied to a pneumonia detection task, SHAP-based models achieved comparable diagnostic accuracy (94%) to black-box models (95%), but with the added benefit of generating visual explanations that helped clinicians validate predictions [16].

Similarly, Ribeiro et al. [32] demonstrated how LIME can explain predictions of complex classifiers, such as convolutional neural networks, without significantly compromising model performance. For example, in a study involving tumor classification in MRI scans, LIME-based models provided heatmaps highlighting relevant regions, enabling radiologists to confirm predictions with greater confidence. The interpretability offered by LIME resulted in improved usability and trust among clinicians.

However, XAI models often face challenges in achieving the computational efficiency of black-box models. For instance, generating SHAP explanations for high-dimensional datasets can be computationally expensive, limiting their scalability in real-time clinical environments. Nonetheless, advances in model compression and distributed computing are helping mitigate these challenges, bridging the gap between interpretability and performance [320].

The usability of XAI models in clinical practice also provides a critical advantage over traditional AI systems. Studies suggest that the trust and adoption of AI by healthcare professionals significantly increase when interpretable outputs, such as feature attribution scores or visual heatmaps, accompany predictions [321]. This usability factor underscores the importance of developing XAI systems tailored to the needs of end-users, such as clinicians and radiologists.

While XAI models may occasionally sacrifice marginal accuracy compared to their black-box counterparts, their enhanced transparency and usability make them indispensable for high-stakes medical applications where trust and accountability are paramount.

11 Evaluating explainable AI methods: importance and criteria

XAI methods have become increasingly important in domains like medical imaging, where accurate and interpretable model decisions are critical for clinical practice. Techniques such as Grad-CAM, SHAP, LIME, and saliency maps offer diverse ways to explain AI predictions, focusing on areas like feature importance, visualization, and localization. However, the utility of these methods depends not only on their technical implementation but also on their ability to meet specific evaluation criteria that ensure their reliability, usability, and relevance in real-world scenarios. These criteria provide a structured framework to assess the effectiveness of explanation methods and their suitability for high-stakes applications like healthcare, ensuring that they align with both technical and clinical needs [64, 322]. In this, we will list many, but not all, of the evaluation criteria for XAI methods, as you will see below. These criteria emphasize precision, clinical relevance, and usability for healthcare professionals.

– Clinical relevance: is critical in medical imaging, as explanations must align with established medical knowledge to support diagnostic and treatment decisions. Explanations that focus on biologically

significant areas, such as lesions or abnormalities, are more valuable than those highlighting irrelevant features [323] Methods like Grad-CAM [15] contribute to this goal by generating heatmaps that highlight regions in medical images influencing model predictions. For instance, in a chest X-ray used to diagnose pneumonia, Grad-CAM highlights inflamed areas of the lungs, aligning with clinical expectations and enhancing clinician trust and utility [65, 324]

- Fidelity to the model: ensures that the explanation accurately reflects the decision-making process of the AI system, which is crucial in medical contexts where clinicians depend on the reliability of the AI's reasoning [322] High-fidelity methods, such as gradient-based saliency maps, effectively communicate how the model interprets regions in an X-ray or MRI that influence its classification [325]. An example of such a method is SHAP (Shapley Additive exPlanations) [16], which assigns importance scores to input features, such as pixel regions in an image, to reveal how each part of an MRI contributes to diagnosing a tumor.
- Localization accuracy: evaluates whether the explanation correctly highlights the regions of interest in medical images, which is crucial for tasks like detecting pneumonia in chest X-rays, where the explanation must correspond accurately to the affected lung regions [326, 327]. This is particularly important for tasks such as tumor segmentation, where precise localization directly impacts patient care. An example method is LIME (Local Interpretable Model-agnostic Explanations) [65], which perturbs image regions to assess their contribution to the model's output. For instance, in melanoma detection from skin images, LIME can identify specific areas of the lesion responsible for the model's classification.
- Interpretability: refers to the simplicity and clarity of explanations, ensuring they are easily understood by medical practitioners who may not be AI experts. Intuitive methods, such as heatmaps overlaid on original images, are preferred over abstract representations, as they bridge the gap between complex AI systems and practical clinical workflows [328]. An example method is saliency maps [33], which visualize the most critical areas of an image influencing the model's decision. For instance, saliency maps can highlight tumor boundaries in an MRI scan, providing radiologists with clear and actionable insights.
- Robustness: ensures that explanations remain consistent even when there are variations in input data or noise. This is particularly important in medical imaging, where minor variations can arise from equipment differences or patient movement, making robustness essential for trustworthiness [65]. For example, explanations for brain tumor diagnoses should remain stable despite slight image perturbations. An example method is SmoothGrad [329], which enhances robustness by averaging saliency maps generated from multiple noisy versions of the input image. This approach provides stable and reliable explanations, which are crucial in dynamic imaging scenarios.
- Transparency: ensures that the process of generating explanations is clear and understandable, fostering trust among clinicians by making the underlying mechanisms of the AI system more accessible [330] For instance, explaining the creation of a saliency map, such as through gradient-based techniques, enhances the system's reliability and user confidence. An example method is rule-based models, such as decision trees, which are particularly useful in medical imaging classification tasks. For example, they can predict disease severity based on explicit pixel intensity thresholds, providing transparency through easily interpretable rules that clinicians can follow.
- Quantitative evaluation: involves the use of objective metrics to assess the quality of explanations, such as localization error or feature importance correlation. These metrics ensure that explanation methods are reliable and meet clinical requirements [322]By quantifying the effectiveness of explanations, they help validate methods for domain-specific needs. An example metric is IOU (Intersection Over Union), used with Mask R-CNN to evaluate its object detection process. Mask R-CNN generates segmentation masks, and the IOU metric measures how well these masks align with ground-truth annotations, providing a quantitative assessment of explanation quality.
- Ethical and regulatory compliance:is a crucial criterion in healthcare, requiring explanations to be interpretable and accessible to diverse audiences, including non-technical stakeholders, while adhering to regulations such as HIPAA or GDPR [331]. Compliance ensures trust in AI systems within highly regulated environments. An example method is Explainable AI using counterfactual explanations, which provide alternative inputs that lead to different model predictions. For instance, in diagnosing diabetic retinopathy, the model could illustrate how changes in pixel intensities might alter the diagnosis, ensuring the explanation is both interpretable and aligned with ethical standards.
- Trust and confidence building: focus on creating explanations that instill confidence among clinicians, making AI diagnoses more reliable and acceptable. Clear and interpretable justifications, such as combining textual descriptions with visual heatmaps, play a key role in enhancing trust [325]. This is particularly important for AI adoption in critical domains like

healthcare. An example method is Visual Question Answering (VQA) in medical imaging, which answers clinician-specific queries about a diagnosis. For instance, if a radiologist asks, "Why is this area marked abnormal?", the system could respond with an explanation such as, "High density in this region suggests potential malignancy," fostering trust through tailored and interpretable responses.

- Temporal consistency: is crucial for dynamic imaging modalities like MRIs or ultrasounds, where tracking changes over time is essential for accurate diagnosis and patient monitoring. Explanations must remain consistent and meaningful across temporal sequences to support clinical workflows effectively [326]. An example method is Explainable RNNs, which analyze sequential imaging data, such as echocardiograms, to identify consistent features over time. For instance, when monitoring cardiac function, the explanation highlights persistent abnormalities in heart motion across multiple frames, ensuring reliability and clinical relevance.

In summary, the evaluation criteria for XAI methods in medical imaging-such as clinical relevance, fidelity to the model, localization accuracy, interpretability, robustness, and ethical compliance-provide a comprehensive framework for assessing the effectiveness and reliability of AI explanations. These criteria ensure that XAI methods not only meet technical standards but also align with clinical needs, fostering trust and usability in real-world healthcare settings. However, to fully understand the robustness and universal applicability of these criteria, it is essential to examine their grounding in established standards, their adaptability across diverse tasks and modalities, and their alignment with ethical and regulatory requirements. In the following subsection, we delve deeper into these aspects, providing a detailed analysis of the robustness 11.1, universal applicability 11.2, Standardized Benchmarking Frameworks for Evaluating XAI Models 11.3,Empirical Comparisons of XAI Evaluation Metrics in Real-World Medical Applications 11.4,and justification of the evaluation criteria 11.5. Additionally, a summary table 13 is provided to concisely describe the criteria, their definitions, and their applicability across medical imaging tasks and modalities.

11.1 Robustness of the criteria

The evaluation criteria for XAI methods in medical imaging are designed to be robust, ensuring that they are reliable, measurable, and consistently applicable across diverse scenarios. Robustness is achieved through the following:

- **Alignment with Established Standards:** The criteria are grounded in established standards from both AI and healthcare domains. For example, clinical relevance is validated through clinician feedback and alignment with medical guidelines, ensuring that explanations are meaningful and actionable in real-world scenarios [64, 322].
- **Quantitative Metrics:** The use of objective metrics, such as Intersection Over Union (IOU) for localization accuracy and feature importance correlation for fidelity, provides measurable and reproducible assessments of explanation quality. These metrics ensure that the criteria are not subjective but based on empirical evidence [322].
- **Adaptability to Variability:** The criteria are designed to handle variability in input data, such as noise, differences in imaging equipment, or patient movement. For instance, robustness is enhanced by methods like SmoothGrad, which averages saliency maps over multiple noisy versions of the input image to provide stable explanations [329].
- **Validation in Real-World Studies:** Validation in real-world scenarios is essential for establishing the efficacy and trustworthiness of XAI methods in medical imaging. The robustness of these criteria is supported by their validation in peer-reviewed studies and real-world applications. For example, studies have demonstrated the effectiveness of Grad-CAM in highlighting clinically relevant regions in chest X-rays for pneumonia diagnosis, aligning with radiologists' expectations [40]. Similarly, SHAP has been employed to explain AI predictions in brain tumor detection, with clinician feedback utilized to iteratively refine its outputs [43]. Another study highlighted the effectiveness of saliency mapping techniques in breast cancer diagnostics by generating heatmaps that emphasize malignant tissue regions, validated through clinical trials [4]. Incorporating feedback loops where clinicians evaluate explanation clarity and relevance can further improve the trustworthiness of XAI models. These iterative loops allow the refinement of explanation formats, ensuring they better align with clinical reasoning and usability requirements. For example, Grad-CAM heatmaps have been evaluated by radiologists in real-time clinical environments, demonstrating their potential to align AI outputs with human expertise. Furthermore, dynamic XAI systems capable of adapting explanations based on user expertise have shown promise in fostering trust among healthcare providers. Tailoring explanations for different users, such as radiologists or general physicians, ensures that the outputs are both understandable and actionable, thereby enhancing trust and adoption [34, 43].

Table 13 Summary of evaluation criteria for XAI in medical imaging

Criterion	Definition	Applicability
Clinical relevance	Explanations align with medical knowledge and highlight biologically significant areas	Applicable to all imaging tasks (e.g., classification, detection) and modalities (e.g., X-rays, MRIs)
Fidelity to the model	Explanations accurately reflects the AI model's decision-making process	Universally applicable, but may require task-specific validation (e.g., segmentation vs. classification)
Localization accuracy	Explanations correctly highlight regions of interest in medical images	Critical for tasks like tumor detection and pneumonia diagnosis
Interpretability	Explanations are simple and clear for clinicians to understand	Applicable across all tasks and modalities, but may require adaptation for dynamic imaging
Robustness	Explanations remain consistent despite input variations or noise	Universally applicable, but particularly important for noisy or low-quality images
Ethical compliance	Explanations adhere to regulations like HIPAA and GDPR	Applicable in all healthcare settings, but may vary by region or institution
Quantitative evaluation	Use of objective metrics (e.g., IOU, correlation) to assess explanation quality	Universally applicable, with task-specific metrics for segmentation, classification, etc
Trust and confidence	Explanations instill confidence among clinicians through clarity and relevance	Applicable across all tasks and modalities, particularly in high-stakes scenarios
Temporal consistency	Explanations remain consistent across dynamic imaging sequences	Essential for dynamic modalities like echocardiograms and functional MRIs

11.2 Universal applicability of the criteria

The proposed evaluation criteria are universally applicable across a wide range of medical imaging tasks and modalities. This universality ensures that the criteria can be used to assess XAI methods in diverse clinical settings:

- **Applicability Across Tasks:** The criteria are relevant for various medical imaging tasks, including classification (e.g., diagnosing diseases), detection (e.g., identifying tumors), and segmentation (e.g., outlining organ boundaries). For example, localization accuracy is critical for both tumor detection in MRIs and pneumonia diagnosis in chest X-rays [326].
- **Applicability Across Modalities:** The criteria are applicable to different imaging modalities, such as X-rays, MRIs, CT scans, and ultrasounds. For instance, temporal consistency is essential for dynamic imaging modalities like echocardiograms, where tracking changes over time is crucial for accurate diagnosis [326].
- **Task-Specific Adaptations:** While the criteria are universally applicable, certain adaptations may be required for specific tasks or modalities. For example, segmentation tasks may rely more heavily on metrics like IOU, while classification tasks may prioritize interpretability and clinical relevance [322].

11.3 Standardized benchmarking frameworks for evaluating XAI models

While the evaluation criteria discussed above provide a foundation for assessing XAI models, the lack of standardized benchmarking frameworks makes it difficult to compare explainability methods across different medical imaging applications. Benchmarking frameworks aim to establish reproducible, quantifiable, and clinically validated comparisons between XAI techniques by defining: Common Benchmark Datasets,Quantitative Performance Metrics and Comparative Evaluation Pipelines.

1. Benchmark Datasets for XAI Evaluation in Medical Imaging: Several publicly available medical imaging datasets have been adopted as benchmarks for evaluating the interpretability and effectiveness of XAI models:

- NIH ChestX-ray14: Frequently used to benchmark explainability techniques for pneumonia and lung disease classification [332].
- CheXpert Dataset: Provides annotated chest radiographs, enabling evaluation of XAI saliency maps for model transparency [333].
- BraTS (Brain Tumor Segmentation): Used to assess the localization accuracy of XAI methods in detecting glioblastoma [334].
- HAM10000 (Dermatology Dataset): Evaluates the reliability of XAI techniques in skin lesion

classification [304]. These datasets serve as common baselines for benchmarking explainability methods, ensuring that XAI models generalize across diverse medical imaging tasks.

While these datasets serve as common baselines for benchmarking explainability methods, one major limitation is their lack of demographic diversity. Studies have shown that bias in training datasets can affect AI explanations, leading to variability in model interpretability across different patient populations [303]. This emphasizes the need for bias-aware benchmarking datasets that ensure fair and generalizable XAI models in clinical practice.

2. Performance Metrics for Benchmarking XAI:

Standardized benchmarking frameworks also rely on quantitative metrics to assess the quality of AI-generated explanations. The following are commonly used in medical XAI evaluation:

- Intersection Over Union (IoU):Measures how well AI-explained regions overlap with ground-truth annotations (used in tumor segmentation).
- Pointing Game Accuracy:Evaluates whether XAI saliency maps correctly localize disease regions (e.g., Grad-CAM in chest X-ray classification).
- AUC for Explanation Consistency:Tests how stable AI-generated explanations remain across multiple patient scans.
- Fidelity Score:Determines whether an explanation faithfully represents the model's decision-making process without introducing misleading artifacts.

Although these metrics provide objective evaluation standards, they do not fully capture clinical usability. Many AI explanations may score highly on technical benchmarks but fail to be useful in real-world clinical decision-making [43]. This has led to efforts in developing clinician-centered evaluation frameworks, where explainability methods are assessed based on their practical impact on clinical workflows. By integrating these considerations, researchers can systematically compare XAI methods across different clinical applications, ensuring that evaluation metrics align not only with technical performance but also with real-world adoption and usability requirements [325, 335].

3. Existing XAI Benchmarking Frameworks:

To address the lack of standardized explainability benchmarks, several research initiatives have proposed systematic evaluation pipelines for XAI models:

- XAI-Bench [336]: A large-scale benchmarking suite for comparing saliency-based XAI methods in medical imaging.
- Quantus Framework [337]: Provides an open-source library for assessing trustworthiness, robustness, and fidelity of XAI explanations.
- ExplainBench [301]: Focuses on benchmarking model-agnostic explainability techniques like LIME and SHAP.

These benchmarking efforts establish standardized evaluation pipelines, ensuring that XAI models are assessed using consistent datasets and reproducible metrics.

11.4 Empirical comparisons of XAI evaluation metrics in real-world medical applications

While evaluation metrics such as fidelity, transparency, robustness, and clinical relevance provide a structured framework for assessing explainability, their impact on real-world AI adoption is best understood through empirical comparisons across medical imaging applications. Several studies have benchmarked XAI techniques using these metrics to determine their effectiveness in practical settings.

1. **Fidelity and Transparency: Comparing Post-Hoc XAI Methods in Chest X-ray Diagnosis** A study by Tjoa and Guan [34]compared SHAP, LIME, and Grad-CAM in pneumonia detection using the NIH ChestX-ray14 dataset. The results showed that:

- Grad-CAM achieved higher fidelity (80 % overlap with radiologist-annotated regions) but lacked fine-grained feature attribution.
- SHAP provided pixel-level feature importance but required significantly higher computation time, affecting real-time usability.
- LIME produced inconsistent explanations across similar cases, raising concerns about robustness.

This empirical comparison highlights that while Grad-CAM aligns well with radiologists' expectations, its interpretability depends on qualitative clinical validation rather than absolute fidelity scores.

2. **Robustness and Clinical Usability: Evaluating Explainability in Brain Tumor Classification**

A study on brain tumor detection in MRI scans [301] compared different XAI techniques based on robustness:

- SmoothGrad and Integrated Gradients produced stable explanations across minor image perturbations.
- Saliency maps and vanilla Grad-CAM were highly sensitive to small changes in input, reducing trust in AI explanations.
- Robustness testing showed that models relying on non-pathological features (e.g., scanner artifacts) generated misleading explanations.

These findings emphasize that robust XAI methods must remain stable across different scans, particularly in high-stakes imaging tasks.

3. **Clinical Relevance: Adoption of XAI in Pathology and Dermatology**

Real-world clinical adoption of XAI has been influenced by how well explanations align with pathologist or dermatologist interpretations:

- In dermatology AI, studies using HAM10000 dataset found that SHAP and LIME were more effective for lesion classification, as they provided clear feature attribution at the pixel level [304].
- In pathology AI, Grad-CAM and Class Activation Maps (CAMs) were preferred, as they highlighted diagnostically relevant tissue structures rather than pixel-level attribution.

These studies illustrate that different evaluation metrics are more effective in specific medical imaging tasks, reinforcing the need for task-specific XAI benchmarking.

4. **Transparency and Ethical Implications: AI for Diabetic Retinopathy Detection**

A study evaluating AI models for diabetic retinopathy screening found that:

- Models trained on imbalanced datasets produced explanations that disproportionately favored certain demographic groups, raising fairness concerns.
- Counterfactual explanations helped detect potential biases by showing how altering patient attributes (e.g., age, ethnicity) affected AI predictions [319].
- Clinicians expressed higher trust in AI models when fairness-aware transparency metrics were included, reinforcing the need for standardized evaluation [303].

This highlights the importance of combining fidelity with fairness-aware evaluation metrics to ensure ethical AI deployment in healthcare.

5. **Real-Time XAI and Decision Support in Emergency Care**

Emergency care applications require explanations that are both interpretable and fast:

- A study on stroke detection AI in emergency settings showed that Grad-CAM provided intuitive heatmaps, allowing clinicians to quickly validate AI predictions [305].
- SHAP-based models were rejected in triage settings due to computation time constraints, making them impractical for real-time decision support.
- Hybrid XAI models combining rule-based reasoning and Grad-CAM improved adoption, showing that speed and transparency must be balanced in high-risk scenarios [306].

The empirical comparisons highlight that evaluating XAI models solely on accuracy is insufficient for real-world medical adoption. Factors such as real-time usability, robustness across diverse patient populations, and clinician trust play equally critical roles in determining the effectiveness of explainability methods. As AI continues to be integrated into high-stakes medical environments, standardized evaluation frameworks must incorporate these broader dimensions to ensure that XAI systems are not only precise but also interpretable, reliable, and seamlessly deployable in clinical workflows.

11.5 Justification for the choice of criteria

The selection of these criteria is justified by their ability to address the unique challenges of medical imaging and their alignment with both technical and clinical needs:

- **Addressing High-Stakes Requirements:** Medical imaging is a high-stakes domain where accuracy, interpretability, and reliability are paramount. The chosen criteria, such as clinical relevance and fidelity to the model, ensure that XAI methods meet these requirements and provide actionable insights for clinicians [322].
- **Alignment with Ethical and Regulatory Standards:** The criteria emphasize ethical and regulatory compliance, ensuring that explanations are interpretable and accessible to diverse audiences, including non-technical stakeholders. This is critical for adherence to regulations like HIPAA and GDPR, which govern the use of AI in healthcare [331].
- **Supporting Clinician Trust and Adoption:** By focusing on criteria like interpretability, transparency, and trust-building, the chosen framework ensures that XAI methods are not only technically sound but also practical and acceptable to clinicians. For example, combining visual heatmaps with textual descriptions enhances clinician confidence in AI-driven diagnoses [325].
- **Derived from Best Practices and Guidelines:** The criteria are derived from best practices in XAI and healthcare, as well as guidelines from organizations like the FDA and WHO. This ensures that they are relevant, up-to-date, and aligned with global standards for AI in medical imaging [331].

A summary table 13 is presented below to provide a concise overview of the evaluation criteria, their definitions, and their applicability across medical imaging tasks and modalities. This table serves as a quick reference for understanding how each criterion contributes to the effectiveness and reliability of XAI methods in healthcare.

12 Research gaps in XAI

The evaluation of explainable AI (XAI) methods in the preceding sections underscores significant advancements in their development and application, particularly in medical imaging. For instance, XAI has enhanced interpretability for clinicians, providing insights into model decision-making processes that were once considered opaque [338, 339]. However, these advancements are accompanied by persistent limitations and unexplored areas that hinder the widespread adoption and effective implementation of XAI in healthcare [309].

Identifying and addressing these research gaps is essential to bridge the divide between theoretical progress and practical impact. Recent studies emphasize that overcoming these limitations will require innovations in algorithm design, interdisciplinary collaboration, and robust frameworks that account for the complex demands of clinical practice [322]. By tackling these areas, future research can improve the trustworthiness, scalability, and ethical robustness of XAI systems in healthcare. To fully realize the potential of XAI in healthcare, several critical gaps must be addressed:

- **Context-Specific Explainability**: Explainability needs in healthcare vary across domains, such as radiology, cardiology, and oncology. Most existing XAI methods are generalized and fail to address the specific clinical requirements of these fields. For instance, a radiologist analyzing imaging data may need pixel-level explanations, whereas an oncologist might prioritize patient-level risks and treatment options [338, 339]. Tailored approaches that align with specific domain requirements remain an unmet need.
- **User-Centric and Dynamic Explanations:** XAI tools are often static, offering explanations that do not adapt to users' varying expertise levels or situational needs. For example, novice clinicians might require simplified explanations, while experienced practitioners demand in-depth insights. Developing dynamic systems capable of tailoring explanations in real-time is essential for improving usability and trustworthiness.
- **Evaluation Metrics and Standards:** The lack of standardized metrics to evaluate XAI methods impedes consistent benchmarking and adoption. Existing metrics, such as fidelity, comprehensibility, and plausibility, are applied inconsistently across studies [322]. There is a critical need to establish robust, universally accepted evaluation frameworks for assessing the quality and clinical relevance of XAI systems.
- **Integration with Clinical Workflows:** Integration of XAI methods into real-world clinical workflows remains a significant research gap, as it involves overcoming technical, organizational, and regulatory barriers. For example, the lack of interoperability between XAI tools and existing electronic health record (EHR) systems creates challenges in ensuring seamless adoption without disrupting clinician workflows. Additionally, XAI outputs must be presented in intuitive formats, such as visual heatmaps or textual justifications, tailored to the needs of diverse healthcare professionals. Training programs for clinicians are crucial to building trust and understanding of XAI-generated explanations, which are often complex and model-specific. Another key challenge is scalability. Many XAI techniques, such as SHAP and LIME, are computationally intensive, making real-time applications difficult in high-stakes medical environments like emergency care. Efforts to optimize these algorithms for efficiency while maintaining interpretability are essential for successful deployment. Finally, regulatory and ethical considerations must be addressed, as the adoption of XAI tools requires compliance with standards such as the FDA's Good Machine Learning Practices (GMLP) and GDPR's "right to explanation." Addressing these challenges is critical to bridge the gap between research advancements and their real-world impact in clinical settings [34, 43]. Real-world examples also highlight these gaps. For instance, while XAI has demonstrated success in tasks such as generating Grad-CAM heatmaps for radiology, its adoption is limited due to insufficient validation in clinical trials and a lack of clinician feedback mechanisms. Cardiovascular risk prediction models are another promising area for XAI integration. For example, LIME has been applied to explain predictions from machine learning models assessing the likelihood of heart attacks. XAI systems enable clinicians to understand and communicate the rationale behind risk assessments to patients by identifying key risk factors, such as cholesterol levels or blood pressure. These models have been incorporated into hospital decision support systems, helping physicians personalize treatment plans for high-risk individuals [40]. Incorporating iterative feedback loops and conducting large-scale validations in diverse medical environments are necessary steps to ensure the reliability and usability of XAI systems in practice. Addressing these gaps through comprehensive research and practical implementation will ensure that XAI fulfills its potential to enhance clinical decision-making and improve patient outcomes [39, 40].
- **Addressing Bias and Fairness in Explanations:** While there is growing attention to reducing bias in AI models, the fairness of explanations provided by XAI methods is less understood. Explanations may inadvertently reinforce biases, especially when trained

on imbalanced datasets Research must explore methods for detecting and mitigating bias within XAI explanations [319].

- **Scalability and Real-Time Applications:** Many XAI techniques are computationally intensive, making them unsuitable for large datasets or real-time applications required in medical imaging. Methods that balance accuracy, scalability, and speed need further investigation to ensure their adoption in high-stakes environments [339]. One of the major gaps in XAI research is scalability, particularly in processing large-scale medical imaging datasets. Techniques such as SHAP and LIME, while effective for generating interpretable insights, are computationally intensive, making them impractical for high-dimensional imaging data like 3D CT or MRI scans in real-time applications. This computational overhead limits their usability in clinical environments where timely decision-making is critical [16, 32]. Future research must explore scalable solutions to bridge this gap. Promising approaches include: Model Compression: Simplifying models to reduce computational requirements without sacrificing interpretability or accuracy. Approximate Explanation Methods: Developing methods that generate faster, resource-efficient explanations while preserving diagnostic relevance. Distributed Computing: Leveraging GPU and cloud-based systems to handle large-scale data processing efficiently [320]. Additionally, federated learning frameworks have shown potential in improving scalability. By enabling collaborative training across institutions without sharing sensitive data, federated learning combines privacy preservation with computational efficiency, making it well-suited for large-scale, multi-institutional medical imaging studies [290]. The scalability of XAI methods is not just a technical challenge but also a fundamental requirement for their broader adoption in clinical practice. Future research should prioritize the development of hybrid approaches that balance computational efficiency with robust interpretability to meet the demands of modern healthcare.
- **Ethical, Legal, and Interdisciplinary Challenges:** The ethical implications of XAI, particularly in life-critical domains like healthcare, remain underexplored. Issues such as accountability, liability, and patient consent pose significant challenges [340] Furthermore, interdisciplinary collaboration between AI researchers, ethicists, and legal experts is essential to address these concerns effectively.

13 Challenges and future perspectives

Explainable AI (XAI) holds great promise for advancing medical imaging by improving the interpretability and transparency of AI-driven decisions. However, numerous challenges remain that hinder its effective implementation in clinical workflows. These challenges range from data-related issues to ethical, regulatory, and practical barriers, as discussed below.

- **Data Challenges:** The availability of high-quality, annotated datasets is one of the most significant barriers to XAI development in medical imaging. Annotating such datasets requires extensive time, effort, and domain expertise to ensure accuracy. This process is further complicated by stringent privacy regulations such as HIPAA and GDPR[341], as well as institutional data silos and interoperability issues between healthcare systems. Together, these factors limit access to diverse datasets necessary for developing robust and generalizable XAI models. Promising solutions to these challenges include federated learning, which enables collaborative model training without sharing raw data [342], thus preserving patient privacy while enhancing data accessibility [290]. Additionally, synthetic data generation techniques can create realistic, de-identified datasets to mitigate confidentiality concerns while improving model training [40]. Standardization frameworks like Fast Healthcare Interoperability Resources (FHIR) also play a crucial role in improving data interoperability and facilitating smoother integration of XAI models into clinical workflows.
- **The Interpretability-Accuracy Trade-off in Clinical AI:** The interpretability-performance trade-off is a persistent challenge in XAI applications, particularly in healthcare. Traditional machine learning models such as logistic regression, decision trees, and rule-based systems offer high transparency, making their decision-making processes easy to understand. However, these models often lack the predictive power required for high-dimensional medical imaging data, such as MRI, CT scans, and histopathological images. In contrast, DL models, including (CNNs) and transformers, achieve state-of-the-art diagnostic accuracy but suffer from inherent opacity, making it difficult for clinicians to trust their outputs. The growing reliance on such black-box AI models raises concerns about their fairness, as these models may exhibit bias due to imbalanced datasets, leading to systematic errors in underrepresented patient groups [319]. While post-hoc XAI techniques such as SHAP, Grad-CAM, and LIME can enhance interpretability by generating feature importance maps and saliency visualizations, they

introduce additional computational complexity that can delay decision-making in real-time clinical applications. This trade-off is particularly critical in emergency settings, such as:

- Trauma care: AI-powered imaging systems must generate immediate and reliable insights to assist clinicians in triage and patient prioritization.
- Stroke detection: Real-time interpretability is essential in determining ischemic vs. hemorrhagic strokes, as treatment pathways depend on fast yet accurate AI assessments.
- Cardiac arrest prediction: AI-driven ECG analysis must provide explanations quickly, ensuring that automated warnings for critical arrhythmias are both accurate and understandable for cardiologists.

Furthermore, bias in AI models can exacerbate these challenges, particularly in high-stakes medical decision-making. Studies have shown that AI models trained on unbalanced datasets can exhibit demographic bias, where diagnostic accuracy is lower for certain ethnicities, genders, or age groups [303]. The trade-off between accuracy, interpretability, and fairness thus becomes a three-way challenge in clinical AI. Over-prioritizing interpretability may result in oversimplified models that do not generalize well, while over-prioritizing performance can lead to highly accurate but opaque and potentially biased AI systems.

Addressing this trade-off requires multi-faceted solutions, including:

- Developing inherently interpretable deep learning models that offer a balance between accuracy and explainability.
- Integrating fairness-aware training approaches to reduce algorithmic bias while maintaining high performance.
- Deploying confidence-aware AI systems that selectively provide autonomous predictions for high-certainty cases while deferring uncertain cases to human experts.

Incorporating these advancements will be essential in the future of medical XAI, ensuring that AI systems do not sacrifice fairness and interpretability for performance alone but instead achieve a clinically viable balance.

- **Scalability and Time-Sensitive Decision-Making:** Scalability remains a critical challenge for XAI, particularly in large-scale healthcare systems requiring real-time decision-making. Many current XAI methods are computationally expensive, making them impractical for time-sensitive applications like emergency diagnostics. Future research should focus on methods such as model compression, distributed computing, and inherently interpretable models to address these concerns [43].

In real-world clinical applications, the trade-off between model complexity and interpretability varies depending on the urgency and risk of the medical task:

- Emergency Medicine and Intensive Care Units (ICUs): AI-driven tools such as chest X-ray classifiers for pneumonia detection or deep learning models for stroke triage must provide fast, reliable outputs. However, if an explainability model (e.g., Grad-CAM) requires additional processing time, this may introduce delays that impact patient outcomes.
- Oncology and Radiology: In cancer diagnostics, where interpretability is critical, AI-based decision support tools need robust XAI techniques to ensure models highlight diagnostically relevant tumor regions while maintaining accuracy.
- Rare Disease Diagnosis and Personalized Medicine: In specialized cases, AI systems must justify their decisions comprehensively to gain clinician trust, especially in high-risk scenarios such as predicting rare genetic disorders from multimodal data.

A study by Tonekaboni et al [317] found that while clinicians prefer interpretable AI models, there is a threshold where excessive complexity in XAI methods leads to cognitive overload, reducing trust in AI-assisted decisions. This suggests that optimizing the trade-off depends on the clinical setting and the expertise of the end-user.

- **Lack of Standardized Evaluation Metrics:** The lack of standardized evaluation metrics remains a major challenge in the adoption of XAI for medical imaging. Unlike conventional AI models, which are typically assessed using performance metrics such as accuracy, sensitivity, specificity, and AUC-ROC, XAI models require multidimensional evaluation frameworks that account for interpretability, trustworthiness, fairness, and clinical usability. However, there is no universal consensus on how to measure these factors, making it difficult to compare different XAI models and determine their clinical reliability [319, 343].

- Challenges in Evaluating XAI Models: Existing evaluation frameworks predominantly focus on quantitative performance metrics but fail to address critical human-centered and fairness aspects, such as:

  * Interpretability: How well can clinicians understand, trust, and act upon AI-generated explanations?

* Trustworthiness: Are the model's explanations reliable, consistent, and stable across diverse patient groups?
* Fairness and Bias Detection: Do the explanations maintain equitable validity across different demographics, or do they introduce systematic biases that could exacerbate healthcare disparities?

Recent studies have shown that XAI models can generate convincing explanations that still propagate bias, raising concerns about the ethical deployment of AI in medical diagnostics [303]. Without a rigorous fairness evaluation, AI-driven medical systems risk reinforcing pre-existing biases in healthcare, disproportionately affecting underrepresented populations and leading to potential misdiagnosis or unequal treatment.

– **Proposed Strategies for Standardizing XAI Evaluation:** To address this issue, several evaluation strategies have been proposed, categorized into three key areas:

1. Human-Centered Evaluation Metrics:

- Clinician Validation Metrics: Developing structured scoring mechanisms where radiologists and healthcare professionals directly assess the actionability and interpretability of XAI-generated explanations.
- Actionability Testing: Measuring whether XAI explanations lead to improved clinical decision-making and patient outcomes when integrated into real-world medical workflows [317].

2. Fairness-Aware Metrics for XAI:

- Equalized Odds and Disparate Impact Analysis: Ensuring prediction confidence remains consistent across different demographic groups.
- Counterfactual Fairness Testing: Simulating changes in non-medical attributes (e.g., race, gender) to evaluate if AI predictions remain stable and unbiased.
- Bias Attribution Scores: Measuring whether an XAI model relies disproportionately on features correlated with protected attributes [319].

3. Stability and Robustness Metrics:

- Explanation Consistency Tests: Ensuring that small variations in input data do not drastically alter the AI-generated explanations.
- Adversarial Robustness in XAI: Evaluating how XAI models respond to adversarial attacks, which may manipulate explanations and mislead clinicians.

– **The Role of Medical Experts in XAI Evaluation:** A significant limitation in current XAI evaluation frameworks is the limited involvement of medical professionals in the design, validation, and refinement of explainability models. Clinicians play a vital role in defining what constitutes a "meaningful explanation" for AI decisions, yet many existing XAI models rely on proxy interpretability measures that lack real-world usability assessments.

To ensure XAI models meet clinical standards, it is essential to:

- Integrate clinician feedback early in XAI development to align models with real-world diagnostic needs.
- Establish collaborative frameworks between AI researchers and medical professionals to define standardized evaluation protocols tailored for healthcare applications [317].
- Develop global regulatory standards for assessing fairness, interpretability, and actionability in medical XAI models, ensuring AI-driven decisions are explainable, trustworthy, and free from bias.

we conclude that,the absence of standardized evaluation metrics poses a significant challenge to the adoption of XAI in medical imaging. While conventional AI models rely on well-established performance measures, XAI requires a broader framework encompassing interpretability, fairness, stability, and clinical actionability. Addressing these challenges will require collaborative efforts between AI researchers, clinicians, and regulatory bodies to develop standardized, bias-aware evaluation methodologies that ensure trustworthy and equitable AI-driven diagnostics.

– **Real-World case of Interpretability-performance challanges:** Several real-world cases illustrate the trade-offs between interpretability and predictive accuracy in AI-driven medical diagnostics. These examples emphasize the risks associated with over-prioritizing one factor at the expense of the other, leading to clinician mistrust, regulatory concerns, and ultimately limited clinical adoption.

1. IBM Watson for Oncology: The Pitfalls of Opaque Decision-Making IBM Watson for Oncology, once envisioned as a revolutionary AI-driven decision support system, faced significant backlash due to its lack of interpretable recommendations. While the system demonstrated high predictive accuracy, clinicians found its treatment suggestions difficult to justify,

as it failed to provide transparent reasoning behind its outputs. Investigations later revealed that some recommendations were not based on actual patient data but instead derived from expert-curated guidelines, reducing trust in AI-assisted oncology [306].

2. DeepMind's AI for Kidney Injury Prediction: A Case of Unusable Explanations DeepMind's AI system, designed to predict acute kidney injury (AKI) in ICU patients, achieved high predictive performance, identifying deterioration up to 48 h before it became clinically apparent. However, despite its impressive accuracy, the system struggled with real-world clinical adoption because its explanations were not actionable for clinicians. Healthcare providers reported that the AI model often provided predictions without clear indications of the underlying cause, making it difficult to align AI recommendations with existing clinical workflows [305].
3. COVID-19 AI Diagnostic Models: The Risks of Misleading Explanations During the COVID-19 pandemic, numerous AI models were developed for automated chest X-ray and CT scan analysis to detect SARS-CoV-2 infections. While many models initially reported high accuracy rates, subsequent analyses found that several AI systems were relying on dataset-specific artifacts rather than pathology-related features. For example:

– Some models wrongly associated hospital markings and scanner types with COVID-19 presence, leading to artificially inflated accuracy scores.
– Others focused on non-diagnostic regions of the image, such as labels, instead of lung abnormalities, rendering the AI models unreliable for clinical use [344].

These failures reinforced the importance of robust interpretability mechanisms and dataset transparency, demonstrating that explainability is not just a technical concern but a critical factor in preventing unreliable AI deployment in healthcare.

Lessons Learned and Implications for Future XAI Development: Ensuring Clinician-Centric Explanations: AI models must align with medical decision-making workflows to facilitate trust and usability in real-world settings. Avoiding Proxy Biases in AI Models: Explainability tools like SHAP, Grad-CAM, and counterfactual analysis should be applied before deployment to validate AI model decision pathways and avoid non-medical feature dependencies. Stronger Regulatory Oversight for XAI in Medicine: Medical AI systems must undergo robust fairness and transparency evaluations before being approved for clinical use, aligning with FDA's AI/ML-Based Software as a Medical Device (SaMD) Action Plan and EU AI Act guidelines [345, 346].

– **Barriers to Clinical Integration:** The practical implementation of XAI in clinical workflows encounters multiple barriers, limiting its widespread adoption and usability in real-world healthcare settings. These challenges stem from a combination of technical, cognitive, and infrastructural limitations, making it difficult to seamlessly integrate XAI into daily clinical practice.-many of these barriers as follows:

1. Trust and Usability Challenges:

* Many clinicians perceive XAI-generated explanations as overly complex, inconsistent, or misaligned with their decision-making processes.
* Studies indicate that opaque AI-generated saliency maps or feature importance scores may not always align with clinician intuition, reducing their trust in AI-driven decisions.
* Additionally, some explainability techniques fail to provide actionable insights, making it difficult for medical professionals to validate AI recommendations effectively [317].

2. Scalability Issues in High-Dimensional Clinical Applications:

* Computationally intensive methods, such as SHAP and LIME, struggle to scale efficiently in real-time medical settings, particularly when applied to high-resolution MRI, CT, and histopathology images [347].
* The time required to generate interpretable outputs in black-box models increases exponentially with dataset size, limiting XAI's usability for time-sensitive diagnostics (e.g., stroke triage or ICU patient monitoring).

3. Workflow Disruption and Cognitive Load:

* AI models must integrate seamlessly into Electronic Health Record (EHR) systems and clinical decision support tools, ensuring minimal disruption to existing workflows.

* Poorly designed XAI implementations may increase clinician cognitive load, requiring additional interpretation effort rather than simplifying decision-making [303].
* A survey of physicians found that explainability models that introduce excessive information complexity or lack direct clinical relevance are often ignored in real-world practice [317].

4. Fairness Concerns and Mistrust in AI Adoption:

* Many AI-driven medical imaging models exhibit biases due to imbalanced training data, demographic underrepresentation, and feature misalignment [319].
* Studies have shown that AI models trained on datasets dominated by specific population groups (e.g., certain ethnicities or age ranges) may underperform in underrepresented patient groups, leading to potential misdiagnosis and disparities in care [303].
* Clinicians express skepticism toward AI models that lack fairness guarantees, fearing that biased predictions could reinforce existing healthcare disparities rather than improving patient outcomes.
* Regulatory concerns surrounding AI fairness (e.g., FDA guidelines on bias detection and GDPR's "Right to Explanation") have slowed AI adoption in healthcare, as developers must demonstrate fairness and generalizability before clinical approval [345, 346].

– Ensuring fairness in AI predictions is critical for gaining clinician trust and preventing AI-related medical discrimination. Bias-aware explainability techniques, such as counterfactual fairness analysis and demographic performance evaluations, must be integrated into XAI models to enhance transparency and reduce clinician skepticism [319].
– **Hybrid AI Models as a Solution:** To balance interpretability and predictive performance, recent research has explored hybrid AI approaches that combine the strengths of high-accuracy deep learning models with interpretable surrogate systems. These methods ensure both explainability and clinical usability, reducing the trade-offs associated with fully black-box or inherently interpretable models.
  – Confidence-Aware AI Systems: AI models that autonomously handle high-certainty cases while deferring uncertain predictions to human experts. This approach optimizes workflow efficiency while maintaining clinician oversight for complex or ambiguous cases.
  – Surrogate Explainer Models: Post-hoc explainability techniques that approximate the predictions of complex deep learning models using simpler, rule-based or decision-tree models. These surrogate models help bridge the gap between black-box AI and clinician intuition by offering more digestible and interpretable outputs.
  – Multimodal AI for Explainability: Advanced AI models that integrate multiple data modalities (e.g., text, imaging, clinical reports) to generate context-aware explanations. This improves interpretability by linking AI-generated insights with existing clinician knowledge, fostering higher trust in AI-assisted decision-making.

By adopting these hybrid XAI solutions, the healthcare industry can reduce the interpretability-performance trade-off and move towards clinically actionable AI models that enhance medical decision-making rather than complicating it.

– **Strategies for Fairness and Bias Mitigation in Medical AI Training Data:** The fairness of AI-driven medical imaging models is heavily influenced by dataset quality, diversity, and representativeness. Biases in training datasets can lead to systematic errors that disproportionately affect certain patient populations, resulting in unequal healthcare outcomes. Addressing these biases is essential to ensure that XAI models provide equitable, generalizable, and clinically meaningful explanations.

1. Common Sources of Bias in Medical Imaging Datasets
Bias in AI models often arises from imbalanced data distribution, underrepresentation of demographic groups, and systemic errors in data collection. Studies have shown that:

– AI models trained on chest X-ray datasets primarily collected from one demographic group exhibited reduced accuracy when tested on other populations [319].
– Skin cancer detection AI systems trained predominantly on lighter-skinned individuals showed lower sensitivity in detecting melanoma in darker-skinned patients [304].
– AI models for retinal disease screening were found to perform significantly worse on images from certain ethnic groups, leading to delayed diagnoses and treatment disparities [305]. These cases underscore the urgent need for fairness-aware training methods to mitigate bias in AI-driven healthcare applications.

2. Strategies for Bias Mitigation in Training Data To ensure fairness, multiple bias mitigation techniques can be applied at different stages of model development.

– Pre-Processing Bias Mitigation (Before Training Begins)

Data Collection and Representation:
Ensuring that training datasets are diverse and representative of all patient demographics is the first step in mitigating bias. Initiatives such as the NIH ChestX-ray14 dataset revision have aimed to include a balanced distribution of gender, age, and ethnicity groups [332].

The HAM10000 skin lesion dataset has been revised to include a broader spectrum of skin tones, reducing racial bias in melanoma detection models [304].

Data Augmentation for Minority Classes:

Techniques such as synthetic data generation (e.g., GAN-based augmentation) can be used to create realistic medical images for underrepresented groups, improving model generalization [40].

– In-Processing Bias Mitigation (During Model Training)

Fairness-Aware Training Algorithms:

AI models can be trained using re-weighted loss functions to reduce bias against minority groups. Studies have shown that introducing demographic parity constraints during training improves fairness without significantly impacting accuracy [303].

– Adversarial Debiasing:

A recent study applied adversarial learning to force AI models to ignore demographic attributes while retaining predictive power. This approach has been effective in reducing racial bias in radiology AI models [319].

– Post-Processing Bias Mitigation (After Model Training)

Fairness Metrics and Model Auditing:

AI models should be evaluated using bias detection metrics such as: Equalized Odds (Ensures model performance is equal across demographic groups).

Disparate Impact Ratio (Measures whether one group is disproportionately affected by AI decisions).

Counterfactual Fairness Testing (Checks whether changing a demographic attribute alters model predictions).

Bias Correction Through Explainability Tools:

SHAP-based fairness analysis has been used to detect and correct biased feature attributions, ensuring that AI explanations are not influenced by confounding demographic variables [302].

3. Regulatory Perspectives on Fairness in Medical AI Regulatory agencies have implemented guidelines to promote fairness in AI-driven healthcare applications:

– FDA's Good Machine Learning Practice (GMLP): Requires that AI models be trained and validated on diverse datasets [345].
– EU AI Act: Categorizes biased AI models in healthcare as "high-risk", mandating bias audits before clinical deployment [346].
– GDPR's Right to Explanation: Ensures that AI models provide transparent, non-discriminatory explanations for automated medical decisions [303]. These regulations emphasize the need for fairness-aware XAI models that align with both ethical and clinical standards.

– **Ethical, Legal, and Regulatory Considerations:** XAI adoption raises significant ethical and regulatory concerns that must be addressed to ensure trust and accountability:

1. Algorithmic Bias: Imbalanced datasets can result in biased XAI models, leading to inequitable outcomes for underrepresented populations [331, 348].
2. Transparency: The opacity of some AI models undermines trust, as clinicians and patients may find it difficult to validate AI recommendations [321].
3. Patient Autonomy: Opaque AI outputs can limit patients' ability to make informed decisions, posing risks to shared decision-making processes [38].

– Regulatory frameworks such as GDPR and HIPAA play a crucial role in ensuring that XAI systems prioritize fairness, transparency, and accountability. The EU GDPR, for example, enforces a "Right to Explanation," which requires AI systems to provide interpretable outputs [341]. Similarly, the FDA's Good Machine Learning Practices (GMLP) emphasize the necessity of interpretable AI models in healthcare applications [307]. Recognizing the trade-offs in XAI, regulatory agencies stress the importance of "clinically meaningful explainability," ensuring that models align with medical workflow requirements. Furthermore, the European Union's 2022 AI Act proposal classifies medical AI applications as high-risk, necessitating that

developers demonstrate both accuracy and explainability to gain regulatory approval. These regulatory measures create legal pressures for AI developers to design systems that provide transparent justifications for automated decisions in patient care.

Overall,While XAI faces numerous challenges in medical imaging, there are many current and emerging solutions to mitigate these barriers. Techniques like federated learning, synthetic data generation, fairness-aware algorithms, and standardized evaluation frameworks provide promising avenues for overcoming data, ethical, and scalability challenges. Future research should focus on interdisciplinary collaboration, regulatory alignment, and designing clinician-friendly systems to ensure that XAI systems are not only interpretable but also practical and widely adopted in healthcare. These efforts will pave the way for XAI to fulfill its transformative potential in improving diagnostic accuracy and enhancing patient care.

To overcome the challenges facing XAI, there are some aspects that must be worked on in the future as follows:

- **Enhancing Data Privacy and Accessibility:** Future XAI research must focus on ensuring data privacy and accessibility to enable the development of robust and secure AI models. Federated learning, a privacy-preserving approach that allows collaborative model training without sharing raw data, has emerged as a promising solution [290]. Similarly, synthetic data generation techniques produce realistic yet de-identified datasets that can be shared while maintaining confidentiality [40]. These methods not only address privacy concerns but also promote the availability of diverse datasets crucial for model training. Furthermore, frameworks like Fast Healthcare Interoperability Resources (FHIR) can standardize medical records and improve interoperability, facilitating the seamless integration of XAI systems into clinical workflows.
- **Cross-Disciplinary Collaboration:** The successful adoption of XAI requires close collaboration among AI developers, clinicians, and policymakers. Each stakeholder group brings unique expertise that is vital for addressing the multifaceted challenges of XAI deployment. Developers focus on designing robust and interpretable models, while clinicians ensure these models are practical and aligned with real-world medical needs. Policymakers establish regulations and ethical guidelines to promote fairness, transparency, and accountability in AI systems. Structured initiatives, such as multi-stakeholder workshops and government-funded pilot projects, can foster collaboration and bridge the gap between technical innovation and clinical utility. For example, public-private partnerships like the UK's National AI Lab have demonstrated the potential of interdisciplinary efforts to drive innovation while ensuring alignment with healthcare priorities [349]. Incentivizing cross-disciplinary projects through funding programs and collaborative grants will further accelerate the integration of XAI into medical workflows [350]. Furthermore, successful initiatives like the European Health Data Space demonstrate how harmonized policies can promote data sharing and innovation across borders while maintaining strict privacy protections [351]. Policymakers can also incentivize interdisciplinary research through funding programs, fostering collaboration among AI developers, clinicians, and ethicists. Frameworks for ethical AI, such as AI4People's ethical principles [350], emphasize fairness, transparency, and accountability. Mitigation strategies include the incorporation of fairness constraints during model training and rigorous auditing of AI systems for potential biases [319]. Engaging diverse stakeholders, including patients, clinicians, and ethicists, can also help address ethical dilemmas proactively.
- **Standardized Evaluation Frameworks:** Developing standardized evaluation frameworks is crucial for ensuring the consistency, reliability, and clinical relevance of XAI systems. Such frameworks should go beyond accuracy metrics to include interpretability, trustworthiness, and fairness. Techniques like SHAP and LIME provide quantifiable benchmarks for assessing explanations and can serve as foundational tools for evaluating XAI models [16, 32]. Globally recognized standards, such as ISO frameworks for medical devices[297], should be adapted for XAI to accelerate regulatory approval and foster trust among stakeholders and is essential for benchmarking XAI systems and ensuring consistent evaluations [352].
- **Education and Training:** To ensure the successful adoption of XAI, clinician training programs must be developed to enhance understanding and trust in AI-assisted decision-making. Dynamic XAI systems tailored to user expertise can improve clinical decision-making by offering clear, actionable explanations [38]. Training initiatives should focus on explaining XAI outputs, fostering confidence in clinicians, and addressing concerns related to AI reliability and safety [298, 299].
- **Leveraging Emerging AI Paradigms:** Emerging technologies, such as generative models and multimodal approaches, are paving the way for advanced XAI solutions. Generative AI, such as GANs and diffusion models, enables the creation of high-quality synthetic datasets that address data scarcity while preserving privacy [300, 353]. Multimodal XAI techniques integrate diverse data sources, including imaging, textual, and genomic data, to provide richer and

more actionable insights for clinical decision-making. Additionally, federated learning can enhance the scalability and applicability of XAI by enabling secure, distributed training across institutions [290].

- **Practical Frameworks and Pilot Projects:** Practical frameworks and pilot projects are essential for transitioning XAI from research to clinical implementation. Examples include:
  - Pilot Projects: Deploying XAI systems in diagnostic imaging departments and evaluating their performance using metrics like accuracy, interpretability, and clinician satisfaction.
  - Institutional Review Boards: Validating XAI models for clinical readiness with input from clinicians and ethicists [321].
  - Standardized Metrics: Adopting trust scores, time-to-diagnosis, and error reduction rates to measure success and foster transparency in performance [16].

14 Conclusion

This study reviewed the latest advancements in XAI with a focus on its applications in medical imaging. We examined XAI from three perspectives, as outlined in Sect. 2 to Sect. 4. Three frameworks for XAI were discussed based on the model nature (Specific vs. Agnostic, Global vs. Local, and Pre- vs. Post-hoc models), as detailed in Sect. 3.3. Additionally, we reviewed various XAI techniques for medical imaging, classified into visual, textual, and example-based explanations, as presented in Sect. 4. Evaluation methods for XAI, along with their importance and criteria, were also analyzed in Sect. 11.

This review highlighted key trends in the field. Most publications focus on local explanations rather than global ones, with saliency mapping approaches being the most widely used methods, particularly for CNN-based models. Our categorization of studies by anatomical locations and medical imaging modalities revealed that research predominantly emphasizes chest, brain, and MRI imaging. These trends underscore the current focus of XAI research and the need to expand its scope.

While XAI is critical for fostering trust and adoption in the medical community, significant challenges and gaps remain. Based on the findings of this review, we propose the following actionable recommendations to advance the field:

- **Develop Robust Global Explanation Methods:** Future research should focus on developing global explanation methods that can complement local techniques and provide insights into model behavior at a broader level. Techniques such as attention mechanisms or model-agnostic frameworks could be further explored for this purpose.
- **Address Scalability and Real-Time Constraints:** To enable the integration of XAI in clinical workflows, efforts should be made to optimize XAI methods for real-time applications, particularly for computationally intensive techniques like SHAP and LIME.
- **Enhance Fairness and Bias Mitigation in XAI Models:** Researchers should prioritize developing methods that ensure fairness and reduce bias in AI explanations. This includes using diverse and representative datasets and designing models that can identify and mitigate biases effectively.
- **Strengthen Validation and Standardization Efforts:** To improve trustworthiness, XAI methods should be validated through clinical trials and standardized evaluation frameworks that align with global regulatory guidelines, such as those of the FDA and GDPR.
- **Explore Multimodal XAI Approaches:** Combining data from multiple modalities (e.g., imaging, clinical records, and genetic data) holds promise for generating more comprehensive and interpretable explanations, which can be highly valuable in complex diagnostic scenarios.
- **Increase Collaboration Across Disciplines:** Collaboration between AI researchers, clinicians, and ethicists is essential for designing XAI systems that meet both technical and clinical requirements. Insights from other fields, such as finance and autonomous systems, can also provide transferable best practices for XAI adoption.

In conclusion, XAI is an essential and urgent research topic within AI, especially in the domain of medical imaging. The widespread adoption and acceptability of AI in the medical community will ultimately depend on its ability to provide meaningful, interpretable, and reliable explanations. By addressing the outlined challenges and advancing the methodologies proposed in this review, researchers and practitioners can contribute to the development of more trustworthy, effective, and clinically useful XAI systems.

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Declarations

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Essam H. Houssein (Member, IEEE) received Ph.D. degree in computer science, in 2012. He is currently a Professor of Artificial Intelligence at the Faculty of Computers and Information, Minia University, Minia, Egypt. He is the founder and chair of the Artificial Intelligence Research (AIR) Group, Egypt. He is selected as a Highly Cited Researcher 2023, in 2024 Edition of the Ranking of Top Scientists in the field of Computer Science. He has published more than 300 scientific research articles in prestigious international journals. His research interests include Meta-heuristics Optimization Algorithms, Artificial Intelligence, WSN, Bioinformatics, Internet of Things, Artificial Intelligence, Image Processing, and Data Mining. He serves as a reviewer for more than 120 journals, such as Elsevier, Springer, and IEEE.

Amr M. Gamal holds a bachelor's degree in computer science from the Faculty of Computers and Information, Minia University. I also earned a postgraduate diploma in Computer Science from the Faculty of Computers and Artificial Intelligence at Cairo University. I currently serve as the Director of the Infrastructure and Information Security Department at the General Authority for Healthcare Accreditation and Regulation (GAHAR) in Egypt. Amr has a strong interest in emerging technologies, including Machine Learning, Information Security, and Blockchain. My work reflects a commitment to advancing trustworthy and secure AI applications in healthcare and other critical domains.

Eman M. G. Younis received B.Sc. degree from Zagazig University, Egypt, in 2002. She obtained her M.Sc. degree from Menoufia University, Egypt, in 2007. She gained her Ph.D. degree from Cardiff University, UK, in 2014. She joined NTU, the Computing Department in 2017. He is currently a Professor at Minia University, Faculty of Computers and Information, Information Systems Department. She is interested in affective computing, big data analysis, Machine Learning, Data Mining, Semantic Web, and GIS.

Ebtsam Mohamed is a Lecturer in the Department of Computer Science at Minia University since 2007. I received a B.S. from Minia University in 2007, and an M.S. from Cairo University at Egypt. I received my Ph.D. in Computer Science from Mansoura University in 2020. My research interests include Machine learning, Information Retrieval, Sentiment Analysis and Artificial intelligence.